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Chapter 2. Manuscript frames 140–207. Catalogue snapshot: 5 October 2026.

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Chapter 2

GELFAND-P000140

Gelfand seminar. $5/\overline{\text{IX}}$

1. There will be systematic stat. physics. (Minlos)
2. Status of pseudodifferential operators. (Vishik and Agranovich)
3. Algebraic groups (Kirillov)



Diagram and typed description. At the upper right, page number 5 has a line at its left and below it.

5

Original source details: GELFAND-P000140

Valuable information: Atiyah's lectures on the K-functor require no prior knowledge except the def. of a topological space and of an n-dimensional vector space (In degree of clarity it is not comparable even to Milnor)



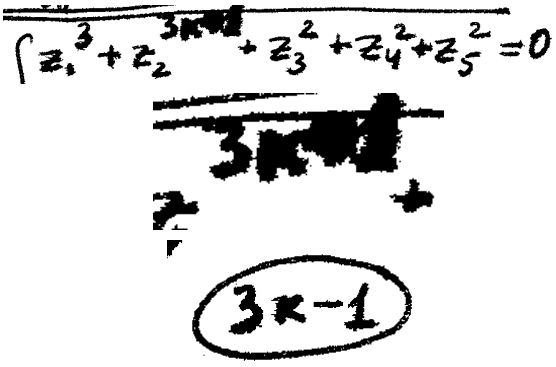
Manuscript notation. A fine stroke joins o and з in «прозрачности» (clarity).

Original source details: GELFAND-P000140

4. Deformations of every kind. (provisional name: nonlinear analysis), including Kodaira and Spencer.

Hirzebruch's result. (Gelfand)

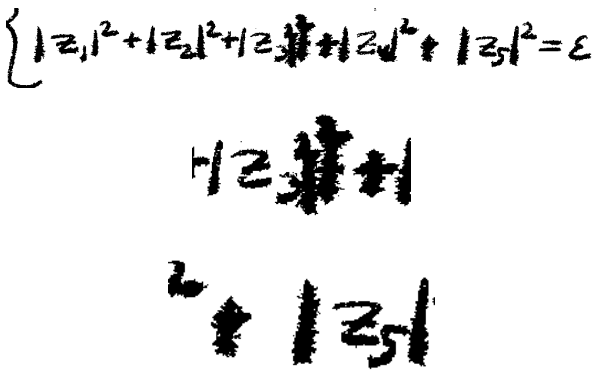
$$\begin{cases} z_1^3 + z_2^{3k-1[?]} + z_3^2 + z_4^2 + z_5^2 = 0 \\ |z_1|^2 + |z_2|^2 + |z_3|^{2[?]} + |z_4|^2 + [?]\,|z_5|^2 = \varepsilon \end{cases}$$



Uncertain reading. The exponent on z_2 begins $3k$; its ending is heavily overwritten: $3k-1$ / another overwritten ending. Separately at the right, $3k-1$ is circled with an oval. The written exponent on z_1 is 3.

$3k-1$

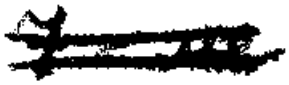
Original source details: GELFAND-P000140



Uncertain reading. In the second line, the right bar and exponent at $|z_3|$ are heavily overwritten; $|z_3|^2$ is given with this qualification. Between $|z_4|^2$ and $|z_5|^2$ the sign resembles $+$ but is dense.

Original source details: GELFAND-P000140

10 unk. and [crossed-out unread fragment] 3 eqns.



Uncertain reading. A short crossed-out fragment after "and": a 7-like sign and further dense strokes; the original content is unreadable.

Original source details: GELFAND-P000140

Milnor: this is homeomorphic to S^7

Hirzebruch: for $k = 1, 2, \ldots, 28$ there are different diff. structures there, and thereafter everything repeats periodically and these are all the Milnor spheres.

S^{11} there are 992 of them and they are obtained by such equations.

GELFAND-P000141

Bass (Kazhdan) rigidity of arithm. subgroups.

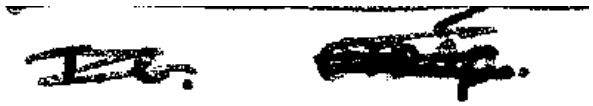


Manuscript notation. After Bass, the parentheses contain Каждан (Kazhdan).

Original source details: GELFAND-P000141

$SL(n, Z) \subset SL(n, R)$ — a rigid subgroup,

[A short opening and a word before "every" are crossed out.]



Uncertain reading. The initial crossed-out designation resembles "II." / Cyrillic "Те."; the following heavily cancelled word is unreadable.

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every deformation, (i.e. a map)

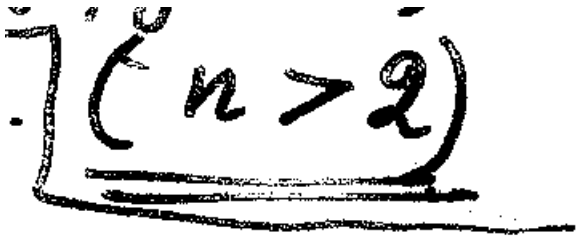


Diagram and typed description. At right, $(n > 2)$ has two underlines. The angular stroke on the left joins the long line above the condition.

$(n > 2)$

Original source details: GELFAND-P000141

$$SL(n, Z) \times [0, 1] \longrightarrow SL(n, R)$$

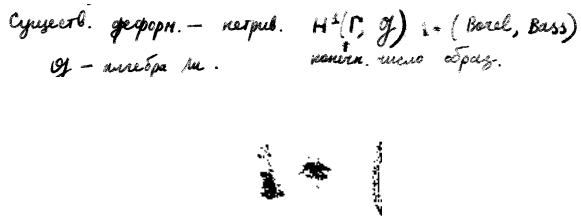
which is continuous and a homomorphism for $\forall t \in [0,1]$, at $t=0$ it is the inclusion) is trivial, i.e. it is an inner automorphism with respect to $g(t)$ (a curve in the group)

$$\varphi(h \times t) = g(t)h \cdot g(t)^{-1}, \qquad h \in SL(n, Z)$$

Weil's theorem is proved only for the case when the quotient space is compact.

The prob. reduces to a problem about congruence subgrps. over a finite field. (If there are two homomorphisms, it is unknown whether they can be joined)

Existence of deformations — nontrivial



Original source details: GELFAND-P000141

Uncertain reading. After $H^1(\Gamma, \mathfrak{g})$ are small colon-like and star-like strokes; their literal reading is not established. Below the expression is "finite no. of generators" with an upward arrow.

$H^1(\Gamma, \mathfrak{g}){:}*[?]$ (Borel, Bass)

↑

finite no. of generators

$\mathfrak{g}$ — the Lie algebra.

In $\mathfrak{g}$ there is a lattice of integer matrices L. (L — a lattice in the Lie algebra)

$$H^1(\Gamma, \mathfrak{g}) = H^1(\Gamma, L) \underset{Z}{\otimes} R$$

Prove that $H^1(\Gamma, L)$ is finite, It suffices to prove that $\exists \underset{\text{prime}}{p}$ such that multiplication by p is an epimorphism

$$H^1(\Gamma, L) \xrightarrow{p} H^1(\Gamma, L) \longrightarrow H^1(\Gamma, L/pL)$$
$$(\text{from } 0 \longrightarrow L \xrightarrow{p} L \longrightarrow L/pL \longrightarrow 0)$$

It suffices to prove that the right-hand term is 0.

One can reduce the matter to the case when Γ is finite: in place of Γ — $\Gamma/[\Gamma_p, \Gamma_p]$, Γ_p — the grp. of elems. congruent to 1 modulo p.

GELFAND-P000142

Thus, it suffices to prove that

$$H^1(\Gamma/\Gamma_p, L/pL) = 0$$

and everything reduces to fin. groups. The same Borel an incredibly difficult proof. Bass — only for split ones.

Smale (Anosov)



Diagram and typed description. Page number 6 at the upper right, with a line at its left and below it.

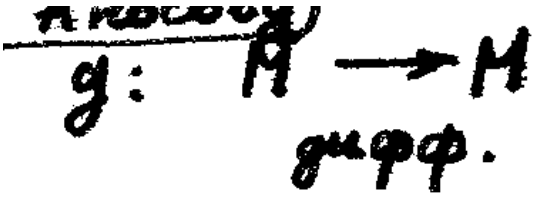
6

Original source details: GELFAND-P000142

Arnold. U-system (according to Anosov)

$$g : M \underset{\text{diff.}}{\longrightarrow} M,$$

if



Original source details: GELFAND-P000142

Manuscript notation. Both letters M have a visible central dip. Beneath the arrow is “diff.”.

there are 2 dirs.



Original source details: GELFAND-P000142

Diagram and typed description. Two separate intersections of horizontal and nearly vertical segments; a gap in the upper part of the first vertical stroke. There are no arrows or letter labels.

along the 1st it stretches, along the other it contracts ($\leq$, $\geq$ estimates on contraction and stretching)



Original source details: GELFAND-P000142

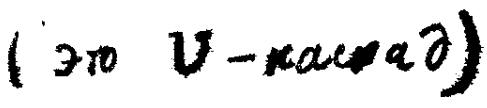
Manuscript notation. After “along the 1st”, stretching is written as «растяг.».

> Example: autom. of the torus, nothing more was known.
> Smale asserts that he has found manifolds besides the torus and their U-automorphisms and all such automorphisms (conjecture).

Precise def.: $\exists TM_x = X_x \oplus Y_x, \forall \xi \in X_x, g^*\xi \in X_{gx}$,
a Riem. metric exists

$$|g^*\xi| < \theta|\xi|, \quad 0 < \theta < 1,$$
$$\forall \eta \in Y_x, \quad |g^*\eta| > \frac{1}{\theta}|\eta|.$$

(this is a V [?]-cascade)



Original source details: GELFAND-P000142

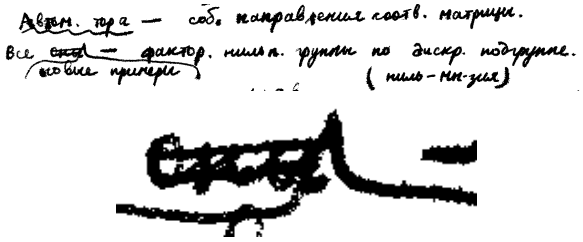
Uncertain reading. The opening heading has U, whereas this sign resembles V: V-cascade / U-cascade.

Autom. of the torus — eigendirections of the corresponding matrix.

All [crossed-out unread word] — a quotient of a nilp. group by a discrete subgroup.

new examples

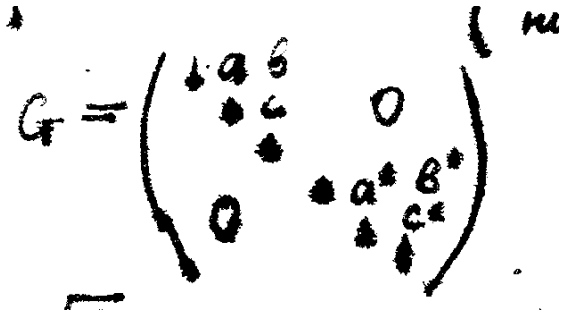
(nil-manifolds)



Original source details: GELFAND-P000142

Uncertain reading. The crossed-out word after “All” has Cyrillic с/ю/д-like remnants; the original word is unreadable. The insertion “new examples” is visible beneath a spanning curved mark.

G/Γ, $G =$



Original source details: GELFAND-P000142

Uncertain reading. A matrix in round parentheses, six diagonal positions, two upper-triangular blocks. At the top a, b, below c; in the lower block a^*, b^*, c^*; two 0 signs outside the blocks. The six dense diagonal marks are shown as dots: 1 / black dots; their exact literal reading is not established.

$$\begin{pmatrix} a & b & \\ & c & 0 \\ & & \\ 0 & & a^* \quad b^* \\ & & \quad\ c^* \\ & & \end{pmatrix}$$

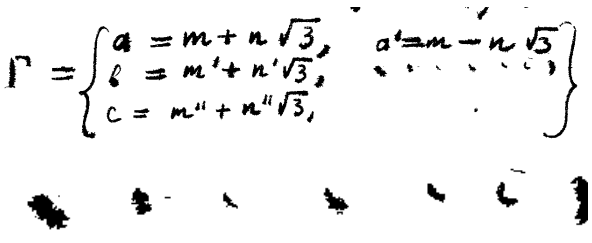
— of order 6

$$\Gamma = \begin{cases} a = m + n\sqrt{3}, & a^{*[?]} = m - n\sqrt{3} \\ b = m' + n'\sqrt{3}, & \text{'} \quad \text{'} \quad \text{'} \quad \text{'} \quad \text{'} \quad \text{'}, \\ c = m'' + n''\sqrt{3}, & \end{cases}$$

G/Γ is compact.

Dim. $3 + 3$

contract. and expand.



Original source details: GELFAND-P000142

Uncertain reading. A short dense mark above the second a: a star / a stroke; a^* is given with this qualification. Six short repetition strokes stand at the right of the b line; their content is not expanded. An encompassing right brace. The horizontal rule beneath “G/Γ is compact.” runs above “Dim. $3 + 3$”.

GELFAND-P000143

Anosov proper.

Cascade — iterations of a diff. homeomorphism (with continuous time — a flow) Describe typical dynamical systems — cascades. (an everywhere dense set, open)

Structurally stable flow — passes to an equivalent one under a small (C^1) perturbation. (equiv. — differ by a diffeom.)

2-dimensional case: Andronov–Pontryagin thm. — structurally stable systems are described (a finite number of "stable" cycles and singular points and saddle separatrices in general position — — do not intersect, do not coincide.) Their set is open, dense.

Andronov–Pontryagin system.

1. A fin. number of equilibria and periodic trajectories, all locally structurally stable (conditions on eigenvalues and multipliers) (in the multidimensional case this is already nontriv.)

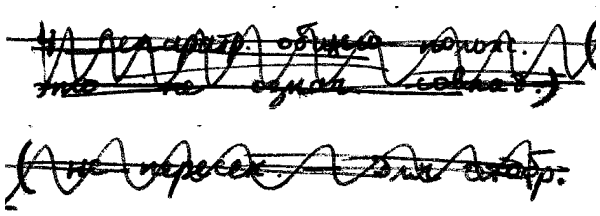
2. Separatrices in general position

(Cond.,
Conditionally stable manifolds,
Asympt. bundles,
horospheres,
horocycles)

(Manifolds of incoming and outgoing — — see Coddington and Levinson)

3. All trajectories as $t \longrightarrow \pm\infty$ tend to equilibria and periodic solutions.

[Two lines are densely cancelled with straight and wavy strokes. Surviving fragments: 4) / ∀ [?], "separatrices in general position", "(do not intersect", 2 [?], Cyrillic "собр."; below, "this does not mean[?] coincidence[?]".]



Uncertain reading. Before "separatrices in general position", the crossed-out sign resembles 4) / ∀. Then "(do not intersect" and several obscured words. The lower line tentatively reads "this does not mean coincidence". The cancellation strokes pass through the letters.

Original source details: GELFAND-P000143

4 assertions.

1. All A.–P. — structurally stable — (?)

2. All structurally stable — A.–P.
3. A.–P. are dense
~~(Cyrillic дру [?]~~
4. Structurally stable are dense

Refuted by Smale.

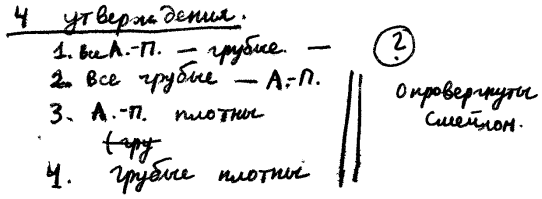
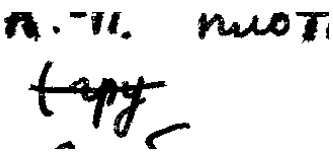


Diagram and typed description. After item 1, a hook-shaped question mark with a separate dot is circled. At the right of items 2–4 are two vertical rules and the words "Refuted by Smale".

Original source details: GELFAND-P000143



Uncertain reading. A short cancelled parenthesis beneath item 3: Cyrillic "дру" / "друг" / unreadable.

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GELFAND-P000144

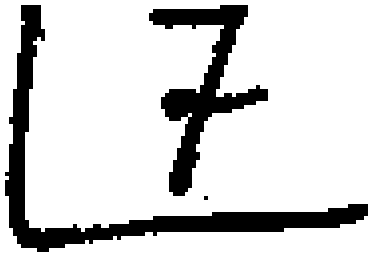


Diagram and typed description. Page number 7 at the upper right, with a line at its left and below it.

7

Original source details: GELFAND-P000144

Structurally stable system: $\dot{w} = f(w)$.

$$\forall \varepsilon\ \exists \delta, \quad \forall g, \quad \|g - f\|_{C^1} < \delta,$$

$\exists\varepsilon$-homeom., taking trajectories of f to trajectories of g.

For a cascade: commutative diagram

($\exists\varepsilon$-homeom. χ)

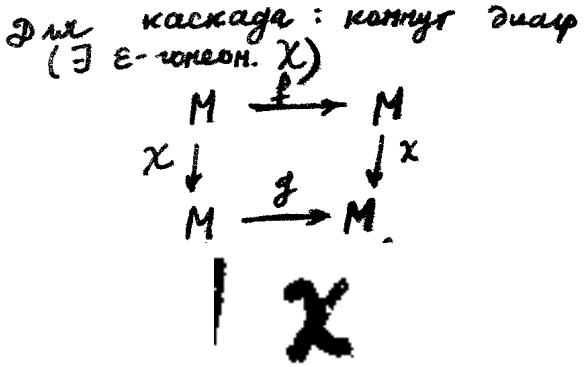
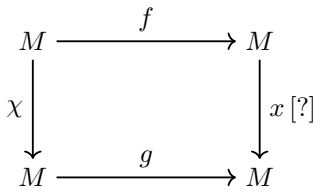


Diagram and typed description. Four M nodes; f above, g below, both horizontal arrows point right, both vertical arrows down. At the left, χ; at the right, a short x- / χ-like sign.



Original source details: GELFAND-P000144

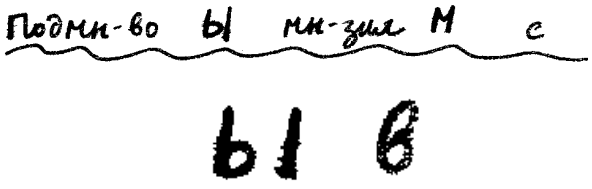
(For flows it is natural to assume that the speed may change; if this is forbidden, almost no structurally stable ones remain)

A subset $b|$ [?] of a manifold M with a smooth cascade is called a U-subset (hyperbolic subset)

compact,

invariant,

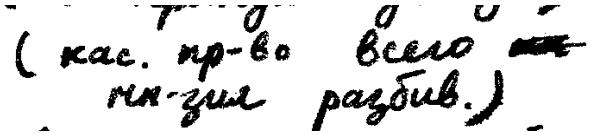
$\exists\Theta$ and a decomposition of the tangent bundle over $b|$ [?] into a direct sum, having the indicated properties. (U ...)



Uncertain reading. Subset designation: $b|$ / b_1 / another sign; a loop and a separate tall stroke. Repeated after “over”.

Original source details: GELFAND-P000144

(the tangent space of the whole [unreadable short cancelled fragment] manifold is decomposed.)



Uncertain reading. After “whole”, a short fragment is densely cancelled; the literal word cannot be read.

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Example: 1) a hyperbolic equilibrium (e.g. stable — — expanding — the whole space)

2) any U-system

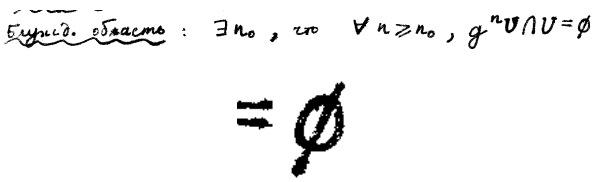
3) There are examples when such a set is a torus, a Cantor set, etc.

Smale systems.

Nonwandering point.

Wandering region: $\exists n_0$ such that

$$\forall n \geq n_0, \quad g^n U \cap U = \varnothing$$

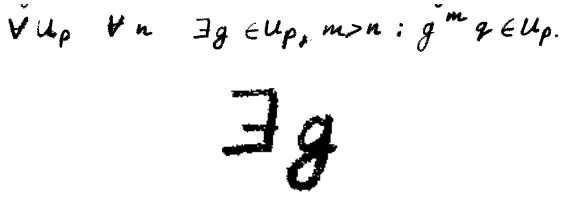


Manuscript notation. The oval to the right of the equality has a diagonal stroke extending past it: $\varnothing$.

Original source details: GELFAND-P000144

A point is called wandering if it has a wandering neighbourhood.
A nonwandering point — a point that is not wandering, i.e.

$$\forall U_p \; \forall n \quad \exists g \in U_p, \quad m > n: \quad g^m q \in U_p.$$



Manuscript notation. The g after $\exists$ has the same lower loop as the base of g^m. The letter after g^m is q.

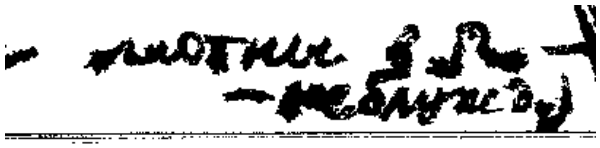
Original source details: GELFAND-P000144

The set of nonwandering points is closed. (denoted — Ω)

Smale system (def)

1. Ω — hyperbolic.

2. $\overline{\Gamma} = \Omega$ (periodic points are dense in Ω — nonwandering [?])

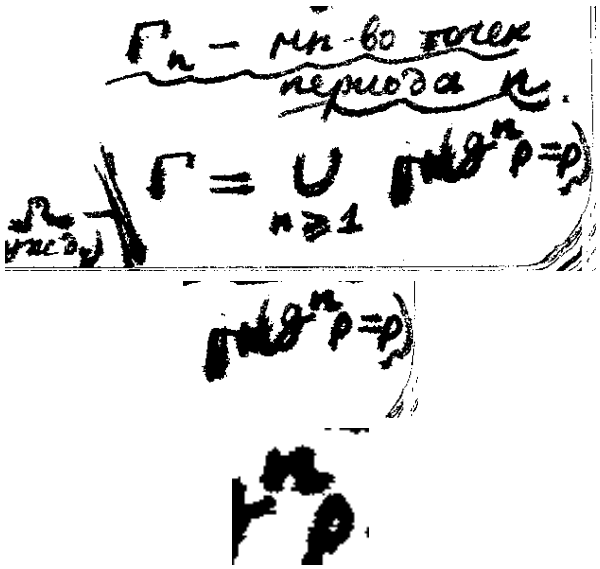


Uncertain reading. The explanation ends at the lower boundary: “— неблужд.” (nonwandering) / a similar contraction; dense strokes.

Original source details: GELFAND-P000144

Γ_n — set of points of period n.

$$\Gamma = \bigcup_{n\geq 1} \Gamma_n\,[?] \quad (g^n p = p)\,[?]$$



Uncertain reading. After the union sign, Γ_n-like signs, the opening parenthesis and g are densely overwritten; the exponent is n-like. The original overwritten signs are undetermined. At the left are two slanting strokes with a short transverse stroke.

Original source details: GELFAND-P000144

GELFAND-P000145

It can happen that all points are nonwandering, but there are no periodic ones (e.g. nil-flows [?])



Uncertain reading. Faint parenthetical entry: “ниль-потоки” (nil-flows) / “нуль-потоки” (null-flows); the middle letter is not quite clear.

Original source details: GELFAND-P000145

Conjectures.

$\{\overline{\Gamma} = \Omega\}$ — everywhere dense — proved by Pugh; 1964

(only in C^1) !!



Manuscript notation. Before “; 1964” the name Pugh is written «Пью».

Original source details: GELFAND-P000145

Smale systems are everywhere dense (?)

A great problem to which it is worth devoting one’s life.

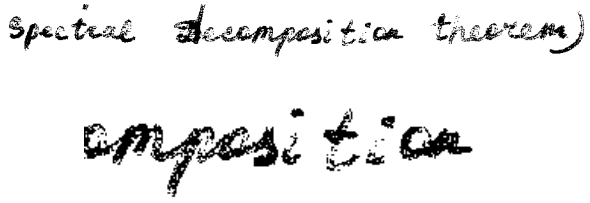
Apparently 2 follows from 1 (Prob. 70%), so it suffices to prove that systems with hyperbolic Ω are dense.

Theorem S. D. T. (Smale) (Spectral Decomposition theorem)



Manuscript notation. The initials S. D. T. precede “(Smale)”.

Original source details: GELFAND-P000145



Manuscript notation. The last o and n in Decomposition are joined.

Original source details: GELFAND-P000145

on the structure of Smale systems.

$$\Omega = \bigcup_{i=1}^{k} \Omega_i$$

Ω_i — invariant, closed and topologically transitive.

All theorems follow from 1 alone.

GELFAND-P000146

$12/\overline{\text{IX}}$ Gelfand sem.



Diagram and typed description. Two wavy lines beneath the name “G. E. Shilov.”

G. E. Shilov.

Original source details: GELFAND-P000146

$$S = \{x \in H,\ |x| \leq 1\}$$

— the unit ball of a Hilbert space.

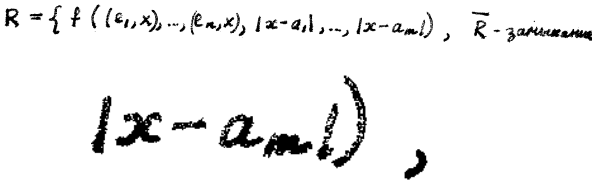
$L(S)$ — ring of all weakly continuous functions.

$$L(S) \subset \overline{R} \subset C(S)$$

$C(S)$ — strongly continuous functions.

$$R = \{f((e_1, x), \ldots, (e_n, x), |x - a_1|, \ldots, |x - a_m|),$$

$\overline{R}$ — closure



Uncertain reading. After the long formula for R, a closing round parenthesis and a comma are visible; a closing brace for the initial { is not visible.

Original source details: GELFAND-P000146





Uncertain reading. The chain $L(S) \subset R \subset \overline{R} \subset C(S)$ is cancelled; the first letters and first inclusion sign are overwritten, and the beginning is not fully established.

Original source details: GELFAND-P000146

PIM of the ring $L(S)$ — the Hilbert ball

PIM of $\overline{R}$ — a bicompact extension.

$$M_\tau(f(x), z) \qquad M^*_\tau(f(x), z)$$



Uncertain reading. The mean's subscript is τ- / r-like; the same looped sign occurs in the later limit formula and squared in its denominator.

Original source details: GELFAND-P000146

$e_1, e_2, \ldots, e_n, \ldots$ — an orthogonal basis, determining an increasing chain of subspaces $E_1 \subset E_2 \subset \ldots$



Diagram and typed description. Two ovals around "over the sphere" and "over the ball".

over the sphere over the ball

Original source details: GELFAND-P000146

The means — the limit of finite-dimensional ones. If they exist, they are equal.

The means are defined on $\overline{R}$.

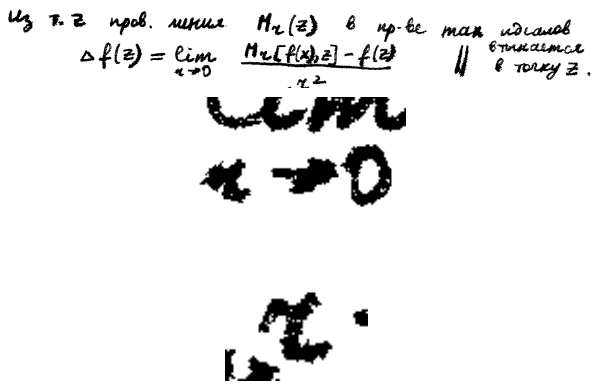
(For R, one can simply write down formulas).

M_τ — a multiplicative functional. This is a max ideal, but it is not a value at a point, but a value in a bicompact extension.

From the point z, lines $M_\tau(z)$ are drawn in the space of max ideals.

$$\Delta f(z) = \lim_{\tau \to 0} \frac{M_\tau[f(x), z] - f(z)}{\tau^2}.$$

$\Big\|$ stick into the point z.



Uncertain reading. The looped sign in M_τ, below the limit and in τ^2 is τ / r / an undetermined letter. At the right of the formula are two vertical rules and "stick into the point z".

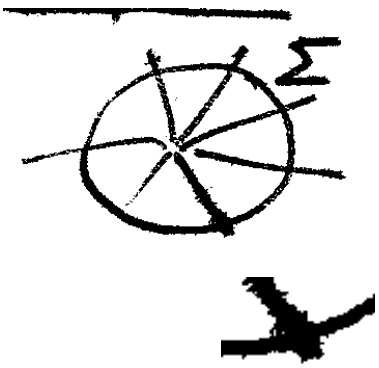
Original source details: GELFAND-P000146

This is a first-order operator!!

$$\Delta(fg) = f \cdot \Delta g + (\Delta f)g$$

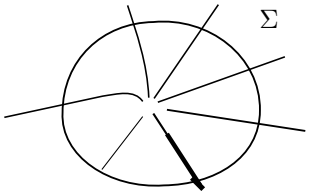
Dirichlet problem.

Σ — a closed surface (star-shaped)
(weak)



Original source details: GELFAND-P000146

Diagram and typed description. An irregular rounded closed contour with Σ at the upper right; seven radiating strokes, with small gaps at the centre. Several pass beyond the contour, the lower-left one ends inside; the left stroke bends before the centre. The lower-right stroke thickens at the contour; a separate arrowhead / point is undetermined.



$$\Delta u(x) = 0,$$
$$u(x)|_{x=\sigma} = f(\sigma)$$

GELFAND-P000147



Original source details: GELFAND-P000147

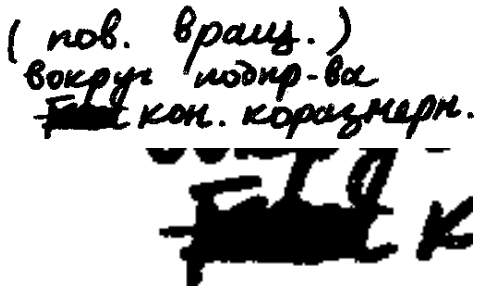
Diagram and typed description. At the upper right, page number 10 has a line at its left and below it.

10

There is uniqueness and continuous dependence for $f \in \overline{R}$ and surfaces of the form

$$\sum_{k=m+1}^{\infty} \xi_k^2 = \varphi(\xi_1, \ldots, \xi_m)$$

(surface of rotation) about a subspace [unreadable short cancelled fragment] of finite codimension.



Original source details: GELFAND-P000147

Uncertain reading. Before “of finite codimension”, a short fragment is densely cancelled; the literal word is undetermined.

and bounded curvature.

The solution formula — the angular mean of the boundary functional.

$\Delta u = g \in \overline{R}$ — also explicit

Next, $\Delta u = \varphi(x, u(x))$ is solved by the fixed-point method.



Diagram and typed description. Two wavy lines with cross-strokes beneath the name “I. M. Gelfand.”

I. M. Gelfand.

Original source details: GELFAND-P000147

$u = (u_1, \dots, u_n)$



Manuscript notation. The final component is u_n. The n is shorter than the m in the formula above.

Original source details: GELFAND-P000147

$$\frac{\partial u}{\partial t} = A(u)\frac{\partial u}{\partial x}$$

Hyperbolic (at $\forall$ point, $A(u)$ has real eigenvalues)

Is there uniqueness? Shock waves — the solution may be nonunique.

$n = 1$ — always uniqueness if $f(u)$ is convex

Hydrodynamic equations — [cancelled: “nonunique” / “may” / unreadable] uniqueness too. ($n = 2$)



Uncertain reading. Before “uniqueness too”, a word is cancelled: “может” (may) / “неод.” (nonunique) / another short fragment; the cancellation crosses the letters.

Original source details: GELFAND-P000147

In the general case it is unclear where there is uniqueness and where there is not.

Magn. hydrodyn. — uniqueness too.



Manuscript notation. Before “hydrodyn.” the abbreviation is «Магн.» (Magn.).

Original source details: GELFAND-P000147

A purely mathematical question: at $\forall$ point u, draw the eigenvectors — 3 families of lines — — characteristics.

Convexity: possibly, growth of λ along the characteristics

Better:

$$\frac{\partial u}{\partial t} = \frac{\partial f(u)}{\partial x}$$

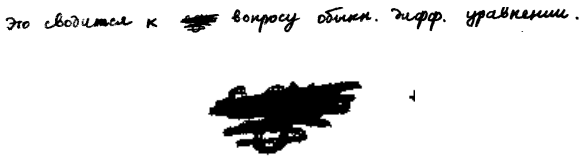
(divergence form)

There is work by Godunov, but it concerns a narrow class of systems.

When is there uniqueness upon adding viscosity:

$$\frac{\partial u}{\partial t} = A\frac{\partial u}{\partial x} + \varepsilon\frac{\partial^2 u}{\partial x^2} \quad ?$$

This reduces to the [unreadable cancelled word] question of ordinary differential equations.



Uncertain reading. In the source, between “к” (to) and “вопросу” (question), a densely cancelled word with a looped lower stroke; the literal reading is undetermined.

Original source details: GELFAND-P000147

GELFAND-P000148

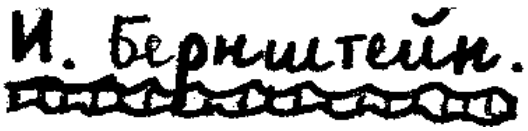


Diagram and typed description. Double wavy underline with cross-strokes under “I. Bernstein.”

I. Bernstein.

Original source details: GELFAND-P000148

Part of what Atiyah spoke about.

g — a compact Lie group. R_g — the ring of finite-dimensional representations (direct sum, tensor product and formal differences), commutative.

An embedding

$$R_g \longrightarrow K(B_g)$$

B_g — the classifying space, K — the K-functor.

A representation corresponds to a vector bundle on B_g (associated).

(I.) If the ring R_g is completed at the ideal of representations of dimension zero, there will be an isomorphism

$$\widehat{R}_g \longrightarrow k(B_g)$$

(The dimension of an element of the form of a difference of two representations — the difference of their dimensions)

Example.

$$g = S^1, \qquad R_{S^1} = \sum_{i=-\infty}^{+\infty} n_i x^i,$$

n_i — integers, finitely many nonzero.

The ideal is generated by the element $x - 1 = y$.

$$\widehat{R}_{S^1} = Z[[y]]$$

(formal power series)

Only positive powers

$$\left(\frac{1}{1-y} = 1 + y + y^2 + \dots\right)$$

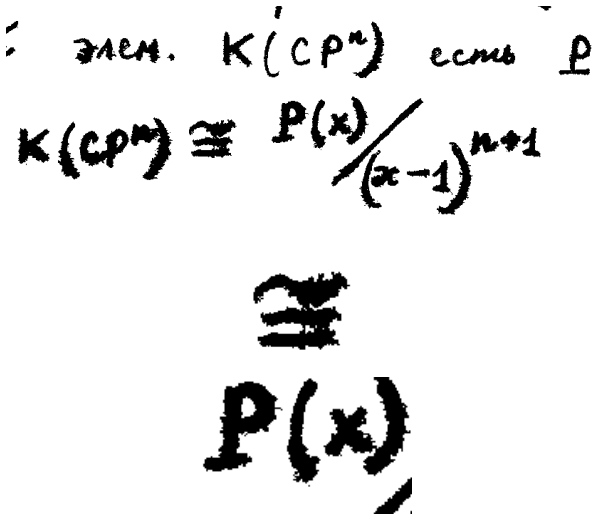
$$B_g = CP^\infty,$$
$$k(CP^\infty) = \lim K(CP^n).$$

On CP^n, the canonical bundle (fibre CP^1), total space CP^{n+1}.

Every element of $K(CP^n)$ is $P(x-1)$.

Precisely

$$K(CP^n) \cong P(x)/(x-1)^{n+1}.$$



Manuscript notation. The sign is ≅, a wave over two horizontal strokes. The numerator is $P(x)$.

Original source details: GELFAND-P000148

Next time, a lecture by Minlos on statistical physics.

GELFAND-P000149

19/IX. Gelfand sem.

I. Bernstein (continuation).

$K(B_g)$ is defined as follows:

Let

$$(B_g)_1 \longrightarrow (B_g)_2 \longrightarrow \dots$$

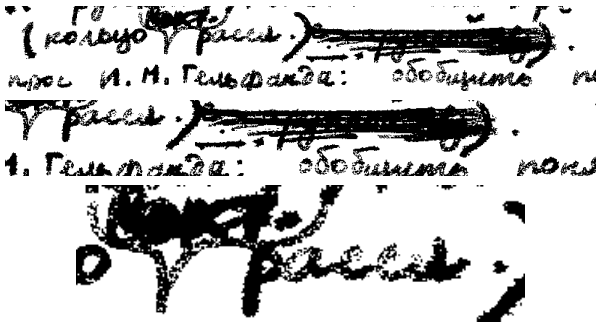
be the skeleta of the corresponding dimensions

$$K_1 \longleftarrow K_2 \longleftarrow \dots$$

One must take the projective limit.

The K-functor of a finite polyhedron is defined in the usual way (ring [overwritten: K-like sign, ?] bundl.) [~~unread long entry~~]) .

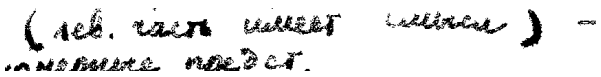
$\longrightarrow$ •



Uncertain reading. After “ring” there is an overwritten K-like sign, then the abbreviation for “bundles” and a closing parenthesis. A dense cancellation follows, with strokes below and another closing parenthesis. Under its beginning is a short rightward arrow and a dot. The cancelled letters remain unread.

Original source details: GELFAND-P000149

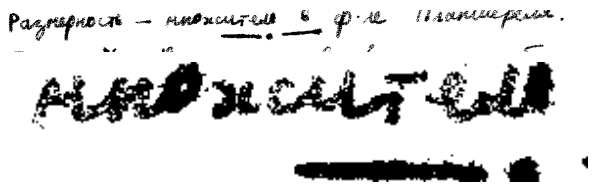
Question of I. M. Gelfand: generalize the notion of the K-functor and classifying space so that the theorem remains valid also for locally compact Lie groups. (the left-hand side makes sense) — — one must also use infinite-dimensional representations.



Original source details: GELFAND-P000149

Diagram and typed description. A wavy line below the heading. The faint parenthesis reads “the left-hand side makes sense”; the abbreviation at the right of the preceding line is “лок.”.

Dimension — a factor in the Plancherel formula.

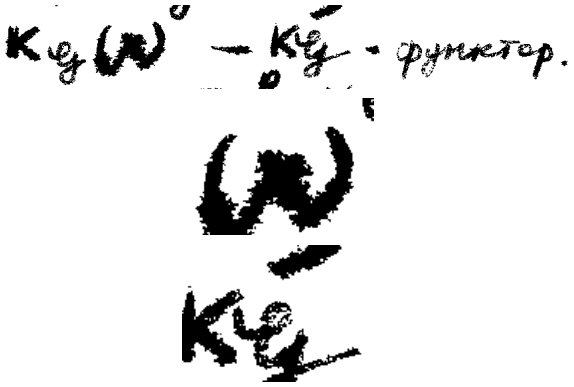


Original source details: GELFAND-P000149

Manuscript notation. After the dash the word reads “factor.” Below are two strokes with a dot between them. The second stroke rises slightly.

Let X be a g-space (i.e. there is a map $g \times X \longrightarrow X$, continuous in both arguments, and there is an action of g on X)

$K_g(X\,[?])$ — the K_g[?]-functor.



Original source details: GELFAND-P000149

Uncertain reading. At the second K, small loop-shaped strokes stand to the right, above and below: a K_g / K^g / K_g^0-like entry; a separate 0 is not established. In the parentheses of the first K_g, an X-like letter is densely overwritten.

A g-bundle $E \xrightarrow{p} X$ — a vector bundle, then E ... — a g-space, g acts compatibly with p. In the fibres g must preserve the linear structure.

In the usual way, sum and tensor product are defined and formal differences are introduced.

Example. K_g — the functor of a point — — linear spaces with representations of g, this will be exactly R_g.

~~[Cyrillic H-like beginning, ?]...~~



Original source details: GELFAND-P000149

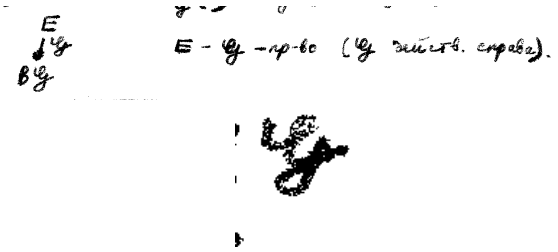
Uncertain reading. Before “Generalization”, a short cancelled fragment with a Cyrillic H-like beginning; the remaining letters are not established.

Generalization. Instead of a point — a g-space X.

The left-hand side $K_g(X)$ — already generalized.

$$\begin{array}{c} E \\ \big\downarrow g \\ B_g \end{array}$$

E — a g-space (g acts on the right).



Original source details: GELFAND-P000149

Manuscript notation. At the bottom, E is above B_g with a downward arrow. To its right is the letter g, as in the following line.

GELFAND-P000150

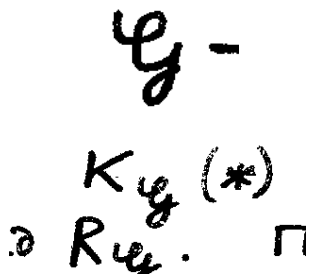


Diagram and typed description. At the upper right, page number 14 has a line at its left and below it.

14

Original source details: GELFAND-P000150

Let X be a G-space. Introduce the space $X_G = E \times_G X$ (in $E \times X$, the orbits of the group G are identified). If X is a point, then $X_G \cong B_G$ — Grassmann.



Manuscript notation. The looped group letter is rendered as G. In the action below, the group element is written differently, with an ordinary g.

Original source details: GELFAND-P000150

Theorem. (a generalization of the one formulated at the beginning)

$$\widehat{K}_G(X) \xrightarrow{\sim} K(X_G).$$

There is a map $K_G(*) \longrightarrow K_G(X)$, so $K_G(X)$ — — a module over R_G. Completion is taken with respect to the ideal J of the ring R_G.

An exact def. can also be given as follows:

$$\widehat{K}_G(X) = K_G(X) \underset{R_G}{\otimes} \widehat{R}_G.$$

———— • ————

1. Reduction to the unitary group U.

Let $G \subset U$. Then $K_G(X) = K_U(U \times_G X)$.

The action of G on U: $g : u \longrightarrow ug^{-1}$. On $U \times_G X$, U acts on the left. Moreover, $X_G = (U \times_G X)_U$.

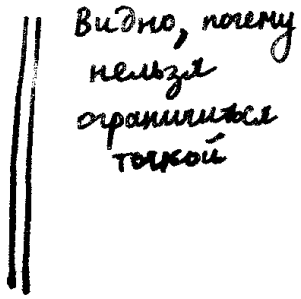


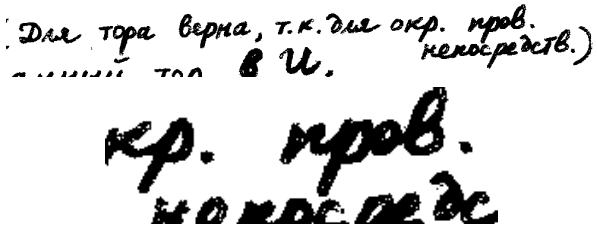
Diagram and typed description. An unfinished four-line note to the right of two vertical strokes.

It is clear why
one cannot
restrict oneself
to such

Original source details: GELFAND-P000150

2. Reduction to the torus case.

(For the torus it holds, since for the circle verif. directly.)



Original source details: GELFAND-P000150

Manuscript notation. After the abbreviation for “circle” is “verif.”, with “directly” abbreviated below.

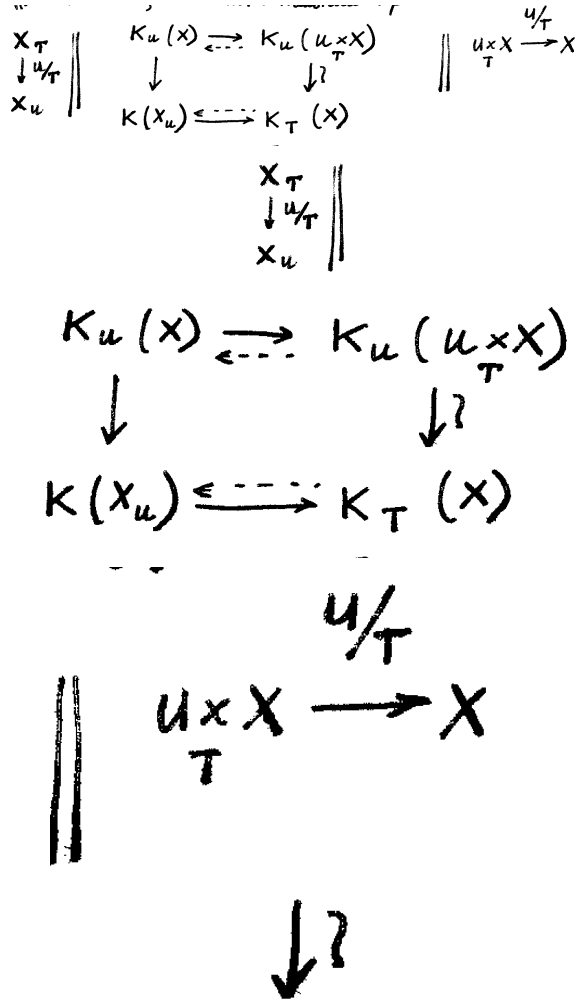
~~[unread cancelled sign]~~.



Original source details: GELFAND-P000150

Uncertain reading. Before $T \subset U$, a short number-like sign is densely cancelled; its original reading is not established. A separate dot at the right.

$T \subset U$, T is a maximal torus in U.

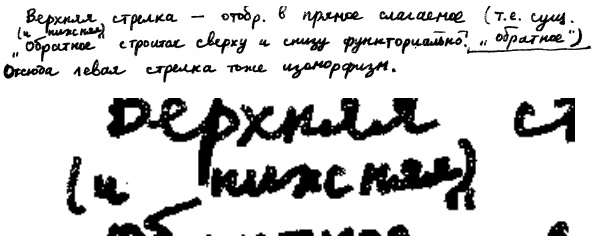


Original source details: GELFAND-P000150

Diagram and typed description. At the left, $X_T \longrightarrow X_U$ downward with U/T. Two double vertical rules separate three parts. In the square, $K_U(X)$ and $K_U(U \times_T X)$ above, $K(X_U)$ and $K_T(X)$ below; solid horizontal arrows rightward, dashed arrows backward, both vertical arrows downward. To the right of the right vertical arrow, a question-like hook with a thickened base; a separate dot cannot be distinguished. Separately at the right, $U \times_T X \longrightarrow X$ with U/T above the arrow.

$$\begin{array}{ccc} X_T & & \\ \downarrow{\scriptstyle U/T} & \| & \\ X_U & & \end{array} \qquad \begin{array}{ccc} K_U(X) & \longrightarrow & K_U(U \times_T X) \\ \downarrow & & \downarrow \;? \\ K(X_U) & \longrightarrow & K_T(X) \end{array} \qquad \begin{array}{ccc} U \times_T X & \xrightarrow{U/T} & X \end{array}$$

The upper arrow is a map into a direct summand (i.e. there exists „inverse (and the lower one) is constructed above and below functorially. „inverse“)



Original source details: GELFAND-P000150

Manuscript notation. The insertion “(and the lower one)” sits above the first “inverse”, beneath the beginning of the line “The upper arrow”.

Hence the left arrow is also an isomorphism.

GELFAND-P000151



Original source details: GELFAND-P000151

Diagram and typed description. A hatched band under the name “R. A. Minlos.”

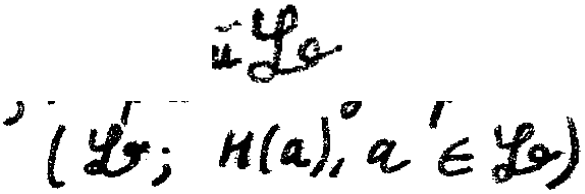
R. A. Minlos.

A phys. system is descr. by a set of states $\mathcal{L}[?]$ and a function $H(a)$ on ~~[unreadable sign]~~, which is called the Hamiltonian fn. (a real-valued fn.). — the pair $(\mathcal{L}[?]; H(a), a \in \mathcal{L}[?])$.



Original source details: GELFAND-P000151

Uncertain reading. After “on” a letter with an upper loop and a lower stroke is densely crossed out; the original reading is not established.



Original source details: GELFAND-P000151

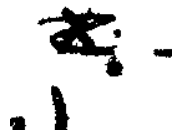
Uncertain reading. The right loop of the state-space sign joins the bottom stroke. It is provisionally set as $\mathcal{L}[?]$. A separate index on this line is unclear.

3 types.

1. Classical system — tuples

$$(p_1, \dots, p_N, x_1, \dots, x_N)$$

All p_i independently range over $\mathbb{R}^\nu$ (in physics $\mathbb{R}^3$) — — the set of momenta.
All x_i range over a region ~~x_i~~ [?] $\Omega \subset \mathbb{R}^\nu$, called the vessel (the reg. is bounded).



Original source details: GELFAND-P000151

Uncertain reading. Before Ω the crossed-out entry resembles x_i. Two separate strokes are visible below its right end.

The tuple x_i — configuration of the particles

$$(\underbrace{p_1, \dots, p_N}_{\mathbb{R}^{N\nu}}, \underbrace{x_1, \dots, x_N}_{\Omega^N})$$

Notation (will sometimes be consid.)

$$(\Omega_\delta)^N : \quad |x_i - x_j| > \delta$$

(the particles — "hard" balls)

This is the set of states.

Hamiltonian fn.:

$$H(a) = \frac{1}{2m} \sum |p_i|^2 + V(x_1, \dots, x_N), \qquad m > 0$$

V is called the potential energy.

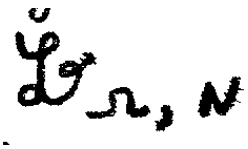
In what follows we consid. only fns. V of the form

$$V(x_1, \dots, x_N) = \sum_{i \neq j} U(|x_i - x_j|)$$

(a system with pairwise interaction)

$U(r)$ — the potential — is def. either for $r > 0$, or for $r > \delta$.

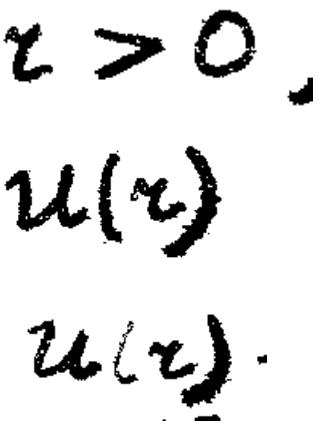
$$\{\mathcal{L}_{\Omega,N}[?], H_{m,U(r)}\}$$



Uncertain reading. Inside the brace is the same connected sign, provisionally rendered as $\mathcal{L}[?]$, with a separate Ω, N below.

Original source details: GELFAND-P000151

Ω — the vessel, N — the number of particles, m — the mass, $U(r)$ — — potential energy.



Manuscript notation. The argument r in $U(r)$, $r > 0$ and $r > \delta$ has the same short upper loop and rightward tail.

Original source details: GELFAND-P000151

GELFAND-P000152



Diagram and typed description. Page number 15 at the upper right, with a line at its left and below it.

15

Original source details: GELFAND-P000152

Identical sys. — differ either in the number of particles or in the vessel. (the same — mass and potential)

δ — is contained in $U(r)$.

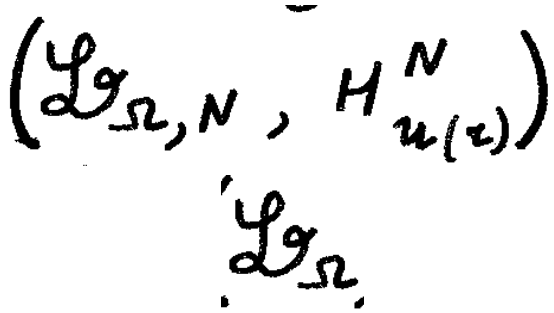
2. Configurational system.

Obt. from the preceding one. A state — $(x_1, \ldots, x_N)$ — the region Ω^N or its subregion $(\Omega_\delta)^N$.

$$H(a) = \sum_{i \neq j} U(|x_i - x_j|)$$

— without kinetic energy.

$$(\mathcal{L}_{\Omega,N}\,[?], H^N_{U(r)})$$



Uncertain reading. The right loop joins the bottom of the sign provisionally rendered as $\mathcal{L}\,[?]$. Below it, Ω, N is clear.

Original source details: GELFAND-P000152

System 1 reduces to system 2. (see below).

3. Discrete model system.

In R^ν a cubic lattice, Ω — a subset of it. A state — a tuple of N lattice sites $(x_1, \ldots, x_N)$, in which no two are equal.

$U(x)$ — an even function on the lattice.

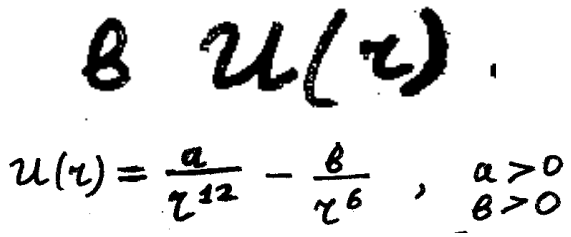
$$H(a) = \sum_{i \neq j} U(x_i - x_j)$$

(Isotropy, e.g. — spherical symm. — is not justified here)

The results of stat. physics depend only on the qualit. properties of $U(r)$, but physicists use only the potential

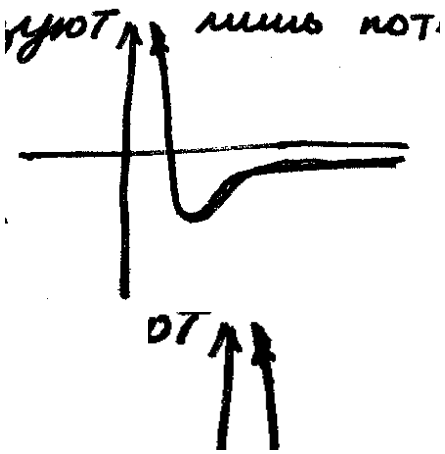
$$U(r) = \frac{a}{r^{12}} - \frac{b}{r^6}, \qquad \begin{matrix} a > 0 \\ b > 0 \end{matrix}$$

What matters to us is the growth at 0 and the decay at ∞.



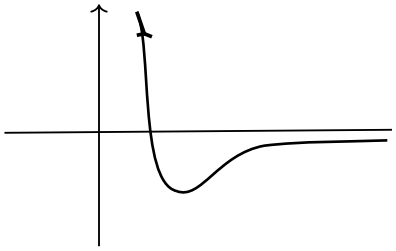
Original source details: GELFAND-P000152

Manuscript notation. The same short looped sign r recurs in $U(r)$ and both denominators.

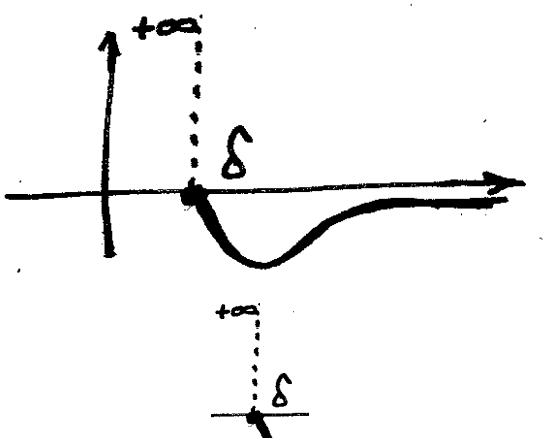


Original source details: GELFAND-P000152

Diagram and typed description. The vertical axis has an upward arrow; no horizontal arrow is established. Neither axis has letters. The branch comes from above to the right of the vertical axis, crosses the horizontal, forms a negative well and approaches the horizontal from below. The upper end of the branch is thick, with short lateral strokes; a separate arrow meaning is not established. No exact scale is given.

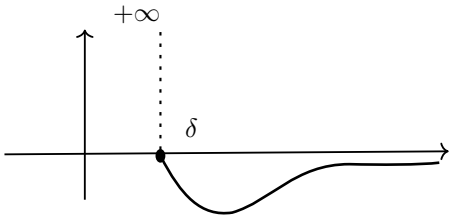


In the presence of δ:



Original source details: GELFAND-P000152

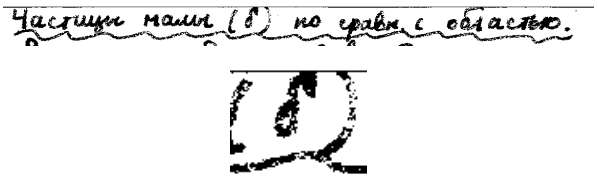
Diagram and typed description. Two axes with arrows. At the right is a dotted vertical; above it $+\infty$, a thick mark at its intersection with the horizontal, and δ to the right. The branch leaves the thick mark for a well and rises below the horizontal axis. The dense ink at the intersection is rendered without an established function value at δ or an assertion of closedness.



Sometimes only a potential of compact support is consid.

GELFAND-P000153

The particles are small (δ) compared with the region.

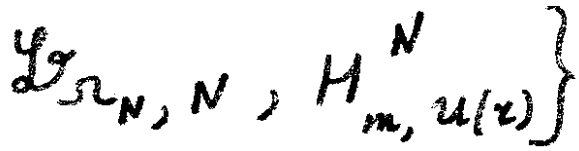


Manuscript notation. The parenthesis contains δ. The top stroke touches the frame edge.

Original source details: GELFAND-P000153

Consid. a seq. of vessels Ω_N which expands and fills the whole space (e.g. similar cubes)
Let us consider

$$\{\mathcal{L}_{\Omega_N,N}[?], H^N_{m,U(r)}\}$$



Uncertain reading. The right loop of the connected state-space sign joins its base. It is provisionally set as $\mathcal{L}[?]$. Below are Ω_N, N.

Original source details: GELFAND-P000153



Manuscript notation. The short looped r in $U(r)$ recurs from the note on the potential.

Original source details: GELFAND-P000153

$$\frac{|\Omega_N|}{N} \longrightarrow v \quad \text{(limiting specific volume)}$$

$|\Omega_N|$ — the volume of the region Ω_N

(Thermodyn. limiting transition)

In most cases everything depends only on v, and the shape of the vessel is immaterial.

Gibbs distribution.

It is assumed that a probability distr. is given on $\mathcal{L}[?]$ In the equilibrium case it is postulated (if everything is independent of time)

$$p(a) = \frac{1}{Q} e^{-\beta H(a)}, \qquad \beta > 0$$

$$Q = \int_{\mathcal{L}[?]} e^{-\beta H(a)}\, da.$$



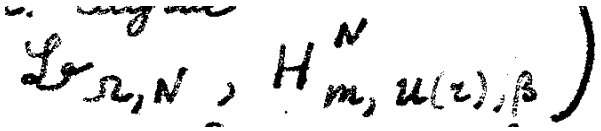
Uncertain reading. The same connected sign appears in the first line and beneath the integral. It is provisionally set as $\mathcal{L}[?]$.

Original source details: GELFAND-P000153

For lattice systems the same formula, but p now gives the probabilities themselves, rather than densities.

A phys. sys. with a given Gibbs distr. is called a small canonical ensemble

$$\begin{pmatrix}\text{in the classical case}\\ \mathcal{L}_{\Omega,N}\,[?],\, H^N_{m,U(r),\beta}\end{pmatrix}$$

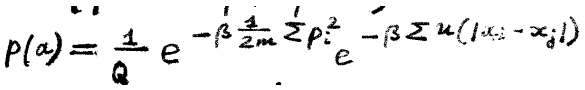


Original source details: GELFAND-P000153

Uncertain reading. The state-space sign is provisionally set as $\mathcal{L}\,[?]$, with Ω, N below. The H has N above and $m, U(r), \beta$ below.

In the thermodyn. limiting transition identical systems are always consid. (β is also fixed)

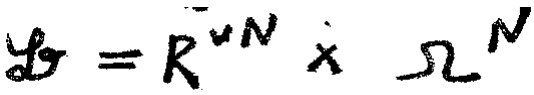
$$p(a) = \frac{1}{Q} e^{-\beta \frac{1}{2m} \sum p_i^2} e^{-\beta \sum U(|x_i - x_j|)}$$



Original source details: GELFAND-P000153

Manuscript notation. The first exponent contains $\sum p_i^2$ and the second $\sum U(|x_i - x_j|)$. Neither sum has written limits.

$$\mathcal{L}\,[?] = R^{\nu N} \times \Omega^N$$



Original source details: GELFAND-P000153

Uncertain reading. On the left is the same connected sign, provisionally set as $\mathcal{L}\,[?]$. On the right is $R^{\nu N} \times \Omega^N$.

GELFAND-P000154

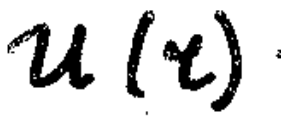


Original source details: GELFAND-P000154

Diagram and typed description. Page number 16 at the upper right, with a line at its left and below it.

16

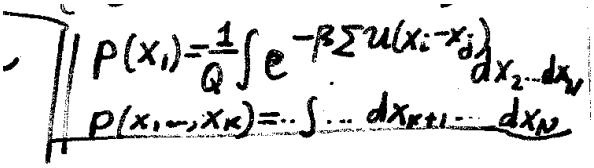
Our distr. is a product of simpler distributions and only the Gibbs distr. in the configuration space is of interest. Where $U(r) = +\infty$, the density will be 0. There are distr. fns. for a subsystem ($\int$ over the others)



Original source details: GELFAND-P000154

Manuscript notation. The r in $U(r)$ has the same form as in the potential on the previous page.

Correlation functions are convenient.

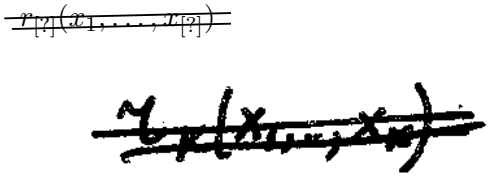


Original source details: GELFAND-P000154

Diagram and typed description. Two formulas to the right of a double rule. The long upper rule extends left over the heading.

$$p(x_1) = \frac{1}{Q}\int e^{-\beta \sum U(x_i - x_j)}\, dx_2 \dots dx_N$$

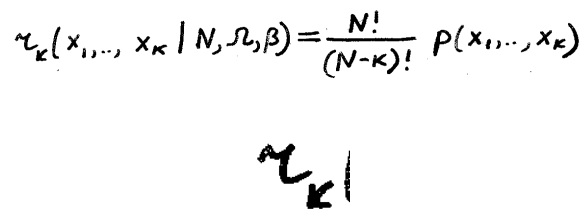
$$p(x_1, \dots, x_k) = \dots \int \dots dx_{k+1} \dots dx_N$$



Original source details: GELFAND-P000154

Uncertain reading. The opening r remains visible above two strokes. The cancelled indices are indistinct.

$$r_k(x_1, \dots, x_k \mid N, \Omega, \beta) = \frac{N!}{(N-k)!} p(x_1, \dots, x_k)$$



Original source details: GELFAND-P000154

Manuscript notation. The r in r_k has the same upper bend and lower rightward stroke as in $U(r)$.

Summatory fn. $F(a)$, $a\,(x_1, \dots, x_N)$

Of the 1st order $F(a) = f(x_1) + f(x_2) + \dots + f(x_N)$

Of the 2nd order $F(a) = \sum_{i \neq j} f(x_i, x_j)$, f — symmetric.

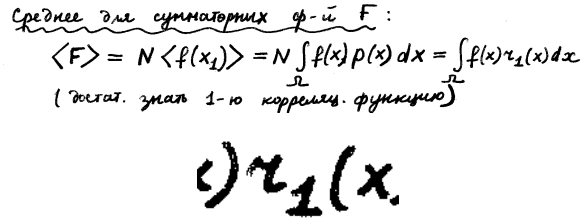
and so on.

Mean for summatory fns. F:

$$\langle F \rangle = N \langle f(x_1) \rangle = N \int_\Omega f(x) p(x)\, dx$$

$$= \int_\Omega f(x) r_1(x)\, dx$$

(it suffices to know the 1st correlation function)

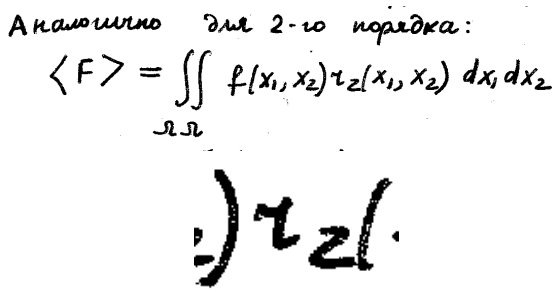


Original source details: GELFAND-P000154

Manuscript notation. The right-hand integral contains $r_1(x)$. Both integration domains are marked Ω.

Similarly for the 2nd order:

$$\langle F\rangle = \int_\Omega\int_\Omega f(x_1,x_2)r_2(x_1,x_2)\,dx_1dx_2$$



Manuscript notation. The integrand contains $r_2(x_1,x_2)$. Each of the two integrals has Ω beneath it.

Original source details: GELFAND-P000154

Probabilistic (phys.) meaning.

$C\subset\Omega$, C — a subregion, $n_C(a)$ — the number of particles that fall in the subregion, $a=(x_1,\dots,x_N)$. This is a summatory quantity of the 1st order, with

$$f(x)=\begin{cases}1, & x\in C,\\ 0, & x\notin C\end{cases}$$





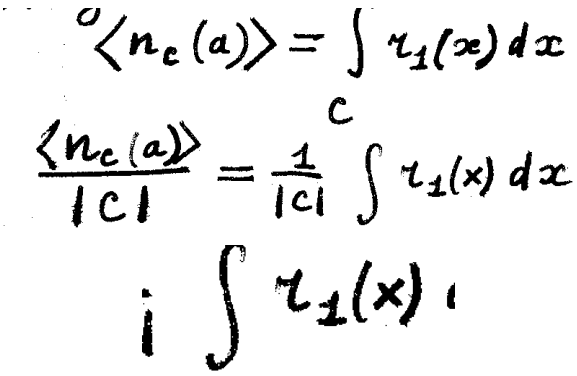
Uncertain reading. Two small angular crossed signs in the left margin; their letter reading and meaning are not established.

Original source details: GELFAND-P000154

Therefore

$$\langle n_C(a)\rangle = \int_C r_1(x)\,dx$$

$$\frac{\langle n_C(a)\rangle}{|C|} = \frac{1}{|C|}\int r_1(x)\,dx$$

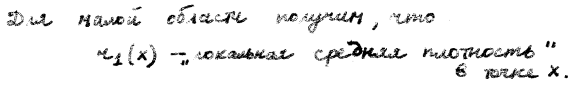


Manuscript notation. The first integral has an explicit C. The manuscript leaves the domain of the last integral unwritten.

Original source details: GELFAND-P000154

GELFAND-P000155

For a small region we obtain that $r_1(x)$ — the "local mean density" at the point x.



Manuscript notation. The manuscript repeats $r_1(x)$ from the previous page.



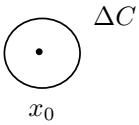
Original source details: GELFAND-P000155

Another probabil. meaning:



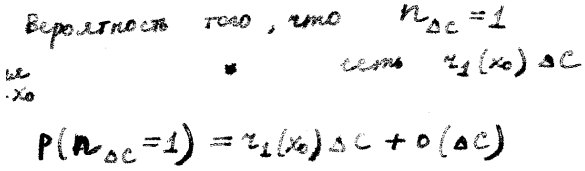
Original source details: GELFAND-P000155

Diagram and typed description. A separate approximate circle with a dot inside; ΔC at the upper right edge and x_0 below.



ΔC — a small nbhd. of x_0.
The probability that $n_{\Delta C} = 1$ is $r_1(x_0)\Delta C$.

$$p(n_{\Delta C} = 1) = r_1(x_0)\Delta C + o(\Delta C)$$



Manuscript notation. Both lines have $r_1(x_0)\Delta C$. The displayed formula ends with $o(\Delta C)$.

Original source details: GELFAND-P000155



Uncertain reading. Between the lines of the first probability explanation is a separate dense ink mark with an irregular outline; no letter reading or meaning is established.

Original source details: GELFAND-P000155

26/IX. Gelfand Sem.
R. A. Minlos. Lecture 2.

(one)
$r(x_1, x_2)\Delta x_1 \Delta x_2$ — the prob. of a particle falling in Δx_1 and a second one in Δx_2.



Manuscript notation. The same r precedes (x_1, x_2). No separate index is written on this letter.

Original source details: GELFAND-P000155



Original source details: GELFAND-P000155

Diagram and typed description. Two separate approximate outlines, with x_1, x_2 below; dense crossed strokes inside, whose separate letter reading is not established. A short stroke above the right outline. Sizes and distance are not given.



$\Omega_N \longrightarrow \infty$ — the thermodyn. limiting transition.

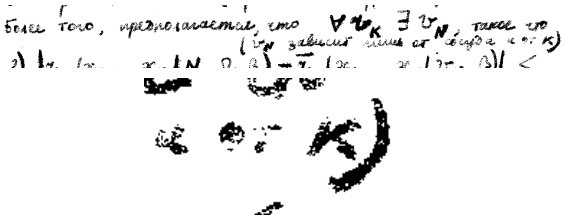
$$\frac{|\Omega_N|}{N} \longrightarrow v = \frac{1}{\rho}, \qquad \rho \text{ — density.}$$

Hypothesis: 1)

$$r_k(x_1, \ldots, x_k \mid N, \Omega, \beta) \longrightarrow \vec{r}_k(x_1, \ldots, x_k \mid v, \beta)$$

This is assumed (limiting correlation function)
Moreover, it is assumed that $\forall r_k \; \exists v_N$ such that

$$(v_N \underset{\longrightarrow}{\text{depends}} \text{ only on the vessel and on } k)$$

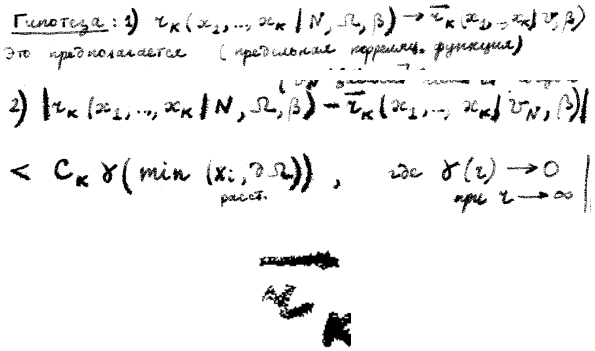


Original source details: GELFAND-P000155

Manuscript notation. Before k is “and on”. A short rightward arrow appears beneath “depends”.

$$2) \quad \left| r_k(x_1, \ldots, x_k \mid N, \Omega, \beta) - \vec{r}_k(x_1, \ldots, x_k \mid v_N, \beta) \right| < C_k \gamma\Big(\min_{\text{dist.}} (x_i, \partial\Omega)\Big),$$

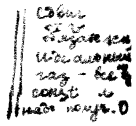
where $\gamma(r) \longrightarrow 0$ as $r \longrightarrow \infty$.

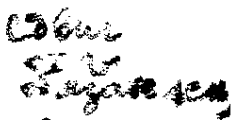


Original source details: GELFAND-P000155

Manuscript notation. A short rightward arrow stands over the limiting function’s r. The estimate has $(x_i, \partial\Omega)$ in parentheses with “dist.” beneath.

Volume [?]
and v [?]
we discuss [?]
Ideal
gas — all [?]
[dense unreadable sign]
const and
need to obtain 0

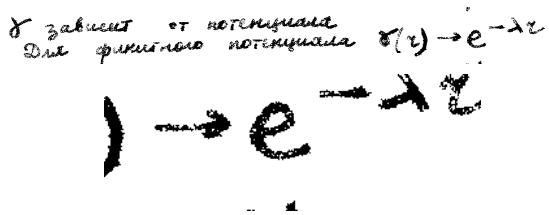




Original source details: GELFAND-P000155

Uncertain reading. Beyond two vertical lines are words resembling “Volume and v we discuss,” then “Ideal gas — all” / “… without …,” const, “and need to obtain 0.” The first connective and endings are faint; a dense unreadable sign follows the gas line. These readings are provisional.

γ depends on the potential. For a potential of compact support $\gamma(r) \longrightarrow e^{-\lambda r}$.



Original source details: GELFAND-P000155

Uncertain reading. A right arrow between $\gamma(r)$ and $e^{-\lambda r}$; separate limit conditions are not repeated in this line.

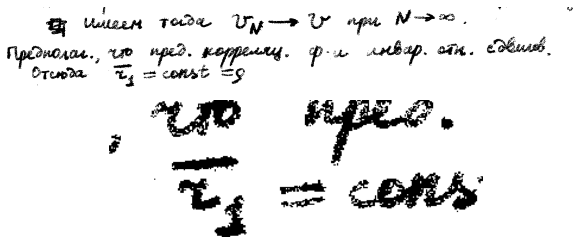
~~[unreadable sign]~~



Original source details: GELFAND-P000155

Uncertain reading. Before “We then have” a short sign or number is densely crossed out; the original reading is not established.

We then have $v_N \longrightarrow v$ as $N \longrightarrow \infty$.
3) Assume that the limiting correlation fns. are invariant under translations. Hence $\vec{r}_1 = \text{const} = \rho$.



Original source details: GELFAND-P000155

Manuscript notation. The last line has $\vec{r}_1 = \text{const} = \rho$. The letters r and ρ have different forms.

GELFAND-P000156

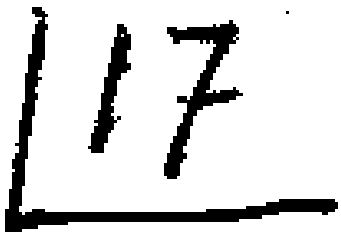


Diagram and typed description. Page number 17 at the upper right, with a line at its left and below it.

17

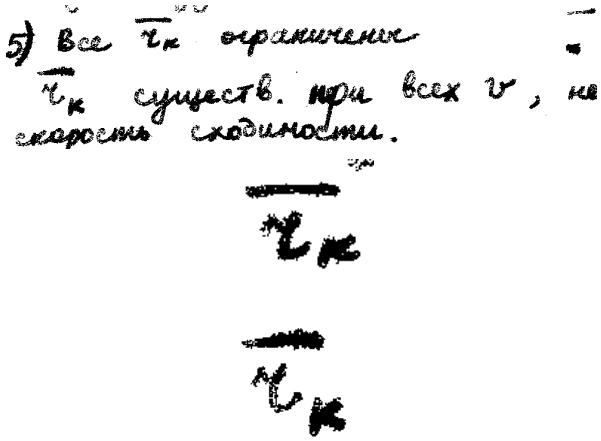
Original source details: GELFAND-P000156

4) Continuity with respect to v and β.

This theorem should hold except at exceptional values of v and β. (assumed)

Proved only for small v and β (under some assumptions). (this will be done later — — Ruelle)

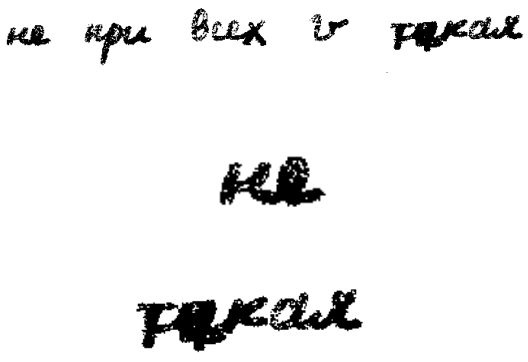
5) All $\vec{r}_k$ are bounded.



Manuscript notation. Both lines of item 5 have $\vec{r}_k$.

Original source details: GELFAND-P000156

$\vec{r}_k$ exist for all v, not for all v with such [?] a rate of convergence.



Uncertain reading. The line reads “not for all.” The opening strokes of “such” are heavily overwritten, obscuring the earlier entry.

Original source details: GELFAND-P000156



Uncertain reading. At the right beside item 5 is a separate small dense mark with an irregular outline; no letter reading is established. Below are two horizontal rules and a separate diamond-shaped central dot; the right end of the left rule is thickened.

Original source details: GELFAND-P000156

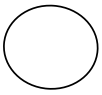
On the Gibbs distribution. (at Gelfand's request)

$$a(p_1, \dots, p_N, x_1, \dots, x_N)$$



Diagram and typed description. One unfilled approximate circle without labels or arrows to the left of the paragraph on motion; no sizes are specified.

Original source details: GELFAND-P000156



Motion on the surface of constant energy $H(a) =$ const. Assume that it is ergodic and there is an invariant measure.

Perhaps any distribution, as the number of particles increases, leads to the Gibbs distribution. (Gelfand)

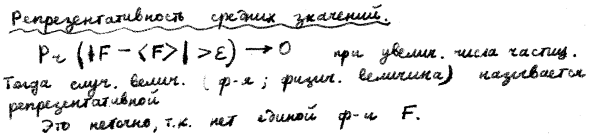
Not any: if there is no ergodic hypothesis (Arnold)

Representativity of mean values.

$$P_r\big(|F - \langle F\rangle| > \varepsilon\big) \longrightarrow 0 \quad \text{as the number of particles increases.}$$

Then the random variable (function; physical quantity) is called representative.

This is imprecise, since there is no single function F.



Manuscript notation. Below P is the same short, right-curved r used in the correlation formulas.



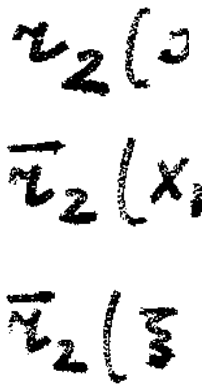
Original source details: GELFAND-P000156

The needed example. (summatory function)

$$F_N(x_1, \dots, x_N) = \sum_{i \neq j} f(x_i - x_j),$$

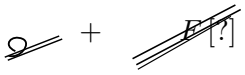
f is one and the same, $\int |f(\xi)|\, d\xi < \infty$.

$$\langle F_N \rangle = \iint r_2(x_1, x_2 \mid N, \Omega, \beta) f(x_1 - x_2)\, dx_1\, dx_2 =$$
$$= \iint \vec{r}_2(x_1 - x_2 \mid v, \beta) f(x_1 - x_2)\, dx_1\, dx_2 + o(|\Omega|) =$$
$$= |\Omega| \int \vec{r}_2(\xi \mid v, \beta) f(\xi)\, d\xi + o(|\Omega|).$$



Manuscript notation. The first line has r_2 and the next two have $\vec{r}_2$. The integration domains are unwritten. Equality signs recur at the line ends and beginnings.

Original source details: GELFAND-P000156



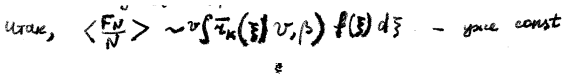


Uncertain reading. To the right of the last equality, a loop-shaped sign, a plus and an F-like form are crossed out. The first reading is not established; F is provisional. The cancellation strokes are diagonal.

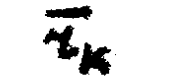
Original source details: GELFAND-P000156

Thus,

$$\left\langle \frac{F_N}{N} \right\rangle \sim v \int \vec{r}_k(\xi \mid v, \beta) f(\xi)\, d\xi \quad \text{— already const.}$$



Manuscript notation. The last formula has $\vec{r}_k$, with the divergent strokes of k. Before the integral is v.





Original source details: GELFAND-P000156

GELFAND-P000157





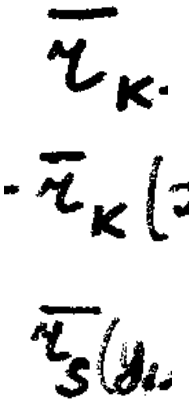
Uncertain reading. In the upper left margin is a densely cancelled loop-shaped sign resembling 6; its original reading is not established. At its right is a separate dot.

Original source details: GELFAND-P000157

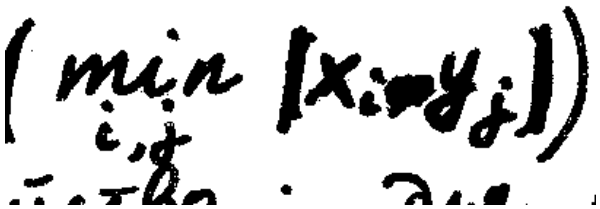
One more hypothesis (but it already follows from these, though with difficulty):

$$6)\quad \left|\begin{matrix}\bar{r}_{k+s}(x_1,\dots,x_k,y_1,\dots,y_s \mid v,\beta)\\ -\bar{r}_k(x_1,\dots,x_k \mid v,\beta)\bar{r}_s(y_1,\dots,y_s \mid v,\beta)\end{matrix}\right| \leq$$
$$\leq C_{k,s}\gamma\big(\min_{i,j}|x_i-y_j|\big).$$

(multiplicative property; for the ideal gas 0).



Manuscript notation. The three functions use r with a horizontal bar. The minimum has i, j, with $x_i - y_j$ between the absolute-value bars. The manuscript repeats $\leq$ across the line break.



Original source details: GELFAND-P000157

We shall prove that

$$\underset{\text{variance}}{DF_N} \sim dN, \qquad d \text{ — const}.$$

Hence $D\frac{F_N}{N} \sim \frac{d}{N}$ and by Chebyshev's inequality we already obtain ~~[unreadable word]~~. representativity of the mean.



Uncertain reading. After "obtain" a word is densely cancelled, with a loop-shaped stroke below it. Its original reading, including a possible repetition of "represent.", is not established. The following words are "representativity of the mean".

Original source details: GELFAND-P000157

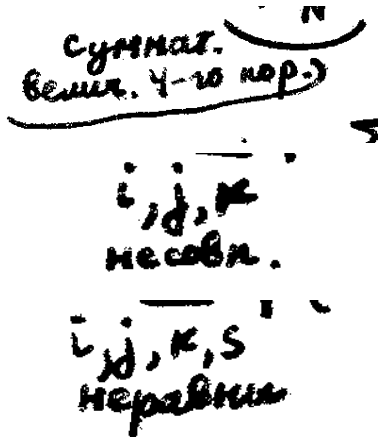
This is proved by direct calculation:

$$DF_N = \langle F_N^2\rangle - \langle F_N\rangle^2.$$

$$\underbrace{F_N^2}_{\substack{\text{summatory}\\ \text{qty. of 4th ord.}}} = 2\sum_{i\neq j} f^2(x_i - x_j)$$

$$+ 4 \sum_{\substack{i,j,k\\ \text{noncoinc.}}} f(x_i - x_j) f(x_k - x_j) +$$

$$+ \sum_{\substack{i,j,k,s\\ \text{unequal}}} f(x_i - x_j) f(x_k - x_s).$$



Original source details: GELFAND-P000157

Manuscript notation. A short arc is below F_N^2. At the left, "summatory quantity of order 4" has a longer curved stroke beneath it. Below the triple sum are i, j, k and the abbreviation for "distinct", below the quadruple sum i, j, k, s and "unequal".

We calculate

$$\langle F_N^2 \rangle = 2 \iint f^2(x_1 - x_2) r_2(x_1, x_2)\, dx_1\, dx_2 + 4 \ldots$$

and use the decay estimates.



Original source details: GELFAND-P000157

Manuscript notation. The final integral contains $r_2(x_1, x_2)$ without an overbar. Then comes $+4 \ldots$.

GELFAND-P000158



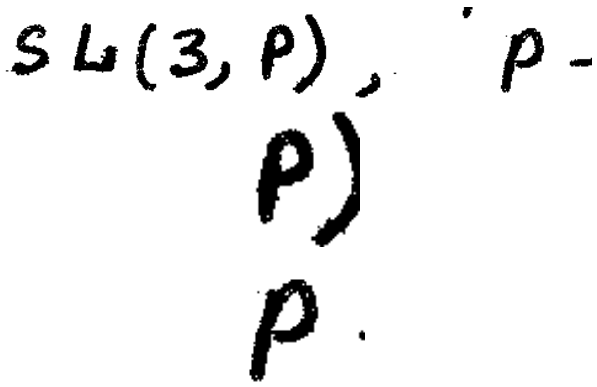
Diagram and typed description. Page number 18 at the upper right, with a line at its left and below it.

18

Original source details: GELFAND-P000158

D. Kazhdan. His results.

1) G is a simple real or p-adic Lie group containing as a subgroup $SL(3,P)$, P is the corresponding field. Γ is a discrete subgroup with a quotient space of finite volume.



Manuscript notation. In $SL(3,P)$ and before "corresponding field" is P, with an upper loop and a long downward stroke.

Original source details: GELFAND-P000158

Then a)

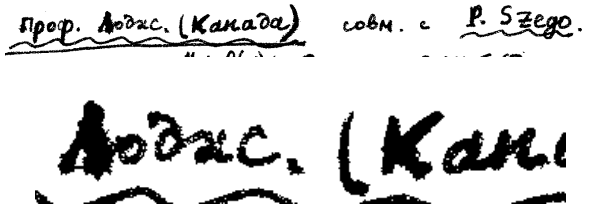
$$\Gamma/\underbrace{[\Gamma,\Gamma]}_{\text{commutator subgroup}} \quad \text{is finite.}$$

b) Γ has a finite number of generators. (even from any set of generators a finite one can be selected)

2) G is an algebraic group over the rational numbers Q, containing $SL(3,Q)$. Something about representations.

The proof uses the isolation of the trivial representation.

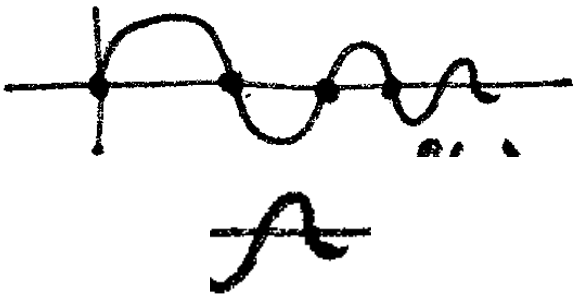
Prof. Lodzhs. (Canada) jointly with P. Szego.



Manuscript notation. The manuscript spells the name "Лоджс.", with a heavy initial stroke. The professor and Canada line has a wavy and a straight underline. P. Szego has a wavy underline.

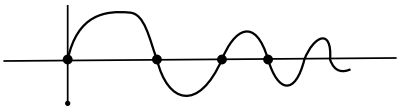
Original source details: GELFAND-P000158

$$y'' + f(x)y = 0, \qquad 0 < x < \infty.$$



Original source details: GELFAND-P000158

Diagram and typed description. A horizontal line, a short vertical at the left, four dense early crossings and a dot at the bottom of the vertical. Three upper waves alternate with lower ones, ending at the right in a short downward curl. No labels, sizes or exact coordinates are given; the contour is approximate.



Sturm: (1836) if $f(x)$ increases, the zeros get closer and conversely.

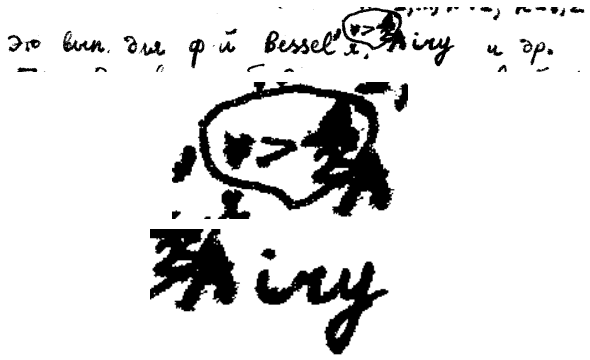
Theorem, if f has a higher degree of monotonicity.
Let

$$(-1)^j f^{(j+1)}(x) \geq 0, \qquad 0 < x < \infty, \quad j = 1, \ldots, N$$

and $f(\infty) > 0$. Then

$$(-1)^n \Delta^{n+1} x_k > 0, \qquad n = 1, \ldots, N+1; \quad k = 1, 2, \ldots$$

This holds for the Bessel functions $\nu > \frac{1}{2}$ [?], Airy and others.



Original source details: GELFAND-P000158

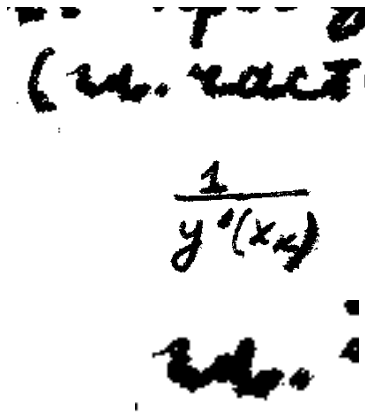
Uncertain reading. The oval insertion after Bessel reads $\nu > 1/2$ [?]. The top of the first sign and the fraction are densely overwritten. In Airy the initial letter is heavy.

The wave areas have the same property.

$$M_k = \int_{x_k}^{x_{k+1}} |y(x)|^\lambda \, dx, \qquad \lambda > -1.$$

$\lambda = 0$ — differences of zeros, $\lambda = 1$ — wave areas,

$\lambda = -1$ — diff. of derivs. (Gelfand's remark) $\frac{1}{y'(x_k)}$.
(princ.[?] parts)



Original source details: GELFAND-P000158

Uncertain reading. Below "diff. of derivs." a parenthesis appears to read "(princ.[?] parts)". The first two strokes are joined. At the right is $1/y\prime(x_k)$.

GELFAND-P000159

I. M. Gelfand.

Repetition of I. Bernstein. (also Atiya)

G is a compact group. K_G is a functor, $K_G(X)$.



Original source details: GELFAND-P000159

Diagram and typed description. E above X, with a downward arrow between them and no additional labels.

$$\begin{array}{c} E \\ \downarrow \\ X \end{array}$$

E, X are G-spaces; a vector bundle. Fiber to fiber — linearly. Formal differences.

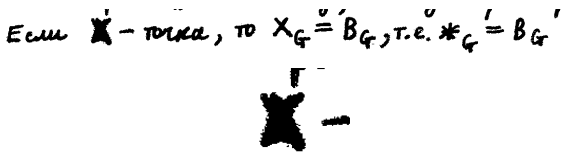
This is a contravariant functor.

Generalization of the notion of a classifying space. Consider

$$E \underset{G}{\times} X = X_G.$$

the orbits of G are contracted to a point; E is the universal space.

If X is a point, then $X_G = B_G$, i.e. $*_G = B_G$.



Uncertain reading. After "If" a dense X is written over earlier strokes. Their original reading is not established.

Original source details: GELFAND-P000159

Theorem. $K_G(X)$ approximately coincides with $K(X_G)$ (already the ordinary K-functor)

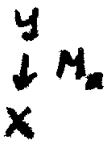
$K_G(X)$ is a ring, $X \longrightarrow \bullet$, hence $R_G \longrightarrow K_G(X)$ and therefore ~~[unreadable word]~~ this is an R_G-module (a ring over a subring)



Uncertain reading. Between "therefore" and "this" a word or group of letters is densely cancelled, with a loop-shaped stroke below it; the original reading is not established.

Original source details: GELFAND-P000159

I. Bernstein. The most essential part of the proof: a construction with elliptic operators.



Original source details: GELFAND-P000159

Diagram and typed description. Y above X, a downward arrow, and M_x to its right. Beside it is the description of M as the fiber and the arrow $K(X) \longrightarrow K(Y)$.

$$\begin{array}{c} Y \\ \downarrow \\ X \end{array} \quad M_x$$

M is a manifold, the fiber; $K(X) \longrightarrow K(Y)$. Sometimes a reverse map can be constructed.

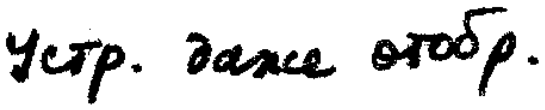
Let there be on X a family of differential operators D_x: acting in a sheaf over M_x (the fibers over x). Let there be over Y 2 sheaves N_1, N_2.

$$D_x : \Gamma(N_1|_{M_x}) \longrightarrow \Gamma(N_2|_{M_x})$$

— an elliptic operator (or a complex)

For example, the de Rham complex.

$K(Y)$ is a $K(X)$-module. Construct even a map of $K(X)$-modules.



Manuscript notation. Between the abbreviations for “construct” and “map” the manuscript reads “even”.

Original source details: GELFAND-P000159

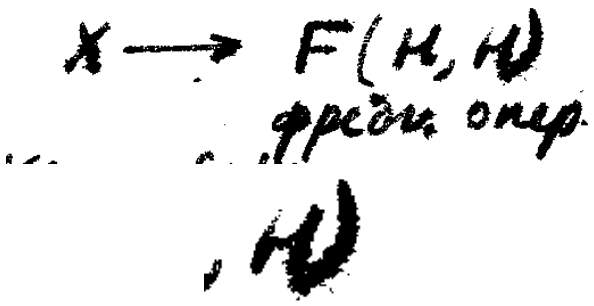
V is a sheaf on Y.

$$M_x : \quad D_x \otimes 1 : \Gamma(N_1 \otimes V)_{M_x} \longrightarrow \Gamma(N_2 \otimes V)_{M_x}.$$

An elliptic operator; $\mathrm{Ker}, \mathrm{Coker} < \infty$. The difference of elements in the K-functor gives the required element.

If there is a map

$$X \longrightarrow \underset{\text{Fredholm ops.}}{F(H, H\,[?])}$$



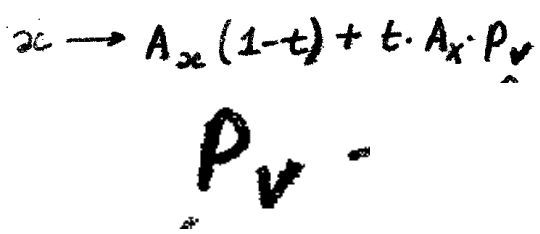
Uncertain reading. The second letter inside $F(H, H)$ is densely overwritten; an H-like sign with lower strokes. The original form beneath them is not established.

Original source details: GELFAND-P000160

then one can def. Ker – Coker as a bundle over X

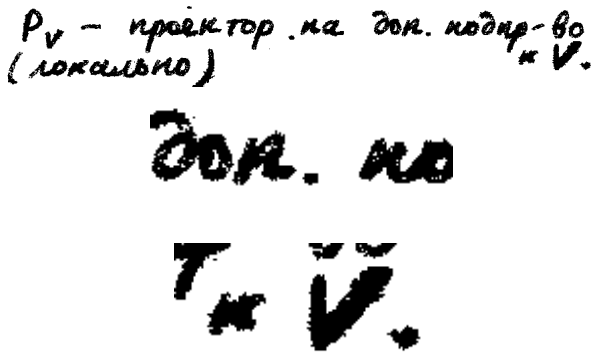
$$x \longrightarrow A_x$$
$$x \longrightarrow A_x(1-t) + t \cdot A_x P_v$$

P_v — a projector. onto a дп. [?] subspace to V.



Uncertain reading. The small subscript of P in the formula and explanation is v- / V-like; below it the subspace is denoted by capital V.

Original source details: GELFAND-P000160



Uncertain reading. Before “subspace” is an abbreviation resembling “дп.” / “дн.”; the second letter has a joined lower stroke. It is provisionally rendered as “дп.” without expansion. Beneath the end of the line, “к V” (“to V”) is written separately.

Original source details: GELFAND-P000160

Already const. dim. of the kernel (locally)

Bundle $\underset{\sim\sim\sim}{ind\, V}$.

Next time, Nirenberg’s report and Minlos.

GELFAND-P000161

3/X. Gelfand's sem.

Nirenberg.

Informal expos. of Seeley and Hörmander's index f-la. The f-la is analytic, without any topology, no connections with topology and how the leading terms disappear in it. However, according to Gelfand, the Atiyah–Singer f-la can be written analyt. [as the degree of a mapping, etc.] The same situation for Pontryagin classes.

L — ellipt. operator on a comp. m-fold. Consid. [?] eigenfns.



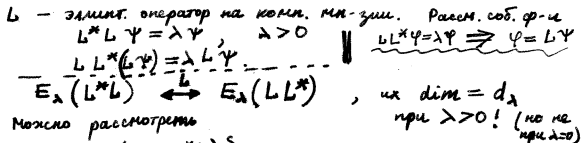
Original source details: GELFAND-P000161

Uncertain reading. Before "eigenfns.", the abbreviation is "Рассм." / "Расм.": the middle joined strokes are not completely separable. Provisionally rendered as "Consid.", without expansion.

$$L^*L\Psi = \lambda\Psi, \qquad \lambda > 0,$$
$$LL^*(L\Psi) = \lambda L\Psi.$$

- -

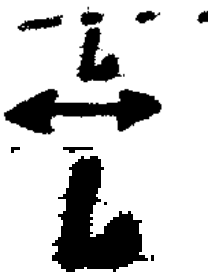
$$\| \quad \underline{LL^*\varphi = \lambda\varphi \Rightarrow \varphi = L\Psi}$$



Original source details: GELFAND-P000161

Diagram and typed description. Below the second formula is a dashed line. The formula on the right is separated by a double vertical line and has a wavy underline.

$$E_\lambda(L^*L) \overset{L\,[?]}{\longleftrightarrow} E_\lambda(LL^*), \qquad \text{their dim} = d_\lambda \quad \begin{matrix}\text{for } \lambda>0!\\ \text{(but not for } \lambda=0)\end{matrix}$$



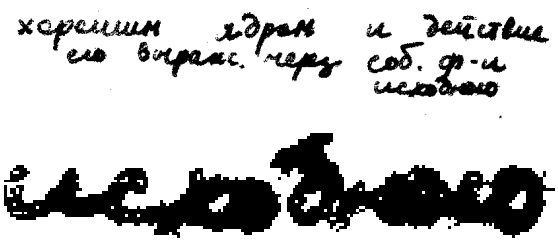
Original source details: GELFAND-P000161

Uncertain reading. Above the two-way arrow is a small sign with a vertical stroke and a lower stroke to the right, resembling L / l; the upper dot meets the level of the dashed line. Provisionally rendered as L.

One can consider

$$(I + L^*L)^s, \qquad \operatorname{Re} s \ll 0$$

Then this is an integr. operator with a good kernel and its action is expressed through eigenfns. [Crossed out: unread word.]



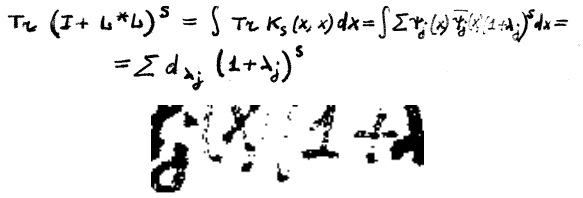
Uncertain reading. Below “eigenfns.” a word resembling “original” is densely crossed out; the internal joined strokes do not permit a confident complete reading.

Original source details: GELFAND-P000161

$$\int K_s(x,y)u(y)\,dy =$$
$$= \int \sum_j \Psi_j(x)\overline{\Psi}_j(y)(1+\lambda_j)^s u(y)\,dy$$

Ψ_j — scalar fns. or sections of a bundle.

$$\operatorname{Tr}(I+L^*L)^s = \int \operatorname{Tr} K_s(x,x)\,dx$$
$$= \int \sum \Psi_j(x)\overline{\Psi}_j(x\,[?])(1+\lambda_j\,[?])^s\,dx =$$
$$= \sum d_{\lambda_j}(1+\lambda_j)^s$$



Uncertain reading. In the second integral the argument after $\overline{\Psi_j}$ and the beginning of the parenthesis $(1+\lambda_j)$ are very faint: x / another faint argument; $1+$ / faint strokes.

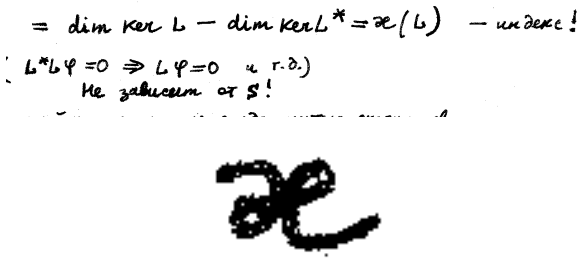
Original source details: GELFAND-P000161

Consider

$$\operatorname{Tr}(I+L^*L)^s - \operatorname{Tr}(I+LL^*)^s =$$

Everything cancels, except the terms with $\lambda = 0$

$$= \dim\ker L - \dim\ker L^* = \text{æ}\,[?](L) \qquad \text{— index!}$$



Uncertain reading. The sign before (L) has two joined loops: χ / x / an “æ”-like sign. Provisionally rendered as “æ”.

Original source details: GELFAND-P000161

($L^*L\varphi = 0 \;\Rightarrow\; L\varphi = 0$ etc.)

Does not depend on s!

What follows — study of the trace of integr. operators.

A — ellipt. of order $m>0$. — $(I + L^*L)$ —

$$|\arg a(x,\xi) - \pi| \geq \delta > 0, \qquad |\xi| = 1$$

|
this is assumed

(a more gen. situation than $I + L^*L$)

$$\sigma(A) = a_m(x,\xi) + a_{m-1}(x,\xi) + \dots$$

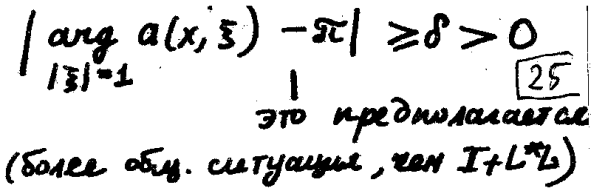
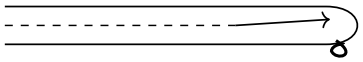


Diagram and typed description. Below "ellipt.": "of order $m > 0$.". Below the angular condition: a short vertical stroke and "this is assumed".

Original source details: GELFAND-P000162

Take $m = 1$ (multipl. by an invertible one)
usual def.:

$$A^s = \frac{1}{2\pi i} \int \lambda^s (\lambda - A)^{-1}\, d\lambda, \qquad \|(\lambda - A)^{-1}\| \leq \frac{C}{1 + |\lambda|}$$



contour, enclos. spectrum.

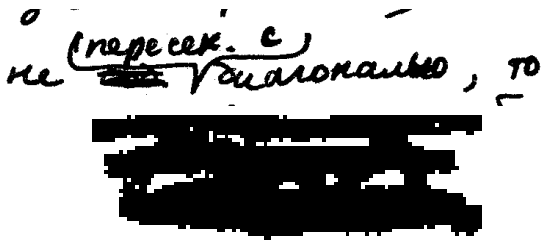


Uncertain reading. Two long lines, open on the left, joined by a rounded turn on the right; inside, a dashed line with an arrow to the right. Thickening at the bottom of the right turn: the direction of traversal is undetermined.

Original source details: GELFAND-P000162

Thus for negat. powers. For posit. powers multiply by a positive integer power.
We must compute $\int \operatorname{Tr} K_s(x,x)\, dx$
$K_s(x,x)$ — kernel of the operator A^s, $\operatorname{Re} s \ll 0$.
(i) $s \longrightarrow K_s$ — holom. mapping of the region $\operatorname{Re}(ms) < -n - k$ into kernels of class C^k on $M \times M$
(smooth dep. of the kernel on s)

(ii) C — comp. in $M \times M$, not [Crossed out: unread word.] [Inserted above with a bracket: intersec. with] the diagonal, then



Uncertain reading. Between “not” and “the diagonal” a word is densely crossed out; above it, “intersec. with” is inserted with a bracket. Beneath the cancellation are separate loops without a confident literal reading.

Original source details: GELFAND-P000162

$s \longrightarrow K_s|_C$ continues [Crossed out: unread short word.] to a holom. mapping



Uncertain reading. After “continues” a short word resembling “all” / “everything” is crossed out.

Original source details: GELFAND-P000162

of the entire plane into $C^\infty(C)$

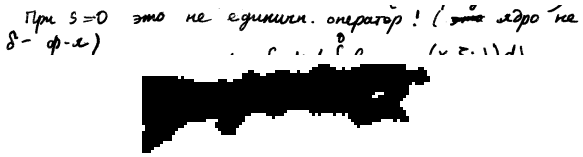
(not analyt. in x, since the coeffs. are infinitely differ.)

(iii) for every x, $K_s(x,x)$ is a meromorphic fn. on the entire plane with poles (simple), possibly only at

$$-\frac{n}{m},\quad \frac{1-n}{m},\quad \frac{2-n}{m},\dots,-\frac{1}{m},\quad \frac{1}{m},\quad \frac{2}{m},\dots$$

$(0$ — regul. point$)$

At $s=0$ this is not the identity operator! ([Crossed out: unread short word.] the kernel is not a δ-fn.)



Uncertain reading. Before “the kernel is not”, short joined strokes are densely crossed out, possibly “this”; the literal reading is undetermined.

Original source details: GELFAND-P000162

$$\text{(iv)}\quad (2\pi)^n K_0(x,x) = \frac{1}{m}\int_{|\xi'|=1} d\xi' \int_{-\infty}^{0} b_{-m-n}(x,\xi;\lambda)\,d\lambda$$

K_0 has no singularities. in calc. through coeffs. [Below “coeffs.”: of oper. A.]

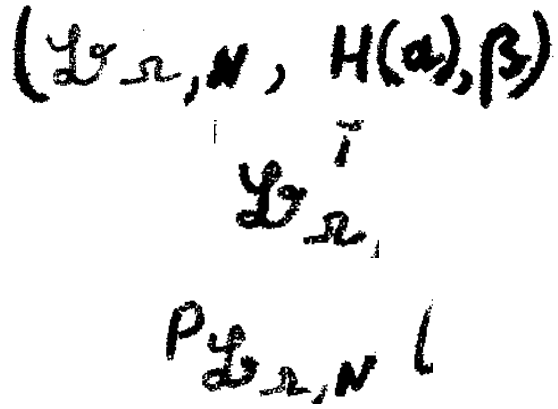
A^s — pseudodiff. oper. and its symbol is obtained in a natural way by subst. the symbol in place of A in the f-la for A^s.

The index — difference of two exprs. of type (iv).

GELFAND-P000163

Minlos. Lecture 3.

$(\mathcal{Y}_{\Omega,N}\,[?], H(\alpha), \beta)$



Uncertain reading. The first sign before Ω, N is calligraphic: a tall looped stem, a left arc and a right loop; $\mathcal{Y}$ / $\mathcal{G}$ / another handwritten sign. The same sign occurs later before the mapping and in the distribution. Provisionally rendered as $\mathcal{Y}$.

Original source details: GELFAND-P000163

Can one specify a "limiting sp." possessing asympt. props.? Probably yes, but the proof is again for low density.

Pass to indistinguishable particles

$$\mathcal{Y}'_{\Omega,N} \underset{x_i \neq x_j}{\longrightarrow} C_{\Omega,N}$$

$C_{\Omega,N}$ — N-point subsets of the vessel.

H can be consid. on C, since it is symmetric

The Gibbs distr. is transferred

$$P_{C_{\Omega,N}}(c) = P_{\mathcal{Y}_{\Omega,N}}\Big(\underset{\text{preimage.}}{\widetilde{c}}\Big)$$

* the compl. of $\mathcal{Y}'$ has measure 0.

C_∞ — finite or countable subsets of R^ν (having no accum. points), $d \in C_\infty$.

A Gibbs distr. can be defined on it.

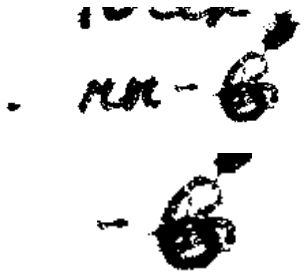
First single out the σ-algebra.

Cylindr. set. $U \subset R^\nu$

$$S_{U,k} \quad (d : n(d,U) = k)$$

the number of points falling in U equals k.

A finite coll. of nonintersect. sets [?]



Uncertain reading. The ending of the abbreviation after "nonintersect." is densely overwritten, with an upper loop and a lower rounded knot; provisionally rendered as "sets". The original strokes underneath the overwriting are undetermined.

Original source details: GELFAND-P000163

$$S_{U_1,U_2,\ldots,U_s;k_1,k_2,\ldots,k_s} = \{d : n(d,U_1) = k_1, \ldots\}$$

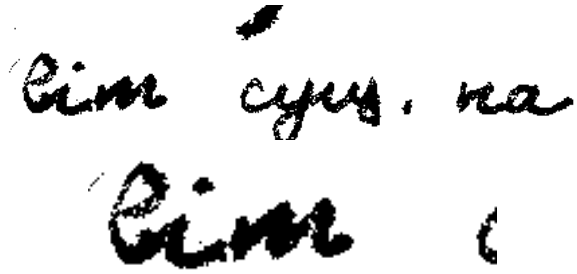
The σ-alg. generated by these cylindr. sets. The usual Gibbs distr. generates a measure on this σ-algebra. The resulting measures satisfy the condition of uniform strong mixing. this is the limiting measure. Its existence is proved for low densities.

$$\mu(S_{U_1U_2,k_1k_2}) - \mu(S_{U_1,k_1})\mu(S_{U_2,k_2}) \longrightarrow 0,$$
$$d(U_1, U_2) \longrightarrow \infty$$

(unif. strong mixing)

uniformly over all pairs U

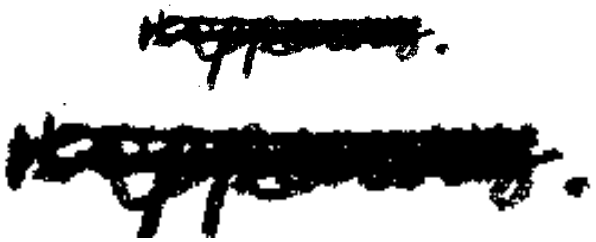
Difficulty — [Unread short word.] exist. on



Original source details: GELFAND-P000163

Uncertain reading. After the dash before “exist.”, a short word: “there is” / “its” / another reading; the joined middle strokes do not permit a confident complete reading.

[Crossed out: unread long word.] cylindr. sets*



Original source details: GELFAND-P000163

Uncertain reading. After “on”, a long word with two descenders is densely crossed out: “correl.” / another word; the readable parts do not permit a confident reconstruction.

This $\longrightarrow$ reformul. of props. of correlation fns.

GELFAND-P000164

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Props. following from this.

1°. For almost all d, the mean density exists —

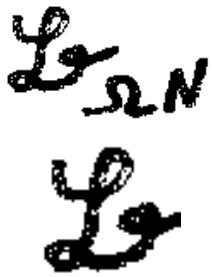
— a consequence of Birkhoff's thm. and ergodicity of the measure.

Gelfand's propos.: start with axioms on correlation functions. How to single them out?

Thermodyn. quantities.

E energy $\quad P$ pressure $\quad S$ entropy $\quad F$ free energy

$$Z = \frac{1}{N!} \int_{\mathcal{Y}_{\Omega N}\,[?]} e^{-\beta H(a)}\, da$$



Uncertain reading. The integration-domain sign matches the calligraphic looped sign in the preceding frame: $\mathcal{Y}$ / $\mathcal{G}$ / another sign; the subscripts here are ΩN without a clearly visible comma. Provisionally rendered as $\mathcal{Y}$.

Original source details: GELFAND-P000164

— normaliz. factor.

They are all expressed through $\ln Z$

We shall prove that $\lim \frac{\ln Z}{N}$ exists

GELFAND-P000165

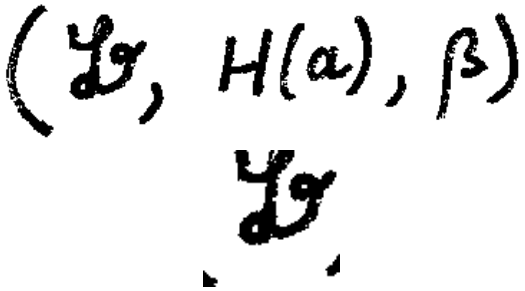
27

$10/\overline{\text{X}}$ Gelfand's sem.

Minlos Lecture 4.

E energy $\quad P$ pressure $\quad S$ entropy

$(\mathcal{Y}\,[?], H(a), \beta)$

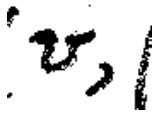


Uncertain reading. In the triple, the first sign is the same looped calligraphic sign: $\mathcal{Y}$ / $\mathcal{G}$ / another sign. Provisionally rendered as $\mathcal{Y}$.

Original source details: GELFAND-P000165

1°. $E = \langle H(a)\rangle$. For $H(a) = \sum U(|x_i - x_j|)$ representativeness was proved, i.e.

$$\langle H(a)\rangle \approx \underset{N,\Omega\longrightarrow\infty}{Ne}, \qquad e = e(v\,[?], \beta)$$



Original source details: GELFAND-P000165

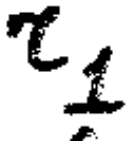
Uncertain reading. The argument in $e(v, \beta)$ is written in a v- / ν-like form; the Latin / Greek sign is undistinguished.

$$\langle H(a)\rangle = \frac{\int H(a) e^{-\beta H(a)}\, da}{Z} = -\frac{\partial}{\partial\beta} \ln Z$$

2°. $p(x) = \tau_1(x)\,[?]$. $\quad P = \dfrac{1}{S}\int p(x)\, d\sigma$

(Before the lim. transition)





Original source details: GELFAND-P000165

Uncertain reading. The sign with subscript 1 after $p(x) =$ has an upper arc and a lower hook: τ_1 / r_1; it recurs in the parentheses below.

There is a hypothesis that $p(x) \longrightarrow const$ in the thermodyn. lim. transition. this does not follow from the earlier hypotheses (that $\tau_1 \longrightarrow const$, since this is on $\forall$ finite segment)

$$p = \frac{\partial \ln Z}{\partial V};$$

3°. Entropy of the Gibbs distr.:

$\forall$ distr. $\pi(x)$

$$S = \int \pi(x) \ln \pi(x)\, dx; \qquad \frac{S}{N} = \frac{1}{N}\ln Z - \frac{1}{\beta}\frac{\partial}{\partial\beta}\frac{\ln Z}{N}$$

The lim. of entropy per 1 particle exists.

———— · ————

V. I. Arnold. On Maslov's works.

R^{2n} — phase sp. $p_1, \ldots, p_n, q_1, \ldots, q_n$, if 3 structures are introduced

1) metric $p^2 + q^2$ and scalar prod. (Eucl. structure) (x, y)

2) compl. structure $z = p + iq$, $R^{2n} = C^n$,

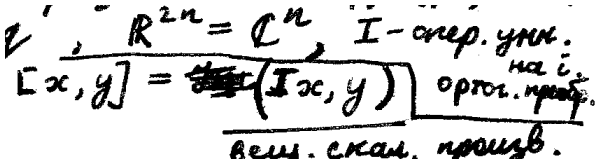
I — oper. of multipl. by i.

3) symplect. structure

$[x, y] =$ [Crossed out: unread sign.] (Ix, y)

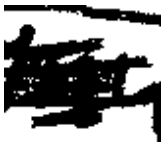
real scalar prod.

orthog. [unread abbrev.]



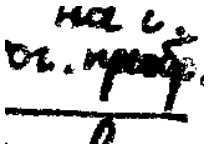
Original source details: GELFAND-P000165

Diagram and typed description. The structures are numbered 1), 2), 3). Beside (Ix, y) is a stepped outline, with the lower caption “real scalar prod.”; on the right, an abbreviation after “orthog.” and a pointer from I.



Original source details: GELFAND-P000165

Uncertain reading. Before (Ix, y) a short mathematical sign / term is densely crossed out; the separate strokes do not permit a confident reading.

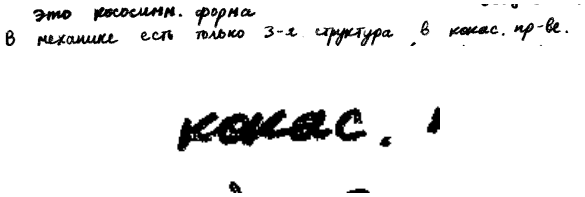


Original source details: GELFAND-P000165

Uncertain reading. After “orthog.” the abbreviation is densely overwritten: “transf.” / “prod.” / another reading; the beginning and a descender are visible.

this is a skew-sym. form

In mechanics there is only the 3rd structure in [unread abbrev.] sp.



Original source details: GELFAND-P000165

Uncertain reading. Before “sp.” the abbreviation looks like “кас.” / “касс.” / another term; the exact reading is undetermined.

Preserv. of these structures: $O(2n)$, $GL(C, n)$, $Sp(n)$

It is easy to prove that the pairwise intersecs. equal the triple one (from [3])

this is $U(n)$.

A subsp. R^n is called Lagrangian, if the 3rd form is 0 on it $(IR^n \perp R^n)$. E.g., the real and purely imaginary ones.

$$[R^n, R^n] = 0$$

GELFAND-P000166

Example. $p = p(q)$ — Lagrangian m-fold $\iff \exists S(q)$, such that $p = \dfrac{\partial S(q)}{\partial q}$

$n = 1$ — every curve is a Lagr. m-fold

$n = 2$ — not every one.

Lagr. planes are real-like, $U(n)$ acts transitively

$O(n)$ leaves the real plane in place.

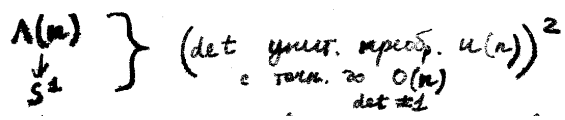
Lagr. planes $\Lambda(n) = U(n)/O(n)$ — Lagr. Grassmann m-fold

(Lagr. planes have dim. $\leq n$, but here only those of dim. n)

M — Lagrangian m-fold (the tang. sp. at $\forall$ point is Lagrangian)

Construct the mapping

$$\left.\begin{array}{c}\Lambda(n)\\ \downarrow\\ S^1\end{array}\right\} \quad (\text{det of a unit. transf. } U(n))^2$$

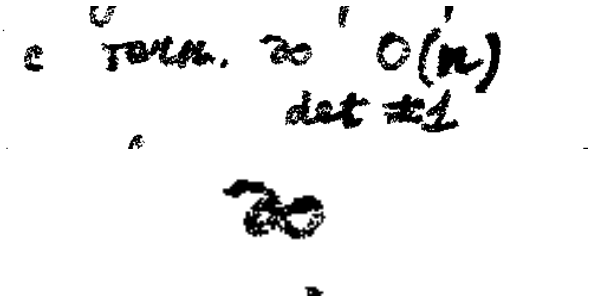


Original source details: GELFAND-P000166

Diagram and typed description. The vertical mapping $\Lambda(n) \to S^1$ is joined by a right brace to the square of the determinant.

up to [?] $O(n)$

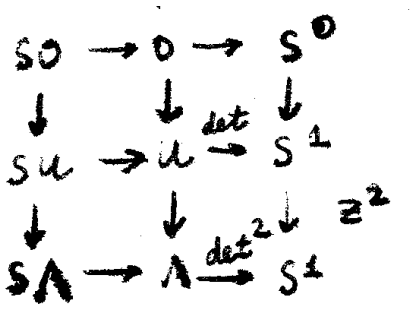
det $= \pm 1$



Original source details: GELFAND-P000166

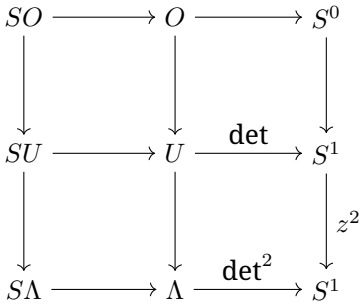
Uncertain reading. After “with precision” are two joined looped strokes resembling “до” (“up to”) / the sign $\approx$; provisionally rendered as “up to”.

$U(n)$ takes the real plane to our Lagrangian one.



Original source details: GELFAND-P000166

Diagram and typed description. Three rows: SO, O, S^0; SU, U, S^1; $S\Lambda, \Lambda, S^1$. Arrow labels: det, $\det^2$, z^2.



$S\Lambda$ — fibre of the corresp. bundle.

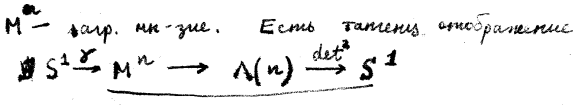
This is a diag. of bundles.

M^n — Lagr. m-fold. There is a tangent mapping
[Overwritten unread sign.]



Original source details: GELFAND-P000166

Uncertain reading. Before the chain $S^1 \longrightarrow M^n$ is a densely overwritten sign: $\forall$ / a crossed-out expression; the original sign is undetermined.

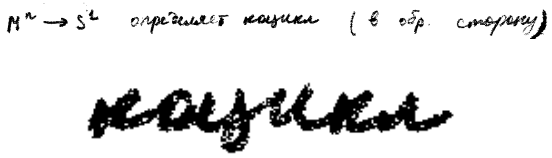


Original source details: GELFAND-P000166

Diagram and typed description. Below the chain from M^n to the last S^1 is a long line; γ is on the first arrow, $\det^2$ on the last.

$$S^1 \xrightarrow{\gamma} M^n \longrightarrow \Lambda(n) \xrightarrow{\det^2} S^1$$

$M^n \longrightarrow S^1$ defines a katsikl [?] (in the reverse direction)

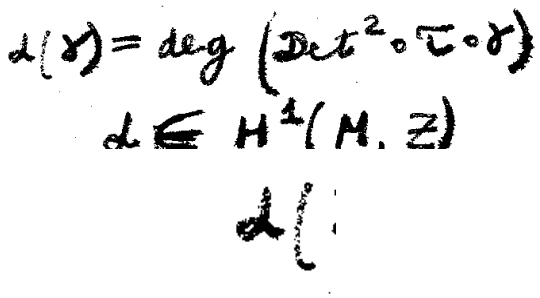


Original source details: GELFAND-P000166

Uncertain reading. The word after “defines” looks like “кацикл” / “коцикл” / “нацикл”; the initial letter and first vowel are not confidently distinguished. Provisionally transliterated as “katsikl”.

γ — a curve in M^n

$$\ell\,[?](\gamma) = \deg(\mathrm{Det}^2 \circ \tau \circ \gamma),$$
$$\ell\,[?] \in H^1(M, Z)$$



Original source details: GELFAND-P000166

Uncertain reading. The sign before (γ) and before $\in H^1$ has a tall stem and a lower loop: an l- / ℓ- / μ-like sign. Provisionally rendered as ℓ.

This is the Maslov index, occurring in asymptotic f-las, in particular, the Morse index is obtained (the number of conjug. points on a geod.)

There is another geom. def. of Maslov’s.

GELFAND-P000167

$17/\overline{X}$ Gelfand Sem. Minlos. Lecture 5.

Phase transitions.

I. M. Gelfand. Tannaka—Krein duality thm.

G — finite group or compact.

Given irreduc., finite-dimensional reps. in the spaces $A_1, \ldots, A_n$

$$A_i \otimes A_k \cong \sum A_j$$

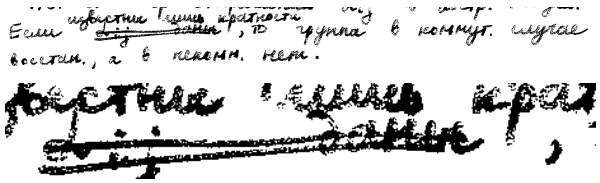
(Tensor prod. decomp. into irreducible representations)

Consider $\sum A_j$ (each 1500 times)

$$A_i \otimes A_j \longrightarrow \sum A_j;$$

i.e. there is a collection of maps d_{ij} into an abstr. module.

If [crossed-out unreadable entry; inserted above: properties of these multiplicities], then the group can be recov. in the finite case, but not in the compact case.



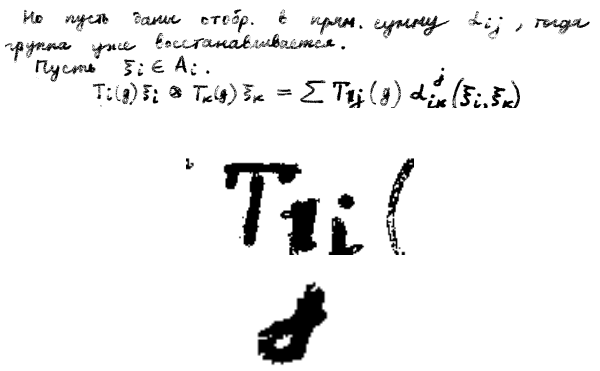
Original source details: GELFAND-P000167

Uncertain reading. After “If” a long entry is crossed out; “properties of these multiplicities” is written over / above it. The two lower strokes and separate lower marks do not give a complete reading of the cancelled entry.

But let the maps d_{ij} into the direct sum be given, then the group can now be reconstructed.

Let $\xi_i \in A_i$.

$$T_i(g)\xi_i \otimes T_k(g)\xi_k = \sum T_{ij}\,[?](g) d^j_{ik}(\xi_i, \xi_k)$$



Original source details: GELFAND-P000167

Uncertain reading. The subscript on the right-hand T has joined or overwritten strokes: ij / j; provisionally rendered as ij. The superscript on d has a loop and upper dot, rendered as j.

In place of $T_i(g)$ write arbitrary C_i — lin. transformations.

Let [?] the formula hold:



Original source details: GELFAND-P000167

Uncertain reading. The word before "the formula hold" is densely overwritten; provisional reading "Let", with the internal cancelled strokes unread.

$$C_i\xi_i \otimes C_k\xi_k = \sum C_j d^j_{ik}(\xi_i, \xi_k).$$

This eq. has solutions $C_i = T_i(g)$.

Then $C_i = T_i(g)$. Thus, all solutions of these eqs. will be $T_i(g)$ and everything is recov. from d^j_{ik}.

There is a generalization to $SL(2, R)$.

Nirenberg.

Varadhan.

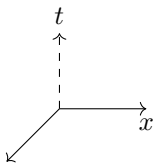
$$\frac{\partial u}{\partial t} = \sum a^{ij}(x) u_{x^i x^j}, \qquad u(x, 0) = \delta(x - y)$$

Study the behavior of the fund. solution.



Original source details: GELFAND-P000167

Diagram and typed description. Three arrows from a common point; the vertical one is dashed with t, the right one has x, the lower-left one is unlabelled.



Then

$$u(x) = e^{-\left(\frac{g(x,y)}{t} + o(\frac{1}{t})\right)}, \quad t \longrightarrow 0$$

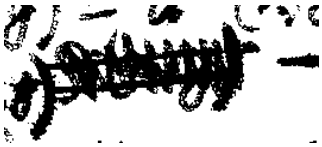
first term of the exp. in x.

For heat cond.

$$u = C\frac{e^{-\frac{(x-y)^2}{4t}}}{t^{n/2}}$$

$$g(x, y) = d^2(x, y)$$

$d(x, y)$ [crossed-out unreadable term] — dist. along geodesics



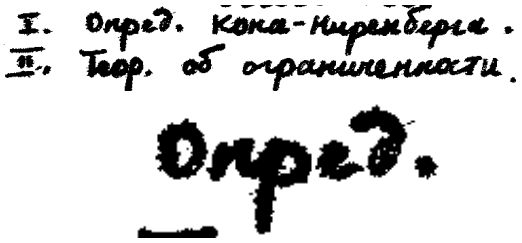
Original source details: GELFAND-P000167

Uncertain reading. After $d(x, y)$ a term is densely crossed out; the internal letter strokes are obscured.

($t^{n/2}$ enters $o(\frac{1}{t})$)
in the metric gener. by a^{ij}.
Also elliptic eigenvalues of the Laplace oper. in a domain in the same spirit.

GELFAND-P000168

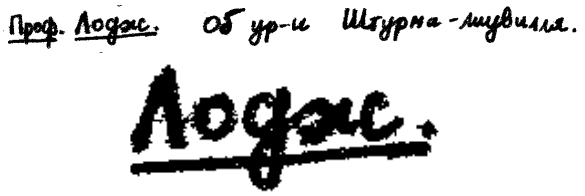
$24/\overline{X}$. Gelfand Sem.
Vishik. Lecture 1.
I. Defn. [?] of Kohn—Nirenberg.
II. Thm. on boundedness.



Original source details: GELFAND-P000168

Uncertain reading. The abbreviation after I looks like “Opr-yo.” / “Opr-e.”; the upper stroke of the final letter is undetermined, provisionally transcribed “Opr-yo.” and translated “Defn.”.

Prof. Lodzhs. [?] On the Sturm—Liouville eq.



Original source details: GELFAND-P000168

Uncertain reading. The surname after “Prof.” has joined middle and final strokes: “Lodzhs.” / “Loszh.” / “Loskh.” / “Lorch.”; provisionally rendered “Lodzhs.”.

$31/\overline{X}$ Gelfand Sem.
Vishik. Lecture 2.

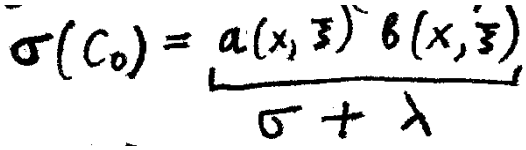
$$\sigma(A) = a(x,\xi) \longleftrightarrow A$$

$$A.Bu = C_0u + \ldots + C_{s-1}u + T_su,$$

where

$$\sigma(C_k) = \sum_{|\alpha|=k} \frac{1}{\alpha!}\partial^\alpha a(x,\xi) D^\alpha b(x,\xi)$$

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial\xi^\alpha}, \qquad D^\alpha = \left(\frac{1}{i}\frac{\partial}{\partial x}\right)^\alpha$$



Original source details: GELFAND-P000168

Diagram and typed description. A straight bracket with short raised ends under the product $a(x,\xi)b(x,\xi)$; label $\sigma+\lambda$.

$$\sigma(C_0) = \underbrace{a(x,\xi)b(x,\xi)}_{\sigma+\lambda}$$

T_s — oper. of order $\sigma + \lambda - s$.

$$\zeta(\xi)a(x,\xi) = h(x,\xi), \qquad \zeta(\xi)b(x,\xi) = g(x,\xi)$$

$$\widetilde{Au}(\xi) = (2\pi)^{-n/2} \int \widetilde{h}(\xi-\eta,\eta)\widetilde{u}(\eta)d\eta$$
$$\widetilde{Bu}(\eta) = (2\pi)^{-n/2} \int \widetilde{g}(\eta-\tau,\tau)\widetilde{u}(\tau)d\tau$$

$$\widetilde{A.Bu}(\xi) = (2\pi)^{-n/2} \int \widetilde{h}(\xi-\eta,\eta)\widetilde{Bu}(\eta)d\eta =$$

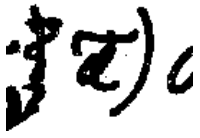
$$= \dots \int \widetilde{h}(\xi-\eta,\eta)d\eta \int \widetilde{g}(\eta-\tau,\tau)\widetilde{u}(\tau)d\tau =$$
$$= \dots \int \left(\int \widetilde{h}(\xi-\eta,\eta)\widetilde{g}(\eta-\tau,\tau)d\eta \right) \widetilde{u}(\tau)d\tau$$

Expand $\widetilde{h}$ by Taylor's formula at $\eta = \tau$

$$\widetilde{h}(\xi-\eta,\eta) = \sum_{|\alpha|\le s-1} \frac{1}{\alpha!}\partial^\alpha \widetilde{h}(\xi-\eta,\tau)(\eta-\tau)^\alpha + h_s(\xi-\eta,\eta,\tau)$$

GELFAND-P000169

$$\int \widetilde{h}(\xi-\eta,\eta)\widetilde{g}(\eta-\tau,\tau)d\eta$$
$$= \int \widetilde{h}(\xi-\eta,\tau)\widetilde{g}(\eta-\tau,\ [?]\,\tau)d\eta +$$
$$+ \sum \frac{1}{\alpha!} \int \partial^\alpha \widetilde{h}(\xi-\eta,\tau)(\eta-\tau)^\alpha \widetilde{g}(\eta-\tau,\tau)d\eta$$



Original source details: GELFAND-P000169

Uncertain reading. In the first right-hand integral, between $\eta - \tau$, and the final τ, is a small densely overwritten sign; a possible cancelled ξ / another sign is undetermined.

30

The 1st convolution corresp. to the product $h(x,\tau)g(x,\tau)$
Everything follows from this.
Defn.

$$\tau_0, \tau_1, \dots, \tau_k, \dots \searrow -\infty$$
$$A_0, A_1, \dots, A_k, \dots$$

A_i homog. of order τ_i with symbols $a_i(x,\xi)$

$$A \sim \sum_{k=0}^{\infty} A_k$$

(In advance $A_i : S \longrightarrow S$)

$$A - \sum_{k=0}^{N} A_k = T_{N+1} \quad \text{— oper. of order } < \tau_N.$$

Symbol [crossed-out formula resembling $\sigma(A)$, reading undetermined]



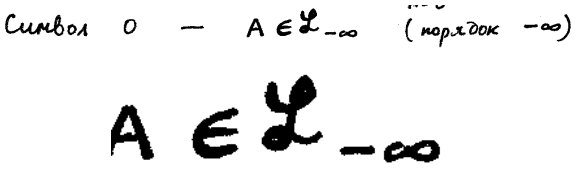
Original source details: GELFAND-P000169

Uncertain reading. After “Symbol” a formula resembling $\sigma(A)$ is crossed out; dense strokes inside the parentheses leave the reading ambiguous.

formal sum

$$\sum_{k=0}^{\infty} a_k(x,\xi) = \sigma(A)$$

Symbol 0 — $A \in \mathcal{L}_{-\infty}$ [?] (order $-\infty$)

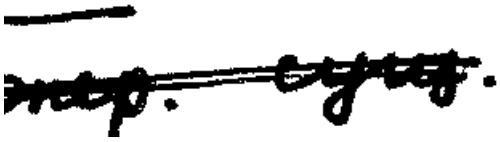


Original source details: GELFAND-P000169

Uncertain reading. The calligraphic letter before $-\infty$ has a tall stem with upper and lower loops: $\mathcal{L}$ / $\mathcal{Y}$ / another handwritten sign. Provisionally rendered as $\mathcal{L}$.

Theorem: the symbol of an oper. is uniquely determ.

$\forall$ [crossed-out unreadable abbreviations] symbol there exists an operator.



Original source details: GELFAND-P000169

Uncertain reading. After $\forall$ two abbreviations are crossed out: “oper. exists” / another entry; lower strokes and a connecting long line are visible.

Pseudodiff. operators form an algebra.

$$\sigma(A.B) = \sum \frac{1}{\alpha!} \partial^{\alpha} \sigma(A) D^{\alpha} \sigma(B)$$

O. A. Oleinik.

One particular problem in mechanics.

Viscous fluids. u, v — veloc. (plane). Navier—Stokes eqs.

$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \nu\Delta u \\ \dfrac{\partial v}{\partial t} + u\dfrac{\partial v}{\partial x} + v\dfrac{\partial v}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial y} + \nu\Delta v \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0 \end{cases}$$

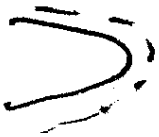
ν — viscosity (small)

$\nu = 0$ — Euler eqs. (ideal fluid).

Even for small ν these terms cannot be crossed out (Prandtl)

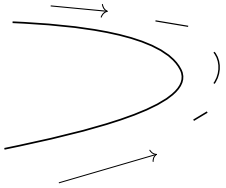
Ideal fluid: bdy. cond. — normal component of veloc. $= 0$

viscous — both comp. $= 0$ (sticking)

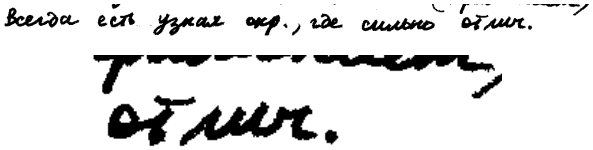


Original source details: GELFAND-P000169

Uncertain reading. Beside the boundary conditions on the left is an arc open to the left and convex to the right. Above and below it are short strokes and arrows pointing right; several separate strokes to the right. No labels; the exact directions of all short strokes are undetermined.



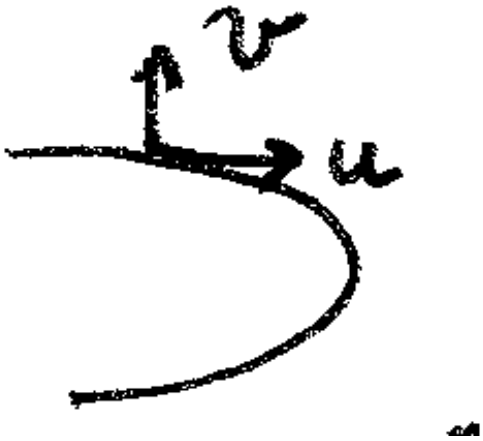
There are always narrow neighb. where strongly otlip. [?]



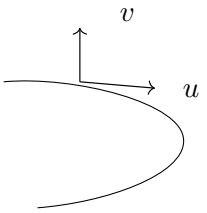
Original source details: GELFAND-P000169

Uncertain reading. The final abbreviation reads “otlip.” / “otlich.” / another ending; the initial letters “ot” are visible, then joined strokes. Provisionally rendered by the transliteration “otlip.”.

GELFAND-P000170



Typeset diagram



Original source details: GELFAND-P000170

v does not change (bdy. conds. preserved).

u changes strongly.

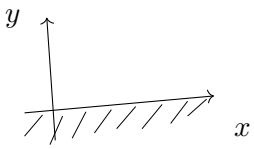
The Prandtl eqs. remain:

$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \nu\dfrac{\partial^2 u}{\partial y^2}, \\ \cancel{\dfrac{\partial p}{\partial y} = 0} \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0. \end{cases}$$

p does not change along the normal and it is consid. to be a known fn., obtained from the eqs. of an ideal fluid. $p = p(t, x)$.

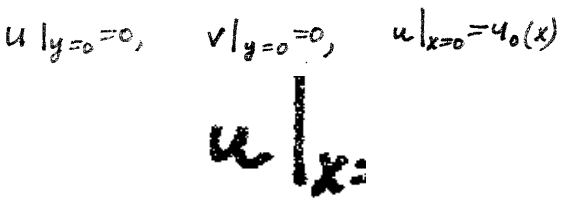


Typeset diagram



Original source details: GELFAND-P000170

$$u|_{y=0} = 0, \qquad v|_{y=0} = 0, \qquad u|_{x=0} = u_0(x).$$



Uncertain reading. The subscript sign in the third condition resembles x: $u|_{x=0} = u_0(x)$. An alternative reading is a y-like sign.

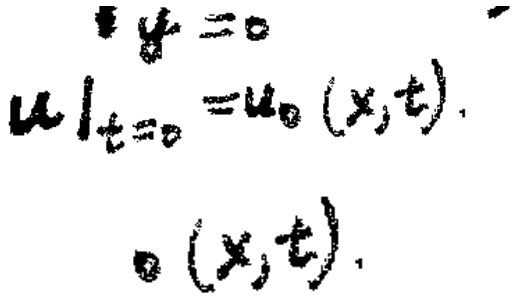
Original source details: GELFAND-P000170

$$u(t,x,y) = U(t,x) \qquad y \longrightarrow \infty.$$

$$\begin{cases} \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = \nu\dfrac{\partial^2 u}{\partial y^2} - \dfrac{1}{\rho}\dfrac{\partial p(t,x)}{\partial x}, \\ \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 0. \end{cases}$$

$$u|_{y=0} = 0, \qquad v|_{y=0} = v_0(x,t), \qquad u|_{x=0} = u_0(x),$$

$$u|_{t=0} = u_0(x,t),$$



Uncertain reading. In the condition at $t = 0$, the second argument on the right resembles t: $u_0(x,t)$. The alternative of a y-like lower loop remains.

Original source details: GELFAND-P000170

$$\lim_{y \to +\infty} u(t,x,y) = U(t,x).$$

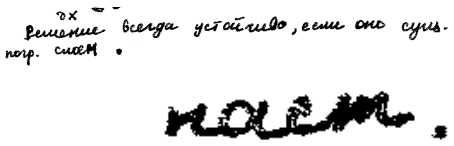
The soln. is unique, if it exists.

One can prove that it exists for an interval $0 \le t \le T$ for the whole body in the flow (separation may occur, but not instantaneously).

Near the beginning of the flow around the body, it exists for $\forall t$.

$$\frac{\partial p}{\partial x} \le 0.$$

The solution is always stable if it exists and is a nast. [?] bdy. layer.



Uncertain reading. The last abbreviation in the first line reads “nast.”; an alternative reading is “nasl.”. The joined final strokes are dense. The abbreviation is not expanded.

Original source details: GELFAND-P000170

GELFAND-P000171

$14/\overline{XI}$. Sem. of Gelfand.

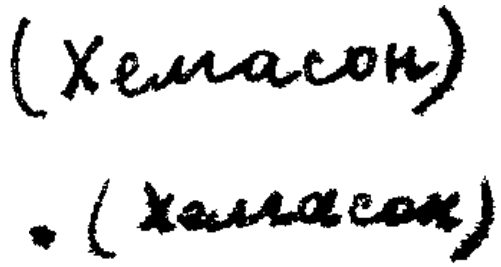
Agranovich. Lecture 3 on pseudodiff. operators.

Problems of diff. geom.

1. There is a complete Riem. manif. of neg. curvature.

$f \in C_0^\infty(\Omega)$, a ball B is given and $\int$ over any sphere containing B, $= 0$. Prove that $f = 0$ outside B.

(Khelgason) Done for const. curvature. (Khelgason)



Uncertain reading. The surname in both parentheses: “Khelgason” / “Khel’gason”; the joined middle strokes are not fully distinct.

Original source details: GELFAND-P000171

2. [Crossed out: illegible sign]

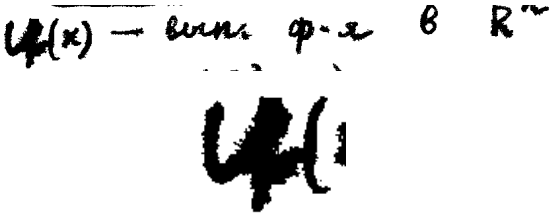


Uncertain reading. Before Calabi is a small densely cancelled sign with a vertical stroke and lower loop: a star / cancelled letter.

Original source details: GELFAND-P000171

Calabi

$U(x)$ [?] — conv. funct. in R^n.



Uncertain reading. The first function letter is reinforced and overwritten: a U- / u-like sign; the internal overlapping strokes cannot be separated.

Original source details: GELFAND-P000171

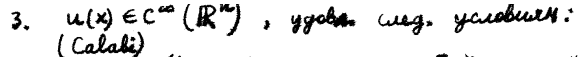
$$\det\left(\frac{\partial^2 u}{\partial x_i \partial x_j}\right) \equiv 1.$$

Prove that U is a polynomial of degree 2.

Proved for $n = 2$ and for $n \leq 5$ Calabi.

3. $u(x) \in C^\infty(R^n)$, satisf. [?] the foll. condition:

(Calabi)

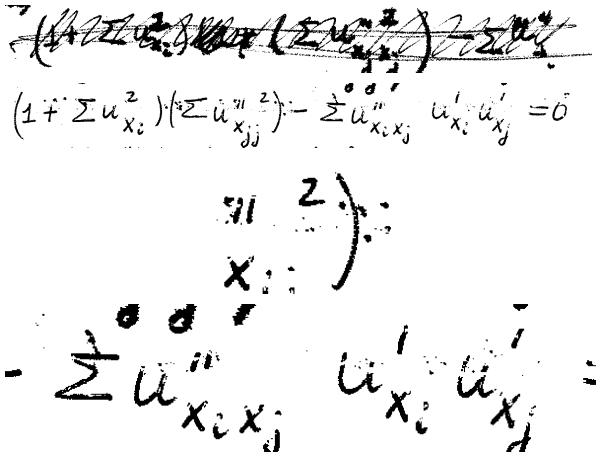


Uncertain reading. The end of the abbreviation translated “satisf.” is reinforced with overlapping strokes; it is followed by “the foll. condition:”.

Original source details: GELFAND-P000171

[Crossed out:]

$$\left(1+\sum u_{x_i}^2\right) \text{ [illegible] } \left(\sum u''_{x_jj}{}^2\right)$$
$$-\sum u''_{x_i}\,[?]$$
$$\left(1+\sum u_{x_i}^2\right)\left(\sum u''_{x_jj}{}^2\right)-\sum u''_{x_ix_j}u'_{x_i}u'_{x_j}=0.$$



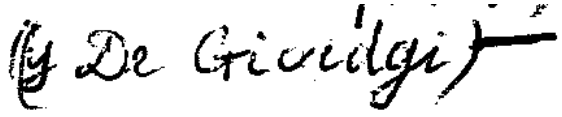
Uncertain reading. The first line is densely cancelled. Between the first closing parenthesis and the next sum are inseparable overlapping strokes; the ending is $-\sum u''_{x_i}$-like, with uncertain indices and upper strokes. In the uncancelled line a separate raised 2 is visible after the double prime on u_{x_jj}; double j / compressed x_jx_j remains uncertain.

Original source details: GELFAND-P000171

Is it true that u is linear?

(These are the eqns. of min. surfaces.)

Done. (there is a reference in De Giorgi [?])



Uncertain reading. The Latin name: De Giorgi / De Gicrigi / De Giccilgi-like joined middle letters. Before the name is the Russian “y” (“in”, here), with additional vertical strokes.

Original source details: GELFAND-P000171

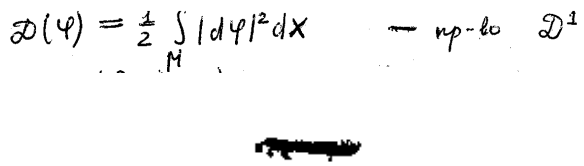
4. M is a comp. Riem. manif.

$$L_1^2(M),\quad \langle\varphi,\psi\rangle=\int_M \varphi\psi\,dx+\int\langle d\varphi,d\psi\rangle\,dx.$$

GELFAND-P000172

[source number: 33]

$$D(\varphi)=\frac{1}{2}\int_M|d\varphi|^2\,dx \quad \text{— sp. } D^1\,[?].$$



Uncertain reading. Before “sp. D^1” is a dense horizontal stroke: dash / arrow, with an overwritten left end; its direction is undetermined.

Original source details: GELFAND-P000172

$$(L_1^2(M),D)$$
$$(L_1^2(M'),D').$$

There is an isometry of $L_1^2(M)$, inducing a diffeom. of M and M'. Prove that M' and M are isometric.

Invariants — eigenvalues of the Laplace oper.

[Crossed out: ~~Er~~ [?]]



Original source details: GELFAND-P000172

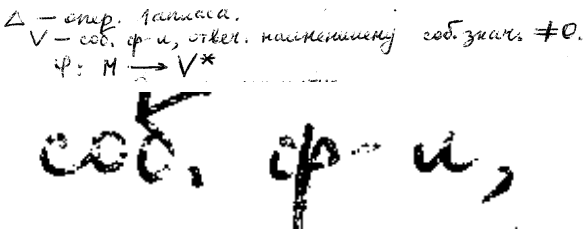
Uncertain reading. Above number 5 is a short cancelled word: “Er” / “Or”; its joined strokes are not fully distinct.

5. M is an n-dim. comp. irreduc. symm. sp.

Δ is the Laplace oper.

V — eigensp. [?], corresp. to the smallest eigenvalue $\neq 0$.

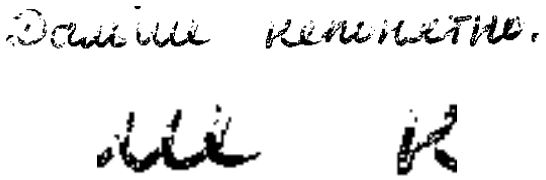
$$\varphi : M \longrightarrow V^*.$$



Original source details: GELFAND-P000172

Uncertain reading. After the abbreviation translated “eigen” is a compressed “pr-i”- / “pr-vo”-like contraction; the final strokes are not fully separate. Before 0 is $\neq$; the map ends in V^*.

If not [?] compact.



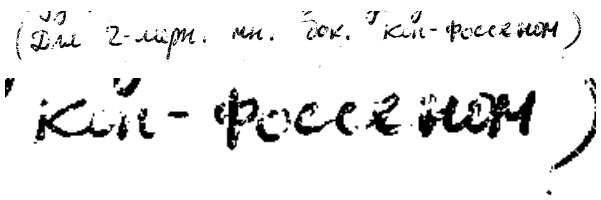
Original source details: GELFAND-P000172

Uncertain reading. Before “compact” are faint joined strokes: “If not compact.” / “If … compact.”.

6. M is a complete noncomp. Riem. manif. (3-dim.) of pos. curvature (Ricci tensor > 0).

Will V here go to ∞.

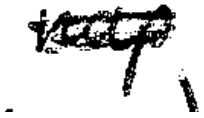
(For 2-dim. manif. prov. by Kn-Fossen [?])



Original source details: GELFAND-P000172

Uncertain reading. The surname is “Kn-Fossen” / “Kon-Fossen”; the middle strokes and possible second “s” are joined.

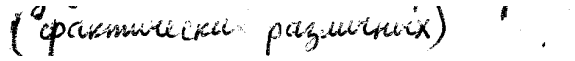
7. M is a comp. n-dimensional manif. Do ∞-many [Crossed out: ~~par~~ [?]] intersect. geodesics exist.



Original source details: GELFAND-P000172

Uncertain reading. Before “intersect.” is cancelled “par” / a short word with a descending stroke; the contraction is not expanded.

(actually [?] distinct)



Original source details: GELFAND-P000172

Uncertain reading. The first long word in parentheses is read as “actually”; some internal strokes are joined. The second word is “distinct”.

3. [?]



Original source details: GELFAND-P000172

Uncertain reading. The number after 7 looks like “3.” / “8.”: a small upper loop and an open lower arc.

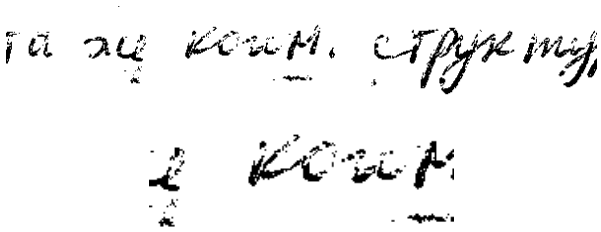
M is a comp. simply connected Riem. manif. in which all geodesics are closed and have the same length. Prove that it is diffeom. to a symm. sp. [Crossed out: $\rho =$ [?] illegible] of the 1st type. (?)



Original source details: GELFAND-P000172

Uncertain reading. Before “of the 1st type” is a cancelled $\rho =$- / $p =$-like sign with an overlapping continuation.

There is an analyt. manif., diffeom. to a sphere, but not isometr.
(Bott). Bott proved that it has the same cohom. [?] structure.



Original source details: GELFAND-P000172

Uncertain reading. The internal strokes in the contraction before “structure” are joined; it is read as “cohom.”. Under it are two short strokes. Bott is written twice, the first time in parentheses.

9. Does a contact structure exist on every comp. M^3?

GELFAND-P000173

$21/\overline{XI}$. Sem. of Gelfand.

Arnold.

Lecture 1.

Theory of singularities of diff. mappings.

$$f : M^m \longrightarrow N^n.$$

$y = x^2$.



Typeset diagram



Original source details: GELFAND-P000173

Stab. [Crossed out: illegible word] sing.

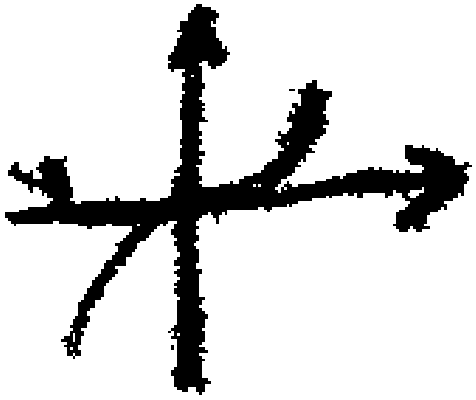


Undeciphered passage. Between “Stab.” and “sing.” is a densely cancelled short word; the letters beneath the cancellation strokes are unrecovered.

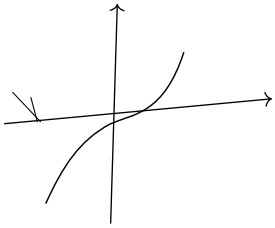
Original source details: GELFAND-P000173

(for a nearby diff. mapping, reduced to the same form)

$y = x^3$.



Typeset diagram



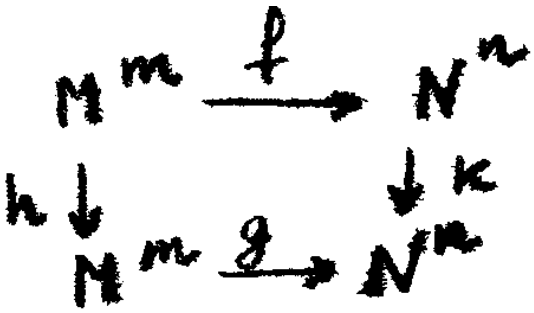
Original source details: GELFAND-P000173

Unstab. sing.



Original source details: GELFAND-P000173

Uncertain reading. At $y = x^2$ is a U-shaped line with an almost vertical internal stroke and a horizontal line below it. At $y = x^3$ are axes with upward and rightward arrows, an S-like line from lower left to upper right, and small oblique strokes left of the crossing. The axes have no letter labels; exact contacts in the dense crossing are undetermined.



Original source details: GELFAND-P000173

Typeset diagram

$$\begin{array}{ccc} M^m & \xrightarrow{\quad f \quad} & N^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ M^m & \xrightarrow{\quad g \quad} & N^n \end{array}$$

Top. stab. $\exists$ homeom. h and k.

(diff.)

$\downarrow$

Diff. stab.

Stability at 0 — the same on neighborhoods.

(germs of a mapping)

(One can change the manifolds so that the mapping is given by the same formulas)

Rang max [?] — stable (nondeg. funct.).

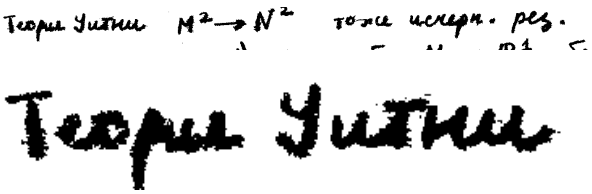


Original source details: GELFAND-P000173

Uncertain reading. The beginning of the line: Rang max / Ring max / min max; the first sign is tall and looped, the final stroke of the first word descends.

Morse lemma — nondeg. quadr. form ($R^n \to R^1$ stab.)

Theorem [?] of Whitney $M^2 \to N^2$ also an exhaustive res.



Original source details: GELFAND-P000173

Uncertain reading. The first source word is "Teorem"-like; its short final strokes are joined. Then Whitney and a map with two upper 2s.

Morse M is comp. 1) stab. mappings $M \to R^1$ form an open everywhere dense set.

2) Conditions for stability (necess. and suff.):

GELFAND-P000174

M.1 All crit. [Crossed out: z [?], illegible] points nondeg.



Original source details: GELFAND-P000174

Uncertain reading. Before “points” is a cancelled short word with a “z”-like beginning; the reading “values” abbreviated / overlapping short strokes.

M.2 All crit. values are distinct.

3) loc. assertion loc. stability $\Longleftrightarrow$ one can choose coords. so that the funct. is written (one of the ways)

$$y = x_1$$
$$y = x_1^2 + \ldots + x_k^2 - x_{k+1}^2 - \ldots - x_m^2.$$

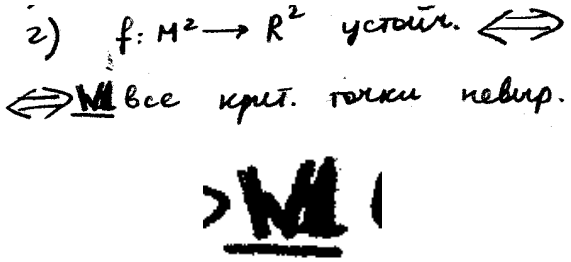
(stab. both top. and diff.)

Whitney's theorem.

1) Stab. mappings are dense.

2) $f : M^2 \to R^2$ stab. $\Longleftrightarrow$

$\Longleftrightarrow$ W1 [?] all crit. points nondeg.



Original source details: GELFAND-P000174

Uncertain reading. The first underlined number is W1 / M1; the angular strokes overlap heavily. The equivalence sign is written twice, again $\Longleftrightarrow$ before the number. The next number is W2.

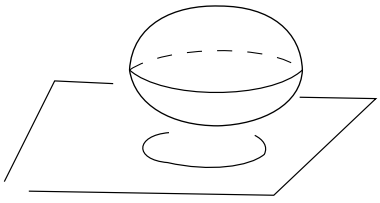
W2 crit. values are transverse (images of folds)

3) Stab. at a given point $\Longleftrightarrow$ the mapping is written by one of the formulas:

1. $y_1 = x_1, \quad y_2 = x_2,$
2. $y_1 = x_1, \quad y_2 = x_2^2$ (folds),
3. $y_1 = x_1, \quad y_2 = x_1 x_2 + x_2^3.$



Typeset diagram



Original source details: GELFAND-P000174

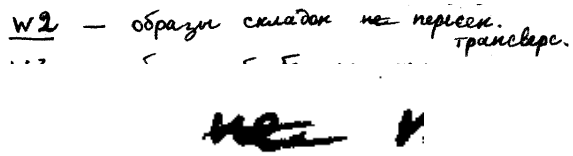


Original source details: GELFAND-P000174

Uncertain reading. A sphere-like outline with a transverse ellipse whose upper arc is dashed; below it are an open oval and a plane with gaps in the upper edge and lower left corner. There are no labels or connecting arrow; the precise projection is unspecified.

2. — fold points, these are curves where the Jacobian $= 0$.

W2 — images of folds [Crossed out: ~~not~~ [?]] intersect. transvers.

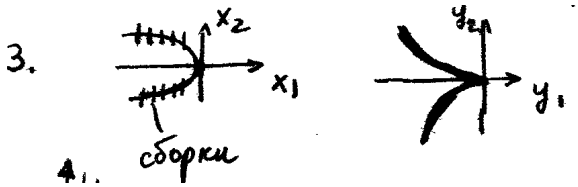


Original source details: GELFAND-P000174

Uncertain reading. Before “intersect.” is a short crossed “not”-like word / dense crossed sign.

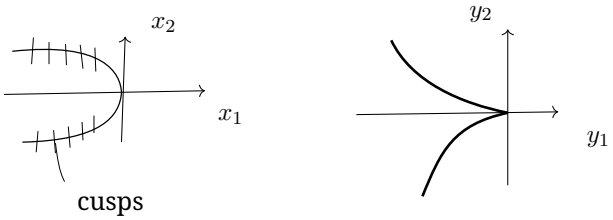
W3 — images of cusps are distinct.

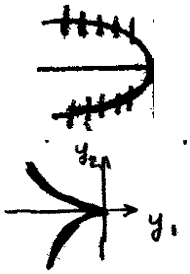
3.



Original source details: GELFAND-P000174

Typeset diagram





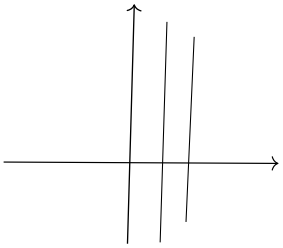
Original source details: GELFAND-P000174

Uncertain reading. At left are axes x_1, x_2, a line with two leftward branches and short strokes, and a “cusps” callout to the lower branch. At right are axes y_1, y_2 and two thick branches meeting near their crossing. Exact stroke placement and contact in the dense crossing remain uncertain.



Original source details: GELFAND-P000174

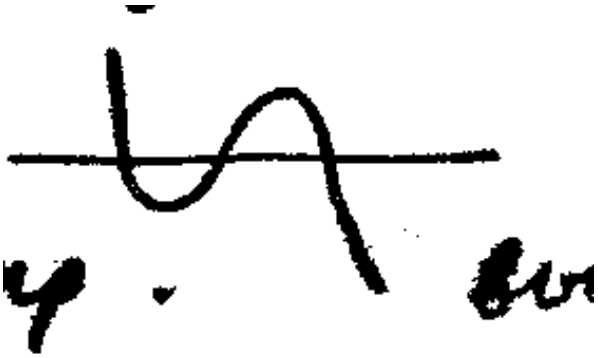
Typeset diagram



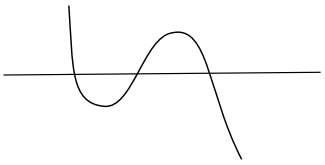
For $x_1 > 0$, monot. on vert. lines.

$$x_1 = \text{const} \Rightarrow y_1 = \text{const}\,.$$

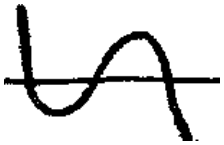
For $x_1 < 0$ there will be



Typeset diagram



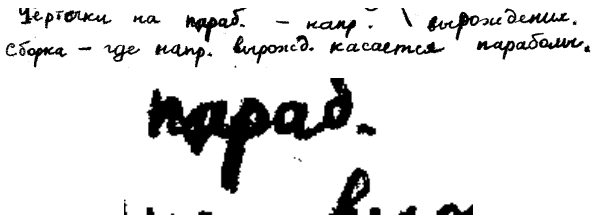
Original source details: GELFAND-P000174



Original source details: GELFAND-P000174

Uncertain reading. An arrowless horizontal line and a curve descending from upper left, then a trough, a rise above the horizontal and a descent at right. There are no letter labels; crossing placement is specified only by the drawing.

Short strokes on the parab. [?] — directions of degeneracy.
A cusp is where the dir. of degen. is tangent to the parabola.



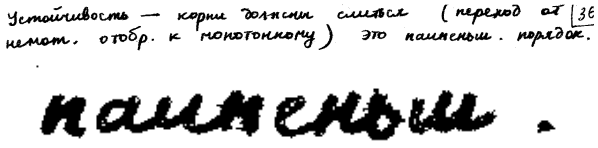
Original source details: GELFAND-P000174

Uncertain reading. The first line has “parab.” / a compressed “prab.”-like contraction with overlapping initial strokes. The last line writes “parabola” in full.

GELFAND-P000175

[source number: 36]

Stability — the roots must merge (transition from a nonmonot. map to a monotone one)
This is odd [?]. order.



Original source details: GELFAND-P000175

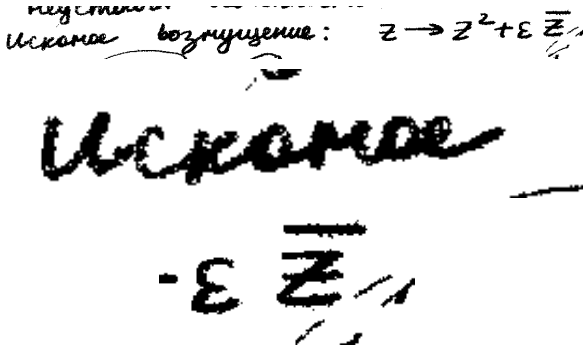
Uncertain reading. The ending of “odd” / the abbreviated Russian word is compressed; a separate dot follows it, before “order”.

$$z \longrightarrow z^2$$

Unstable singularity.

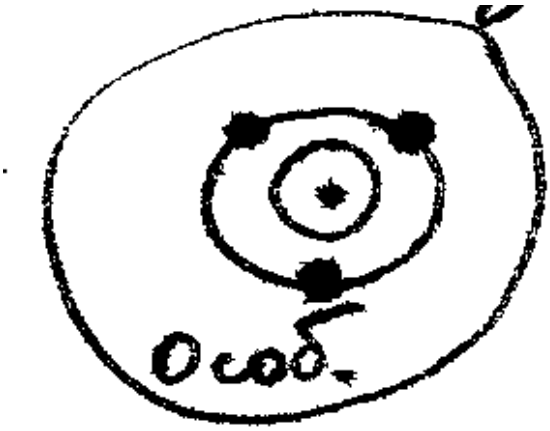
Sought [?] perturbation:

$$z \longrightarrow z^2 + \varepsilon\bar{z}.$$

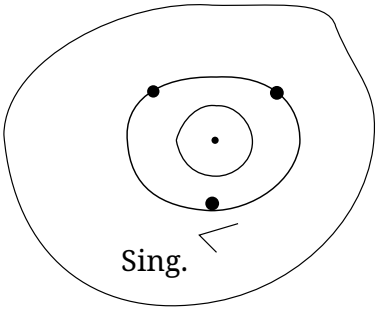


Uncertain reading. The first word has “Iskanoe” / “Iskomoe”-like joined strokes. The second word is “perturbation”; the final z has a horizontal bar. The lower oblique strokes belong to the adjacent drawing.

Original source details: GELFAND-P000175



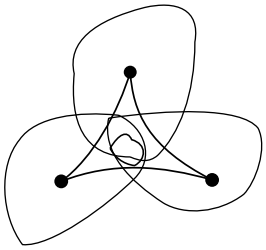
Typeset diagram



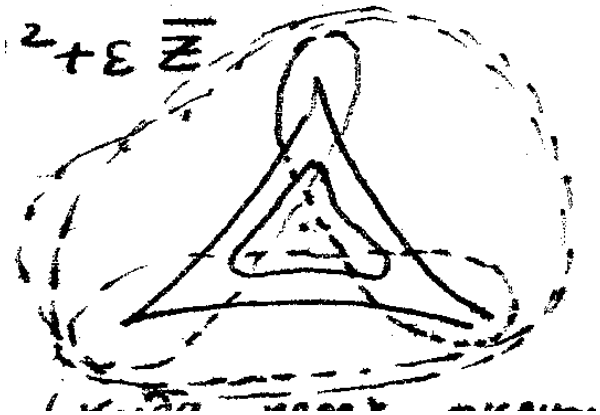
Original source details: GELFAND-P000175



Typeset diagram

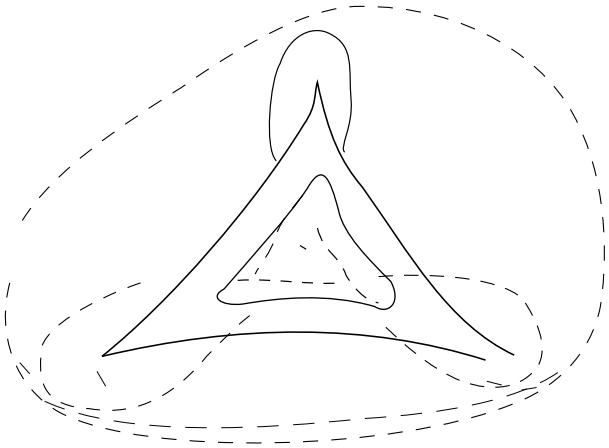


Original source details: GELFAND-P000175

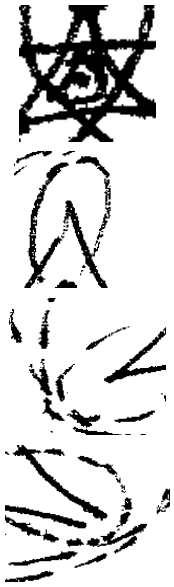


Original source details: GELFAND-P000175

Typeset diagram



Uncertain reading. Three nested circles with a central point and three points on the middle circle; callout “Sing.”. Beside them are three loop-like petals, three heavy points, connecting strokes and a small central contour. At right are concave triangular contours, an upper loop, an outer dashed arc, lower dashed loops and short oblique strokes. Exact connectivity remains uncertain at the dense crossings.

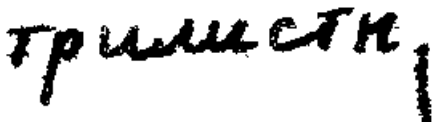


Original source details: GELFAND-P000175

(Where did the circles go?)

Large circles make 2 turns (trefoil [?] knot)

Большие окр. делают 2 оборота (трилистн, узел)



Original source details: GELFAND-P000175

Uncertain reading. “Trefoil” is abbreviated with a fused final letter; a separate descending stroke follows its dot. The number of turns is written 2.

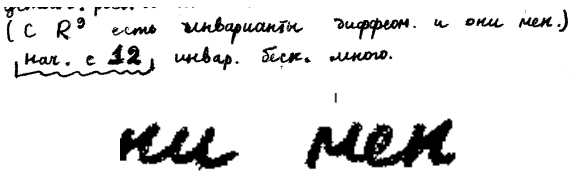
Up to dim. 5-6 there is a classific.

Theorem 1 (Thom)

For maps $R^9 \to R^9$ (and higher), everywhere density of stable germs does not hold.

(from R^9 there are differential invariants and they men.[?])

Starting from 12, there are infinitely many invariants.



Uncertain reading. The Russian abbreviation “men.” / similar strokes remains unexpanded. “Starting from 12” is emphasized below by a wavy line with raised ends.

Original source details: GELFAND-P000175

Thom:

Theorem 2.

There exists a discrete topol. classific. exists!!!

(What it looks like — is unknown)

(If from the space of all maps one discards A_∞ — infinite codim.

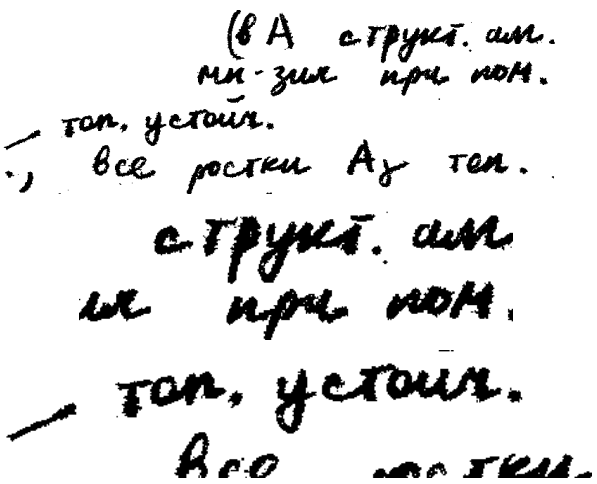
$$A = \bigcup_{\gamma \in \Gamma} A_\gamma + A_\infty.$$

Γ — a countable set.

Each A_γ — finite codim.; all germs of A_γ are topol. equiv. to one another, for different A_γ inequivalent.

(in A jet [?] alg. varieties with topol. [?]

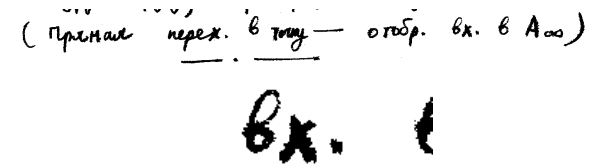
topol. stable



Uncertain reading. The right insertion has “jet” / “struct.” and “topol.” / “that”; an opening parenthesis. Below it, separately, “topol. stable” with an oblique leader from the finite-codimension line; the exact attachment of the insertion is not established.

Original source details: GELFAND-P000175

(A line maps to a point — a map ent. [?] in A_∞)

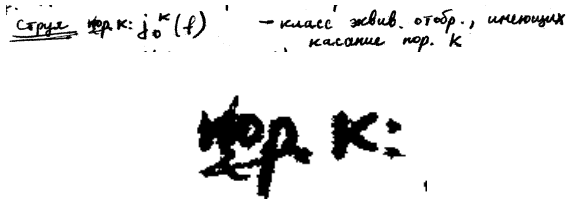


Uncertain reading. The Russian contraction is “ent.” / joined “vl.”-like strokes; “maps” and “to a point” are separately underlined.

Original source details: GELFAND-P000175

$f : M^m \to N^n$ a map germ.

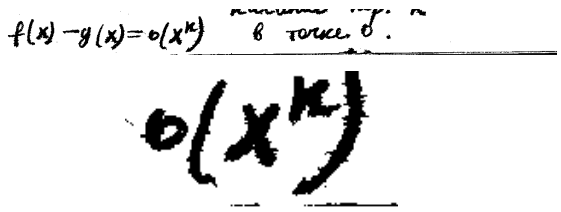
Jet of ord. [?] k: $j_0^k(f)$ — the equivalence class of maps having contact of ord. k



Uncertain reading. “Jet” is doubly underlined. The first Russian “ord.” has superposed strokes; the next line has “contact of ord. k”.

Original source details: GELFAND-P000175

$$f(x) - g(x) = o(x^k) \quad \text{at the point } 0.$$



Uncertain reading. The remainder sign is a small oval o / O; its argument is x^k without absolute-value bars. The line lies at the lower edge of the frame.

Original source details: GELFAND-P000175

GELFAND-P000176

Jets of order 0 — the image of a point
„ 1 — the image and its differential.

$$j_0^k(f) \quad \text{defined invariantly} \quad \text{(def. invar.)}$$

Jets of fixed ord. — an alg. variety (in coordinates — — Taylor series). The bundle of jets of maps from M to N

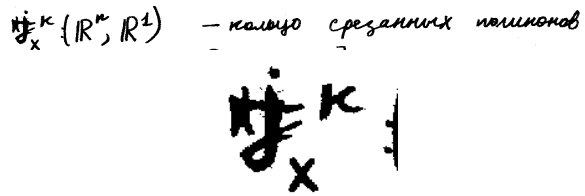
$$J^k(M, N) \quad \text{— a bundle over } M.$$

For $N = R^n$ this is a vector bundle.

(With natural topol., diff. and alg. structure)

$$R^n \longrightarrow R^1.$$

$J_x^k(R^n, R^1)$ [?] — the ring of truncated polynomials



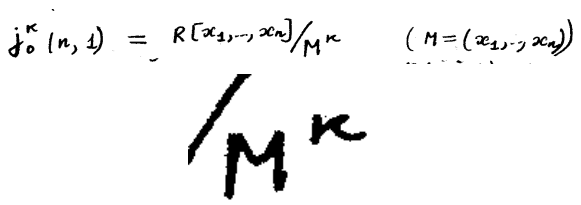
Uncertain reading. Initial letter J / j with dense transverse strokes through its vertical part; subscript x and superscript k are separately visible.

Original source details: GELFAND-P000176

$$R[x_1, \ldots, x_n]$$

$$j_x^k(f) = j_x^k(n, 1)$$
$$(f : R^n \to R^1).$$

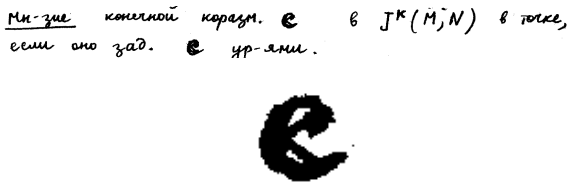
$$j_0^k(n,1) = R[x_1,\dots,x_n]/M^k \qquad (M=(x_1,\dots,x_n)).$$



Uncertain reading. The denominator visibly has M^k; no superscript $+1$ is visible. The M has a long left leg; short neighboring strokes remain unexpanded.

Original source details: GELFAND-P000176

A variety of finite codim. e [?] in $J^k(M,N)$ at a point, if it is given by e [?] equations.



Uncertain reading. The letter for codimension and number of equations is heavily retraced: e / an ε-like form. "Variety" is underlined.

Original source details: GELFAND-P000176

Natural map:

$$f \xrightarrow{j} j_x^k(f).$$

ξ [?]

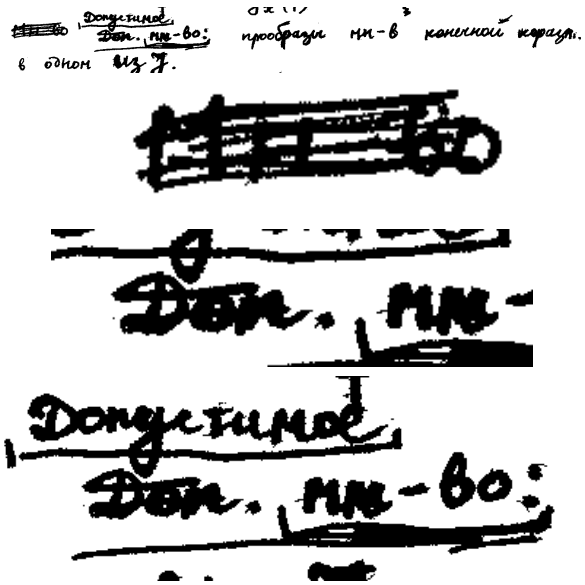


Uncertain reading. At right a separate vertical three-lobed sign: ξ / flourish; no connection to the formula is shown.

Original source details: GELFAND-P000176

Admissible

[Cancelled: illegible] [Cancelled: ~~Adm.~~ [?]] set: inverse images of sets of finite codim. in one of the J.

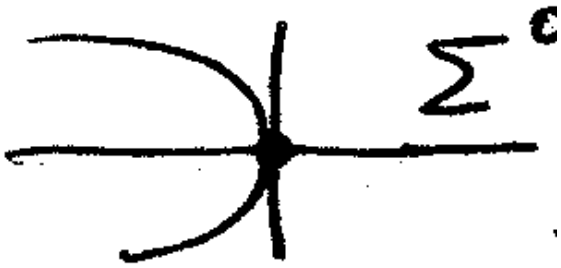


Undeciphered passage. The cancelled short word at left is unrecovered. In the next word, "Adm." / "This"-like superposed strokes; underlined "Admissible" inserted above, followed by underlined "set:".

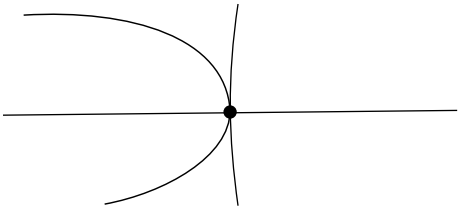
Original source details: GELFAND-P000176

Consider $f : M^m \longrightarrow N^n$.

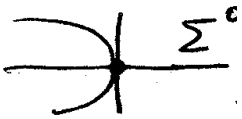
$$x \in \Sigma^i(f), \quad \text{if}$$
$$\dim \operatorname{Ker} df\big|_x = i.$$



Typeset diagram



Original source details: GELFAND-P000176



Original source details: GELFAND-P000176

Uncertain reading. Horizontal and vertical lines without arrows or letters; a heavy point near the crossing, a rounded curve with two leftward branches. Exact contact inside the heavy point remains uncertain.

Σ^0 — all points without the parab.

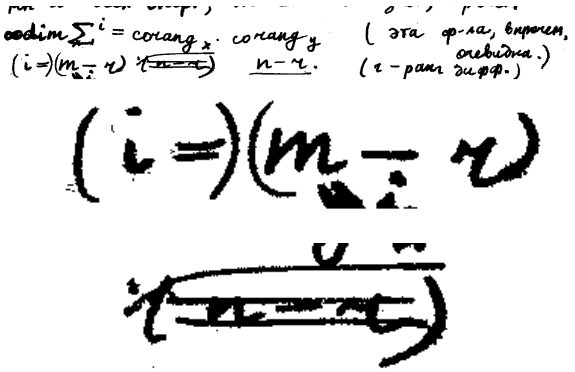
Σ^1 — the whole parabola.

Theorem.

The set of all maps where Σ^i are varieties is dense, and

$$\operatorname{codim} \Sigma^i = \underline{\operatorname{corang}_x} \cdot \operatorname{corang}_y .$$

$$\underline{(i =)(m - r)} \qquad \underline{n - r}.$$



Original source details: GELFAND-P000176

Uncertain reading. The upper formula has a dot between coranks; the first corank is underlined. Below, separately, "$(i =)(m - r)$", a heavily crossed connecting stroke with a right arc, and underlined $n - r$. No clear product sign is recovered between the lower factors; two short heavy strokes lie under m at the edge.

(r — rank of the diff.)

(This formula is, however, obvious.)

GELFAND-P000177

[source number: 37]

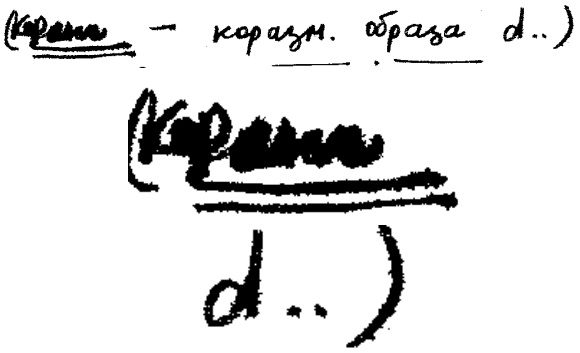
$z \longrightarrow z^2$ is already unstable.

The condition on Σ^i — an algebraic subset in J — the required codimensions (minors $= 0$).

In $J^1(M, N)$ the varieties $\Sigma^1, \Sigma^2, \ldots$ are defined only by germs.

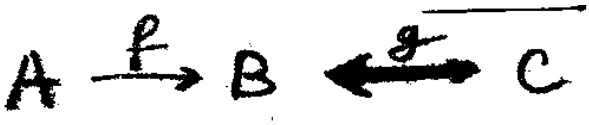
An actual map $\Rightarrow$ a section.

(Corank [?] — codim. of the image d..)



Original source details: GELFAND-P000177

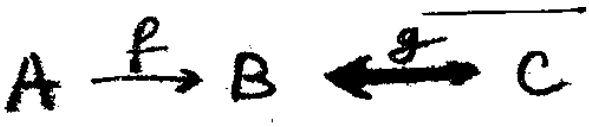
Uncertain reading. The opening “Corank” / “Codim.” is heavily overwritten and doubly underlined. Two small separate strokes follow d; they are not expanded into a map name.



Original source details: GELFAND-P000177

Typeset diagram

$$A \xrightarrow{f} B \xleftarrow{g} C$$



Original source details: GELFAND-P000177

Uncertain reading. At left $A \to B$, with f above the arrow. At right $B \leftarrow C$ with g; the right end of the heavy stroke also has a pointed thickening, so the appearance of two heads remains uncertain.

f, g are transverse if $\forall a \in f^{-1}(b), c \in g^{-1}(b)$ at the point $b \in B$

the relation holds:

$$f_* TA_a + g_* TC_c = TB_b \qquad \text{(may intersect).}$$

2 lines in R^3 are not transverse at even 1 point (if they intersect) (they may intersect).

If a map is transverse to a subvariety, (g — the embedding $C \subset B$), then the inverse image of the variety — is a variety.

Its Codim is the same

$$\operatorname{codim} f^{-1}C\big|_A = \operatorname{codim} C\big|_B.$$

If $f : M \longrightarrow N$ is transverse to Σ^i, then all is well.

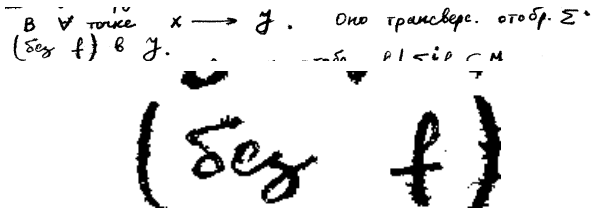
One can perturb $\forall$ map.

Further classific.

$$\Sigma^i f \subset M, \qquad f : M \longrightarrow N.$$

Suppose that the section defined by the map f is transverse to Σ^i in the jets.

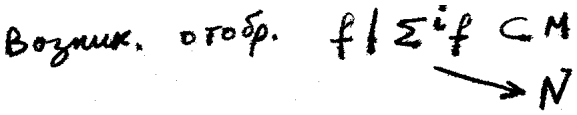
At $\forall$ point $x \longrightarrow J$. It is transverse to the map Σ^i (without f) [?] in J.



Original source details: GELFAND-P000177

Uncertain reading. After “without f”, a closing sign with several bends: a round parenthesis / curled stroke. Both letters are J; the phrase reads “map Σ^i”.

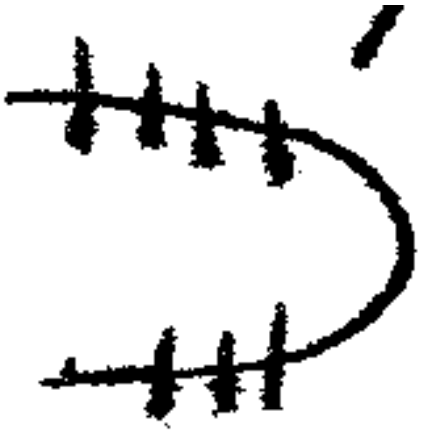
There arises a map



Original source details: GELFAND-P000177

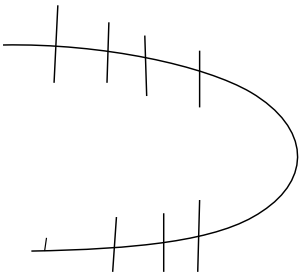
Typeset diagram

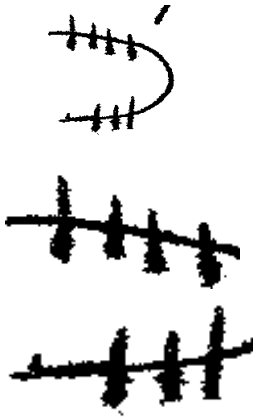
$$f|\Sigma^i f \subset M \longrightarrow N$$



Original source details: GELFAND-P000177

Typeset diagram





Original source details: GELFAND-P000177

Uncertain reading. A rounded curve with a right vertex and two leftward branches, without axes or labels. Four long strokes on the upper branch; three long and one short stroke near the left edge on the lower branch. Exact stroke lengths and contacts remain uncertain.

GELFAND-P000178

The same classific. is applied to it:

$$\Sigma^j f | \Sigma^i f$$

$x \in \Sigma^{ij}(f)$ — a subvariety (Boardman 1965).

(One can perturb) $\forall$ numbers of indices.

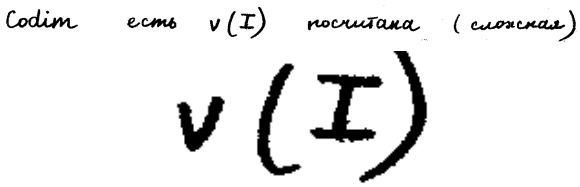
Thm.

1. $\forall k$ in $J^k(M, N)$ algebraic subvarieties are defined

$$\Sigma^I \qquad I = (i_1, \ldots, i_k)$$

(independent of f.)

Codim is $\nu(I)$ [?] has been calculated (complicated).



Uncertain reading. The function sign is ν / a v-like form, followed by argument I in parentheses. “Codim” is in Latin letters.

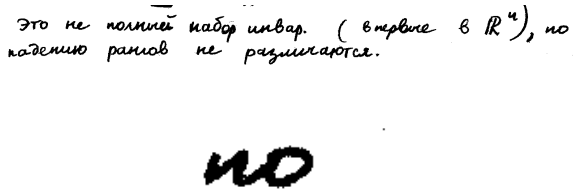
Original source details: GELFAND-P000178

2. Every map is approximated by a map transverse to all these

$$\Sigma^I(f).$$

Classific. Σ^{110} and so on.

This is not a complete set of invariants (first in R^4), but [?] by rank drops are not distinguished.



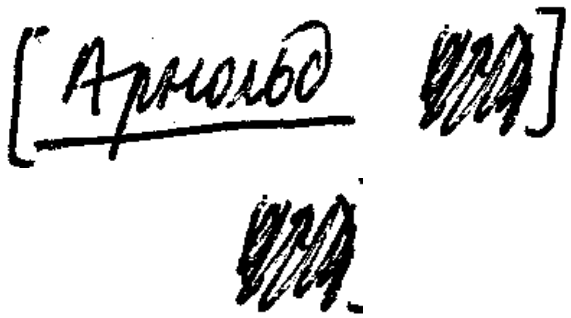
Uncertain reading. The word after the comma is “but” / “by”: the first joined sign has two rising legs. Then “by rank drops are not distinguished”, with no separate subject. The Russian “not complete” is written separately; the exponent on R is4.

Original source details: GELFAND-P000178

GELFAND-P000179

28/XI. Sem. Gelfand.

[Arnold [Cancelled: UR [?]]]



Uncertain reading. After the surname, inside square brackets, two or three heavily cancelled signs: UR-like / Ch-like superposed strokes. The word is not established.

Original source details: GELFAND-P000179

$$f : M^m \longrightarrow N^n.$$

$$\Sigma^i(f) = \{x \in M^m : \dim \ker df_x = i\} \quad \text{— a subset.}$$

if it is a variety, then

$$\Sigma^{ij}(f) = \Sigma^j\Sigma^i(f) = \Sigma^j(f|\Sigma^i f) \quad \text{— Thom's naive def..}$$

(forget)

For $\forall$ multi-index $i_1, i_2, \ldots, i_k = I$ is defined

$$\Sigma^I(f),$$

if varieties are obtained at every step.

There must be

$$i_1 \geq i_2 \geq \ldots$$

$$K_1 = \text{Ker}\, df_x, \qquad K_2 \subset K_1 \quad \text{— next kernel on the subsequent}$$

subvariety.

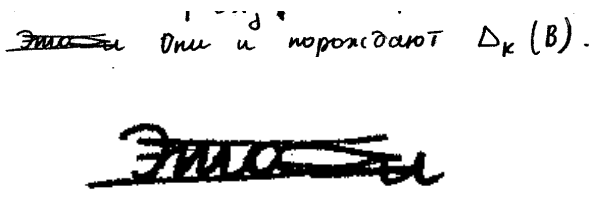
There is a subset independent of the map $\Sigma^I \subset J^k(M, N)$ (is constructed)

A — the ring of functions on a variety with coord. system $x_1, \ldots, x_m$
$B \longrightarrow$ an ideal in A.

The new ideal — the Jacobian extension of B of ord. k $\Delta_k(B)$ — — to the functions from $B \ni \{f\}$ adjoin the minors formed from them

of order k $\left|\frac{\partial f_i}{\partial x_j}\right|$ and adjoin to the ideal.

[Cancelled: ~~This~~ [?]] They also generate $\Delta_k(B)$.



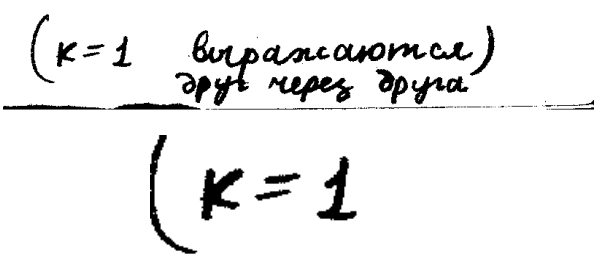
Uncertain reading. Initial cancelled "This..." / similar joined letters with a superposed horizontal stroke. The readable following phrase begins "They also generate".

Original source details: GELFAND-P000179

This does not depend on coordinates (in fact an invar. constr.)

$$\frac{\partial f}{\partial x_i'} = \sum_j \frac{\partial f}{\partial x_j} \cdot \frac{\partial x_j}{\partial x_i'}.$$

($k = 1$ [?] are expressed)
through one another



Uncertain reading. The first sign is k / a κ-like form; $= 1$ is visible. The closing parenthesis follows “are expressed”, while “through one another” is separate below at the frame boundary.

Original source details: GELFAND-P000179

GELFAND-P000180

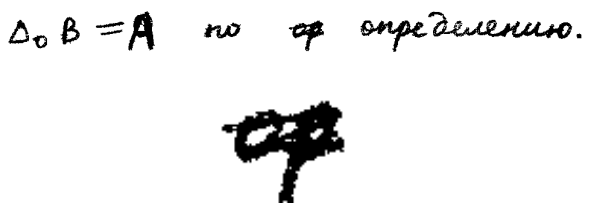
Relation of extensions of different orders.

$k \nearrow \ldots$ ideals $\searrow \ldots$ (decrease)

The extension $\Delta_k(B)$ is critical if k is max for which the ideal is not yet all of B.

$$\Delta_{k-1}B = A, \qquad \Delta_k B \subset A$$

$\Delta_0 B = A$ by [cancelled: illegible] definition.



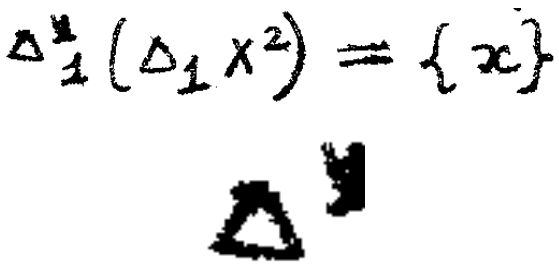
Uncertain reading. Between "by" and "definition" is a small densely overwritten sign with a downward stroke: a φ-like sign / an unidentified contraction. The cancelled text is undetermined.

Original source details: GELFAND-P000180

Example: the ideal generated by x^3.

$k = 1$ $\quad \Delta_1 = \{x^2\}$ $\quad$ critical 1st

$$\Delta_1^2(\Delta_1 x^2) = \{x\}$$



Uncertain reading. The small upper sign on the first Δ_1 resembles 2 / a densely overwritten upper sign. Inside the parentheses, $\Delta_1 x^2$; at right, $\{x\}$.

Original source details: GELFAND-P000180

The extension is performed several times.

Definition: $\Delta^k = \Delta_{m-k+1}$ $\quad (f : M^m \longrightarrow N^n)$

Def. of Boardman:

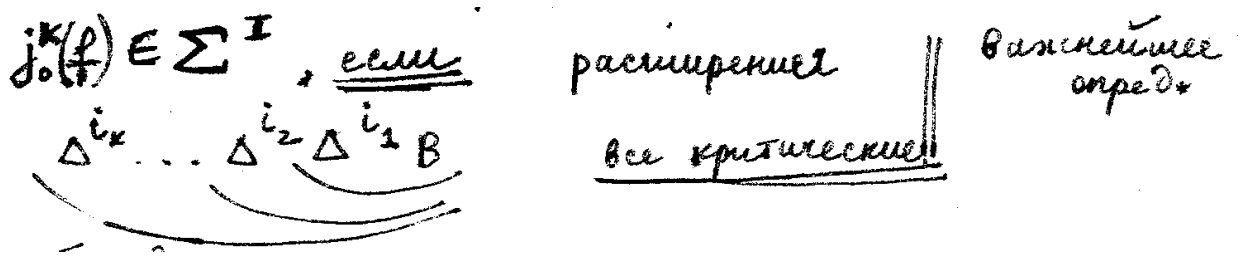
Consider a map f:

$$y_1 = f_1(x_1, \ldots, x_m)$$
$$\ldots$$
$$y_n = f_n(x_1, \ldots, x_m), \qquad f(0) = 0$$

The ring of germs of smooth functions at 0.

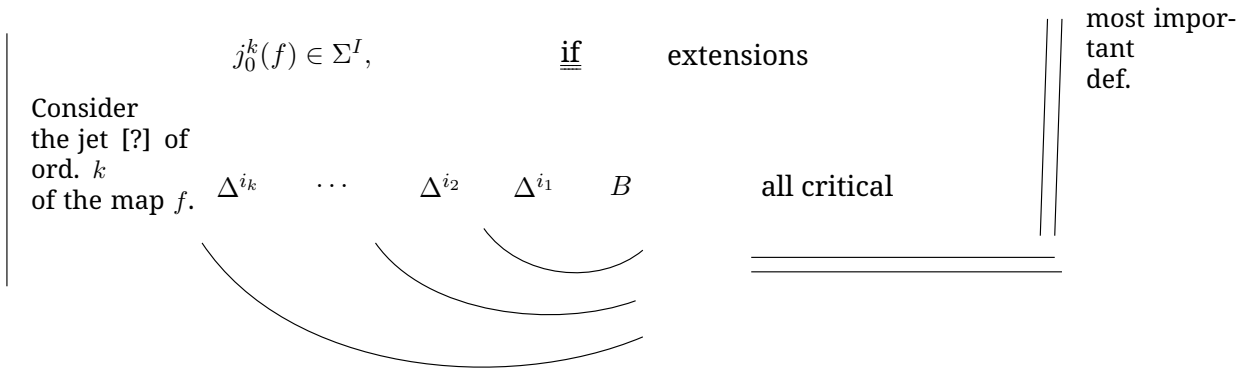
Consider the ideal generated by $f_1, \ldots, f_n$, $\quad B = \{f_i\}$

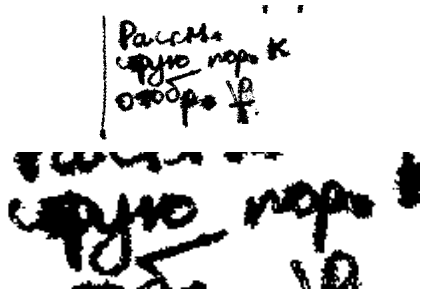
In form the def. depends on coordinates, but in fact it does not.



Original source details: GELFAND-P000180

Typeset diagram





Original source details: GELFAND-P000180

Uncertain reading. The insertion at left reads "Consider the jet of ord. k of the map f": Russian "struyu" / "struya"-like joined and reinforced strokes; the final letter and the beginning of the abbreviation for "map" are not fully separate. A vertical line separates the insertion.

It does not depend on the map f, but only on the jet of ord. k.

The ideal does not depend on X.

Nor does it depend on Y (needs checking).

Main theorem.

This Σ^I is an algebraic variety in the space of jets, and its codimensions have been calculated (functions $= 0$ and $\neq 0$, polynom.)

$$c(I) = (m - n + i_1)\mu(i_1, i_2, \ldots, i_k) - (i_1 - i_2)\mu(i_2, \ldots, i_k) -$$
$$-(i_2 - i_3)\mu(i_3, \ldots, i_k) - \ldots - (i_{k-1} - i_k)\mu(i_k)$$

GELFAND-P000181

[source number: 42]

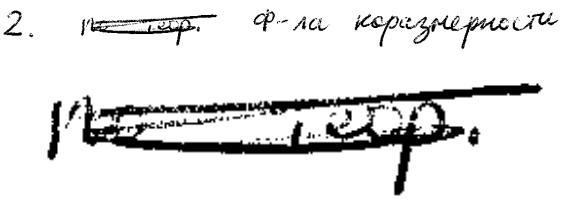
$\mu(i_1, \ldots, i_r)$ is the number of seq. $j_1, \ldots, j_r$ such that

$$1)\ j_1 \geq \ldots \geq j_r$$
$$2)\ i_s \geq j_s \geq 0 \quad \text{for all } \underline{1 \leq s \leq r}.$$

Theorem of Boardman

1. Σ^I is a variety in the space of jets.

2. [Cancelled: illegible] codimension formula.

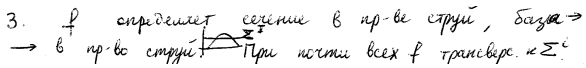


Uncertain reading. Before “codimension formula” several joined letters with a downward stroke are cancelled. The literal cancelled reading is undetermined.

Original source details: GELFAND-P000181

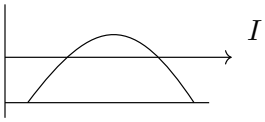
3. f determines a section in the space of jets, base $\longrightarrow$

$\longrightarrow$ into the space of jets



Original source details: GELFAND-P000181

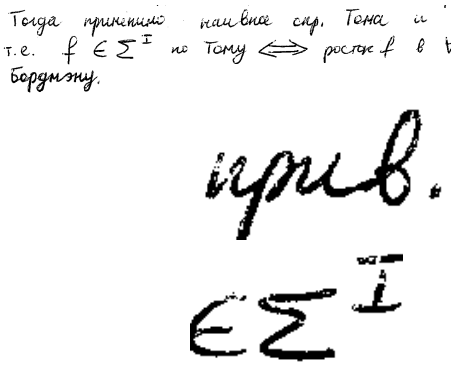
Typeset diagram



For almost all f, transv. to Σ^i

Then Thom’s naive def. and its red. [?] to Σ^i is applicable;

i.e., $f \in \Sigma^I$ in Thom’s sense $\Longleftrightarrow$ the germ of f at $\forall$ point $\in \Sigma^I$ in Boardman’s sense.



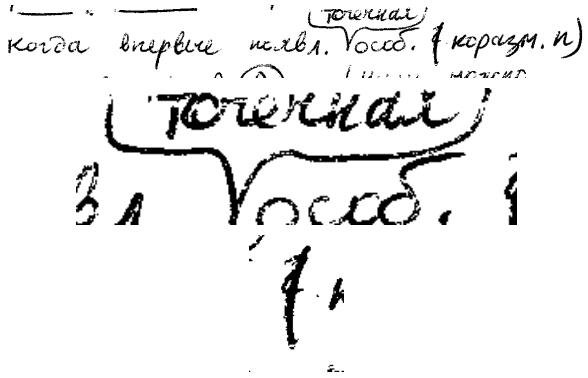
Uncertain reading. After “its” the source contraction reads “priv.” / “pril.”: the final joined stroke forms a loop; the expanded word is undetermined. “Red.” is provisional. The source “is applicable” is singular; on the following line both Σ have a superscript capital I.

Original source details: GELFAND-P000181

Almost all are transv. by the general theorem.

$\underline{R^n \longrightarrow R^n}$.

When do sing. {codim. n) first appear (exact)



Uncertain reading. "Exact" is inserted above the source abbreviation for "sing." and connected to it by a lower brace. Before "codim." a reinforced bracket-like sign: a left brace / an overwritten parenthesis; at right a parenthesis.

Original source details: GELFAND-P000181

n	1	2	3	4	5	6	7	8	(9)
I	1	11	1_3	1_4	1_5	1_6	1_7	1_8	1_9
				2			21		3

x^2 Σ^{11}

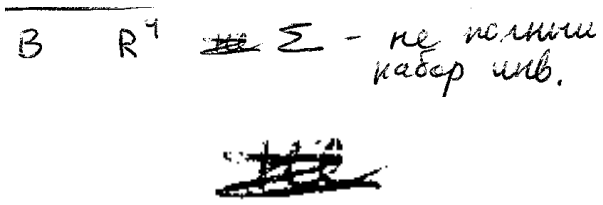
(cusp)

(Need to write out precisely)

(For a general map, more precisely. These are in fact all; further ones are not known.)

10
1_{10}
$2, 2$
$2, 1, 1$

In R^4 [cancelled: illegible] Σ is not a complete set of inv.



Uncertain reading. Between R^4 and Σ a small sign is densely cancelled; the literal reading is undetermined. The source words for "not complete" are written separately.

Original source details: GELFAND-P000181

In dim. 16, sing.: 4 of them.

$n = 9$:

on M^8 — one-dim. folds

M^7 — two-dim. cusps

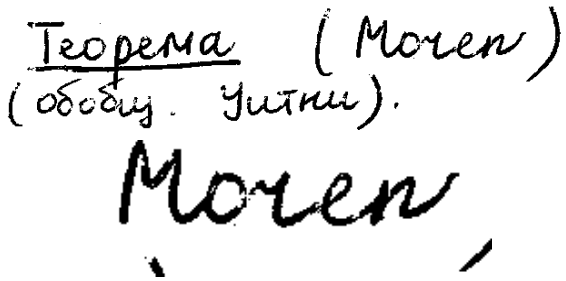
M^5 — sing. of type 2.

The table is easily comp. using the combinat. formula.

1_9: the rank drops by 1 on M^8, then by another 1 on M^7.

GELFAND-P000182

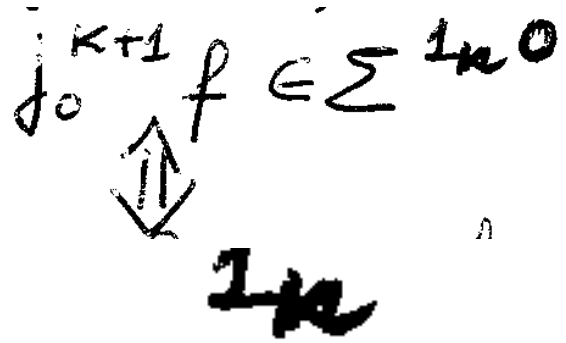
Theorem (Moren [?])
(generaliz. of Whitney).



Original source details: GELFAND-P000182

Uncertain reading. The parenthesized name: “Moren” / “Morin” / Cyrillic “Moren”; the joined third letter and following vowel are not unambiguous. “Theorem” is underlined; below, “(generaliz. of Whitney).”.

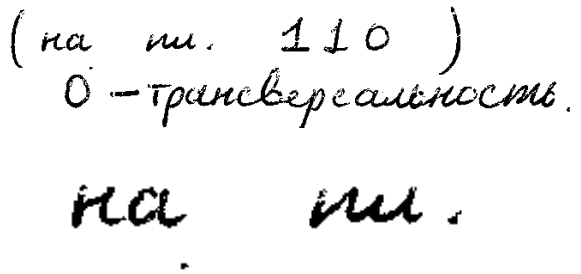
$$j_0^{k+1} f \in \Sigma^{1_n 0}$$
$$\Updownarrow$$



Original source details: GELFAND-P000182

Uncertain reading. The small subscript on 1 in the superscript of Σ: n / k, heavily reinforced. At left, jet order $k+1$; the final 0 is separate.

(e.g. [?] 110)
0: transversality.



Original source details: GELFAND-P000182

Uncertain reading. The right parenthesis before 110 has Russian “na pr.” / joined “napr.”; the abbreviation’s strokes are not fully separate. Beneath is the separate line “0: transversality.”.

$\exists$ coordinates in which

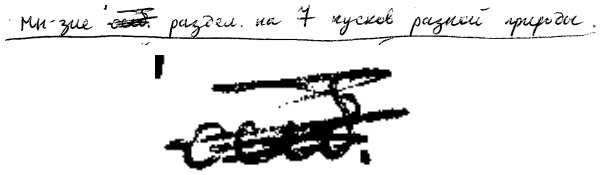
$$x_1 = y_1$$
$$\dots$$
$$x_{n-1} = y_{n-1}$$
$$y_n = x_1 x_n + x_2 x_n^2 + \dots + x_{n-1} x_n^{n-1} + x_n^{n+1}$$

(there, a cubic straight away)
(For classes 1_n, sing. det. up to diffeom.)
For type 20 the class is not a complete inv.

(up to diffeom.)

$$\Sigma^{20} \quad \text{codim. } 4$$
$$f : R^4 \longrightarrow R^4$$

The variety [cancelled: sing.?] is divided into 7 pieces of different nature.



Original source details: GELFAND-P000182

Uncertain reading. After “The variety” a cancelled contraction resembles “sing.” / another cancelled inscription. The joined letters under the strokes do not give a certain full reading. The entire line is underlined.

Quadratic differential of a map.

$f : M^m \longrightarrow N^n$

(In J^k, two diffeomorphism groups of M^m and N^n —

— orbits arise. (at singular points, what orbits))

$$f_x : TM_x \longrightarrow TN_{f(x)}$$

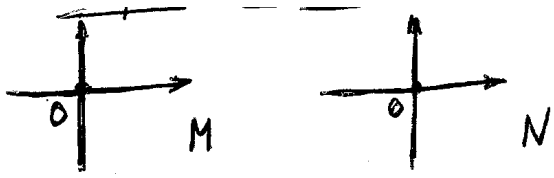
There are $Ker\, f_x \subset TM_x$, $\quad Coker\, f_x = TN_{f(x)}/Im\, f_x$.

$$f_{xx} : Ker\, f_x \longrightarrow Coker\, f_x$$

(a quadratic map, i.e., $\psi : A \longrightarrow B$

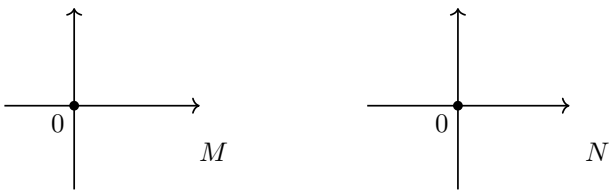
$$A \xrightarrow{\Delta} A \times A \xrightarrow{\overline{\psi}} B$$
$$\Delta \circ \overline{\psi} = \psi)$$

Def. using a system of coordinates.



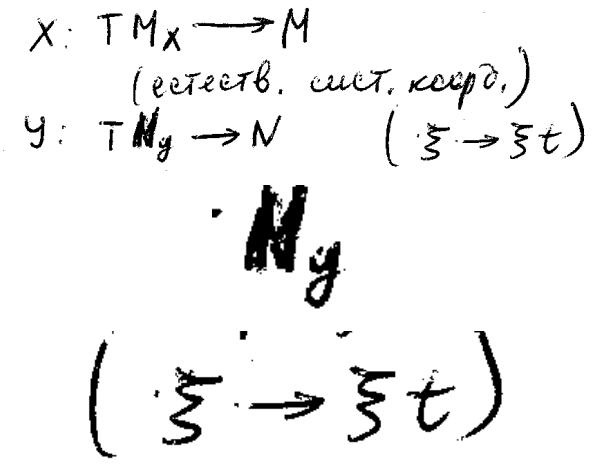
Original source details: GELFAND-P000182

Typeset diagram



$X : TM_x \longrightarrow M$ (natural coord. syst.)

$y : TN_y\,[?] \longrightarrow N \qquad (\xi \longrightarrow \xi t)$



Uncertain reading. The first letter X is capital, the second y lowercase. In TN_y, the N is densely overwritten: N / an M-like lower stroke; the exact underlying form is undetermined. At right, $\xi \longrightarrow \xi t$.

Original source details: GELFAND-P000182

GELFAND-P000183

[source number: 43]

$\xi \in Ker\, f_x$

$X\xi$: a point of the variety

$$\lim_{t\to 0} \frac{(y^{-1} f X\xi t / Im\, f_x)}{t^2}$$

It is easy to prove that this does not depend on the choice of coord. syst.

(The cubic diff. cannot be def. this way)

Look into this!

A bilinear map $f_{xx} \Longrightarrow$ a pencil of quadr. forms.

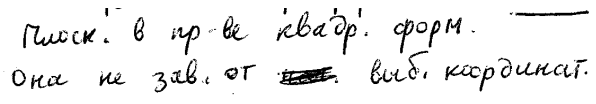
Let $\alpha \in \underbrace{Co\,Coker\, f_0}_{(Coker\, f_0)'} = C'$

αf_{00} is a quadr. form on the kernel.

(already a pure quadr. form)

A plane in the space of quadr. forms.

It does not depend on [cancelled: illegible] the choice of coordinates.



Uncertain reading. Before “the choice of coordinates” a short inscription is densely cancelled; the literal reading is undetermined.



Original source details: GELFAND-P000183

$$C' \longrightarrow \text{Qu.}_{Ker\, f_0}$$

Example. $\quad \Sigma^{20} : R^4 \longrightarrow R^4$

$$2 = dim\, Ker\, f_0,$$

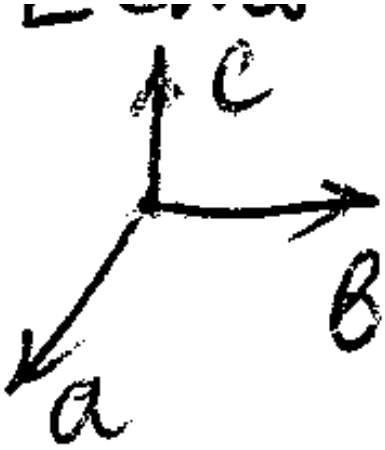
Hence $dim\, Coker\, f_0 = 2$

forms: a 3-dim. space

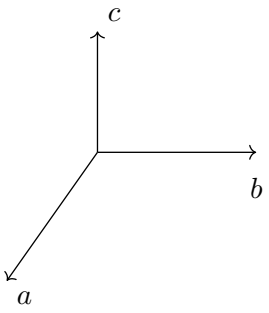
(in 2 vars.)

$$ax_1^2 + 2bx_1x_2 + cx_2^2.$$

There is a classific. by rank



Typeset diagram



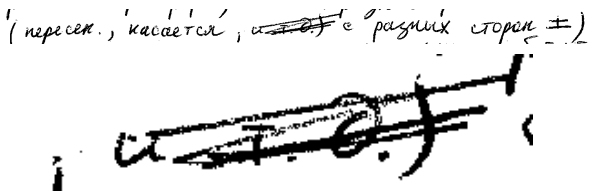
Original source details: GELFAND-P000183

$$\text{Rank} < 2 \quad \Longrightarrow \quad \text{cone } b^2 = ac$$

(inv. def.)

There is also a plane there (a map of a plane into the space of quadr. forms). It can have different positions

(intersects, is tangent, [cancelled: illegible], on different sides $\pm$)



Uncertain reading. After “is tangent” is a cancelled inscription with an oval loop / parenthesis and lower strokes; the literal reading is undetermined. After it, “on different sides $\pm$”.

Original source details: GELFAND-P000183

These positions (a total of 7 pos., including degeneracies), if different, will be different classes of maps.

And there are indeed 7 orbits.

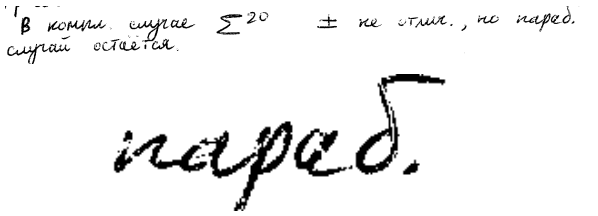
GELFAND-P000184

The maps can be written.

$$y_1 = x_1 x_2$$
$$y_2 = x_1^2 \pm x_2^2 + \underbrace{x_1 x_3 + x_2 x_4}_{\text{added for transversality.}}$$
$$y_3 = x_3$$
$$y_4 = x_4$$

———— · ————

In R^5 there can be a curve Σ^{20} with different sing. along the curve.
In the complex case, Σ^{20} $\pm$ do not differ, but the parab. [?] case remains.



Uncertain reading. After “but”, the contraction reads “parab.” / “pereob.” / “perekhod.”; the joined middle strokes do not give an unambiguous full reading.

Original source details: GELFAND-P000184

Σ^{20}: all sing. are stable, there is a diff. classific.
Σ^{30}: this is no longer so.

———— · ————

Thm. of Thom.

$$\Sigma^{30} : R^9 \longrightarrow R^9$$
$$dim\, Ker\, f_0 = 3$$
$$dim\, Coker\, f_0 = 3$$

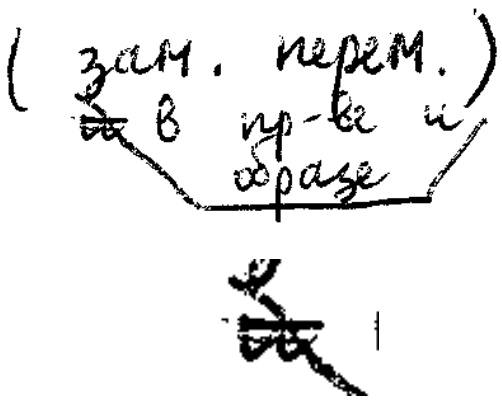
the space of sing. is not discrete (there are moduli)
(Everything is still quadratic)
Lemma. For $n \geq 3$, the dimension of the orbit

$$GL(n, R) \times GL(n, R)$$

in $H(n, n)$: quadr. forms in n vars.

(change of vars.)

[cancelled: ?] in the space and image



Uncertain reading. Beneath “(change of vars.)”, before “in the space and image”, a short sign with joined loops is cancelled. The original inscription is undetermined. A bent stroke connects the parenthesis and interlinear insertion.

Original source details: GELFAND-P000184

$codim$ orbits > 0

(no orbits of full dim.)

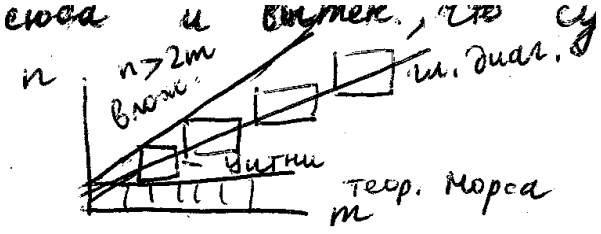
group $2n^2$, forms $\frac{n^2(n+1)}{2}$, and there is always

a one-dim. stationary subgroup ≥ 1

(mult. by a number there and there)

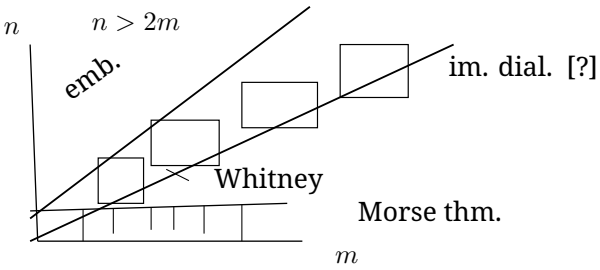
$\underline{n = 3}$ and higher.

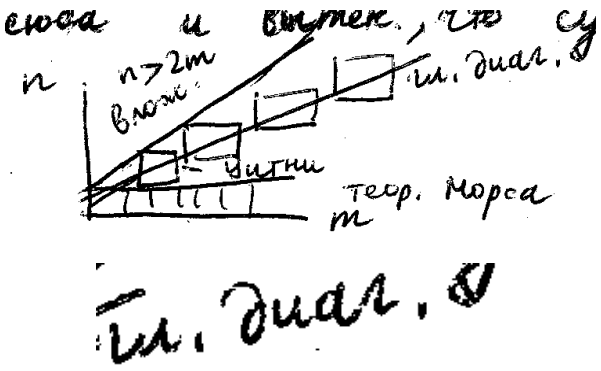
Hence it follows that moduli exist.



Original source details: GELFAND-P000184

Typeset diagram





Uncertain reading. At left a vertical axis n, below a horizontal m, two rising lines, four rectangular outlines, and a lower strip with six vertical strokes. The labels $n > 2m$, “emb.”, “Whitney”, and “Morse thm.” are legible; the small right label is “im. dial.” / “mil. diap.” / an undetermined contraction. Dense crossings at the origin and separate short strokes do not establish additional numerical boundaries.

Original source details: GELFAND-P000184

In cells $9, 9$ and $6, 9$

it has been determined where there are moduli

(*J.* Mather: announced)

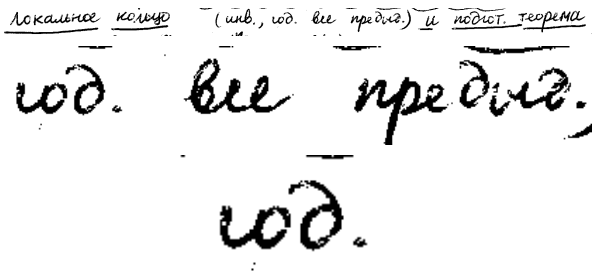
He also: classific. of stable sing.

GELFAND-P000185

[source number: 44]

The only inv. of stable sing. is the local ring.

Local ring (inv., but. [?] all the preceding.) and prepar. theorem

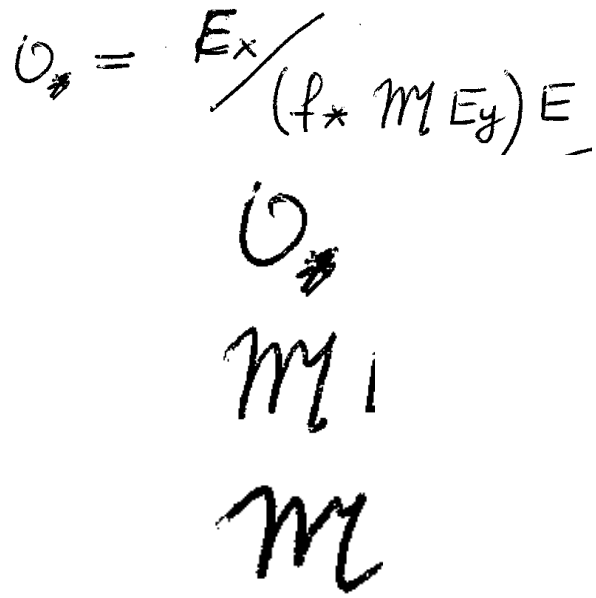


Uncertain reading. The short inscription between “inv.,” and “all the preceding.” reads Russian “no.” / “ne.” / an undetermined contraction. The source spelling is “predid.”; the heading words and “and” are separately underlined.

Original source details: GELFAND-P000185

$$f : M^m \longrightarrow N^n, \qquad f(0) = 0.$$

$$O_* [?] = \frac{E_x}{(f_* \mathfrak{M} [?] E_y) E}$$



Uncertain reading. The initial letter is O without a diagonal tail; the reinforced lower sign is $*$ / x / f. The lower right stroke of $\mathfrak{M}$ is joined to the letter: part of its form / subscript 1; a separate index is undetermined. After the parenthesis, a separate E without a visible subscript, beneath the common quotient slash. The later Q has a separate tail.

Original source details: GELFAND-P000185

E_x: the ring of germs of infinitely diff. functions on M at 0.

E_y: similarly, in y

$\mathfrak{M} E_y$: the max ideal.

f_* is def. naturally.

Meaning: “the ring of functions on the coalesced inverse image.”

Example. $f(x) = x^2$,

$$\mathfrak{M} E_y = y E_y = a_1 y + \dots$$
$$f_* \mathfrak{M} E_y = a_1 x^2 + a_2 x^4 + \dots$$

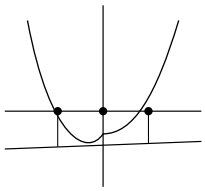
Mult. by E: add everything above the 2nd.

$a_1 x^2 + \ldots$

Take the quotient: $a + bx$



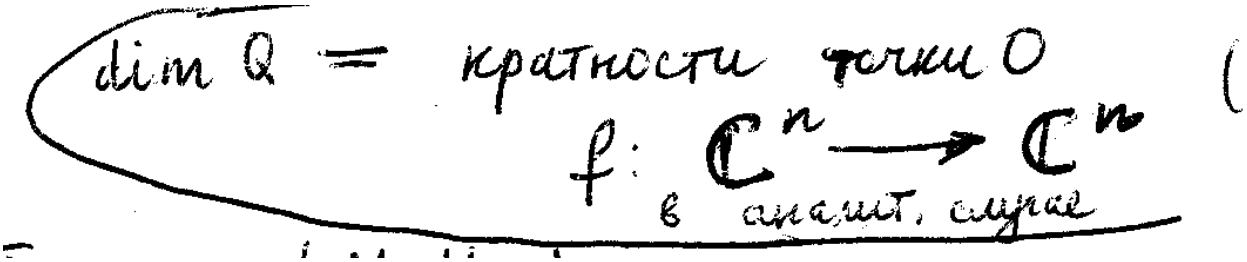
Typeset diagram



Original source details: GELFAND-P000185

the ring of functions on 2 coalesced points (an infinites. pair of points)

Let us move: it will remain so, and let $y \longrightarrow 0$.



Original source details: GELFAND-P000185

Typeset diagram

$dim\, Q =$ multiplicity of the point 0

$f : C^n \longrightarrow C^m$

in the analytic case

(in the general case)

Apparently so.

Theorem: (Mather)

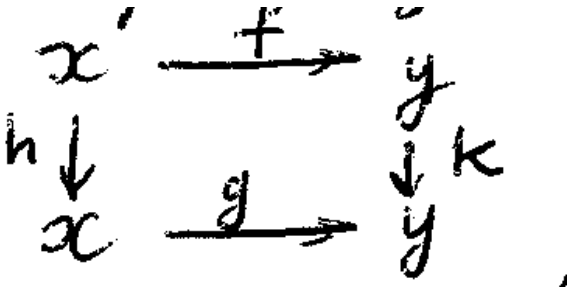
Stable sing. are equiv. $\Longleftrightarrow$ their local rings coincide.

Another theorem. (Mather)

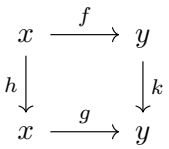
a sing. is stable $\Longleftrightarrow$ it is infinitesim. stable.

The latter means:

At the point: $y = f(x)$

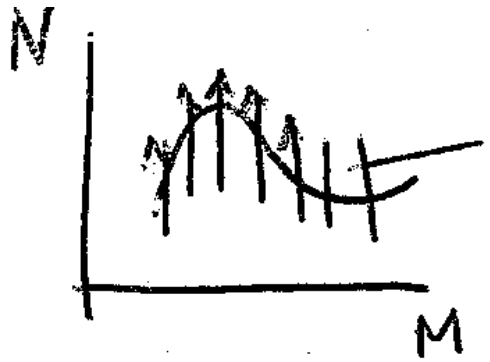


Typeset diagram

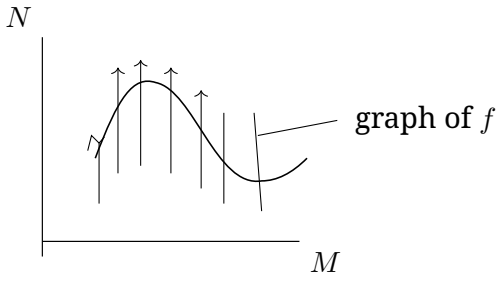


Original source details: GELFAND-P000185

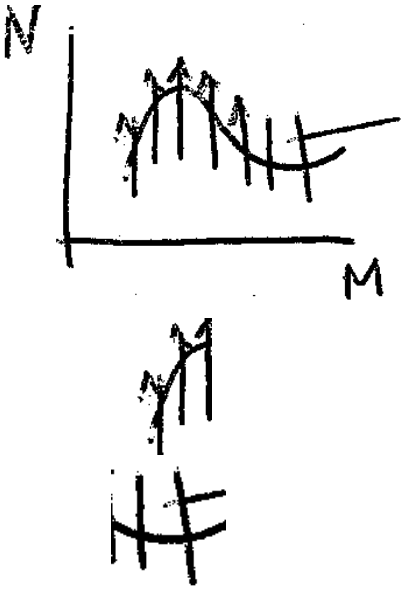
h and k must exist infinitesimally.



Typeset diagram



Original source details: GELFAND-P000185



Uncertain reading. Axes N upward and M rightward; a wavy graph with a maximum and minimum, and seven crossing segments. The four middle segments have distinguishable upper arrowheads; the densely bent far-left stroke does not give a certain direction. Separate arrowheads on the two right segments are undetermined. At right, an oblique stroke toward “graph of f”; exact attachments of dense strokes remain uncertain.

Original source details: GELFAND-P000185

A section of the bundle, parallel to N.

This is also a vect. field (a vertical vect. field).

GELFAND-P000186

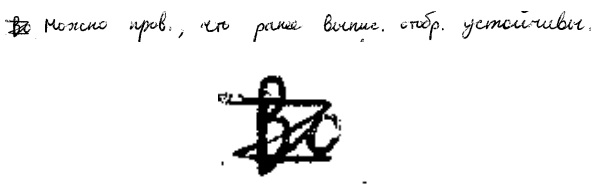
Every vertical field φ must be representable as a sum.

$$\varphi = f_*\xi + f^*\eta$$

(transfer)

$$\overset{\uparrow}{f_*}, \quad \overrightarrow{f^*}$$

[Cancelled: illegible] One can check that the maps written earlier are stable.



Uncertain reading. Before “One can check”, a short sign / contraction with a high upper loop is cancelled; the literal inscription is undetermined.

Original source details: GELFAND-P000186

Weierstrass prepar. thm.

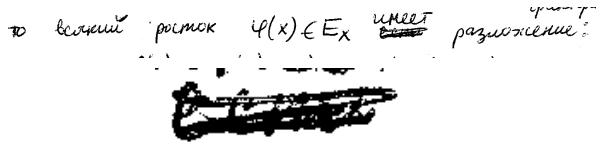
(parametric reduction)

If Q is finite-dimensional $(dim\, Q = r)$ and has generators

$$e_1(x), \ldots, e_r(x)$$

(functions that become generators after quotienting)

then every germ $\varphi(x) \in E_x$ [cancelled: ?] has an expansion:



Uncertain reading. Beneath the raised insertion “has”, a joined word with an initial loop and lower teeth is cancelled. The literal reading is undetermined; then follows “an expansion:”.

Original source details: GELFAND-P000186

$$\varphi(x) = \psi_1(y)e_1(x) + \ldots + \psi_r(y)e_r(x)$$

$dim\, Q$: the correct def. of multiplicity.

Theorem (Mather)

Σ^{20} is divided into parts.

1°. Five series of rings Q

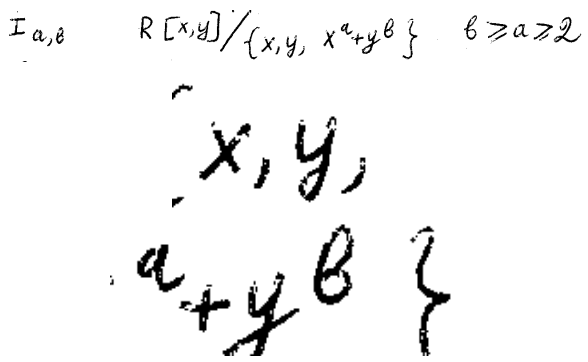
2°. Remainder $codim = \infty$

$f : M^m \longrightarrow N^n$

(need to def. a map of one of the series)

Singularities of the five series are stable:

$$n \geq m1. \quad \mathrm{I}_{a,b} \qquad R[x,y]/\{x, y, x^a + y^b\}\,[?] \qquad b \geq a \geq 2$$



Uncertain reading. In the first denominator, a separate low stroke between x and y resembles a comma: $\{x, y, x^a + y^b\}$ / $\{xy, x^a + y^b\}$. A separate low stroke is visible; the first and second exponents are a, b.

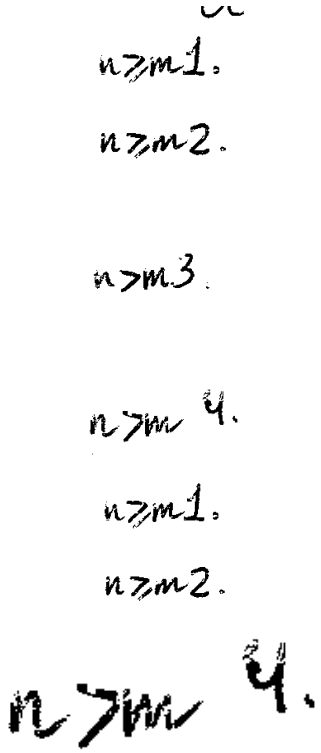
Original source details: GELFAND-P000186

$$n \geq m2. \quad \mathrm{II}_{a,b} \qquad /\{xy, x^a - y^b\} \qquad b \geq a \geq 2 \quad \text{even}$$

$$n > m3. \quad \mathrm{III}_{a,b} \qquad /\{x^a, y^b, xy\}$$

$$n > m4. \quad \mathrm{IV}_a \qquad /\{x^a, xy^{a-1}, x^2 + y^2\} \qquad a \geq 3$$

$$\mathrm{V}_a \qquad /\{x^a, x^2 + y^2\} \qquad a \geq 3$$



Uncertain reading. The four left lines read $n \geq m1.$, $n \geq m2.$, $n > m3.$, $n > m4.$. No separate arithmetic sign is distinguishable between m and the digits; these may be row numbers / part of an abbreviated condition. No separate plus or minus is distinguishable.

Original source details: GELFAND-P000186

The codimensions have been calculated.

(for all series)

The symb. Σ^{20} is strat. into subvarieties

(This is the strongest res. on the classific. of singularities)

Next time $13/II$ — 1967.

GELFAND-P000187

Gelfand sem.

Plan.

1.	Kirillov Harish-Chandra	≈ 10 lectures.
2.	Anosov Maslov and semiclassics	2 lectures.
3.	Kazhdan A. Weil.	6–7 lectures.
4.	Dynkin Index.	3 lectures.
5.	Anosov Pfaff eqs. (1st order)	1 lecture.
6.	KG-functor (Bernstein)	2 lectures.

Next time Kirillov and Anosov.

Kirillov. Lecture 1.

A series of works by H.-Ch., leading to a descr. of all irred. unit. reps. of real semisimple Lie groups.

Today, the scheme of the proof of one theorem (on characters).

G: a real semisimple Lie group. (reductive—for ind.)

On it, consider diff. operators commuting with both left and right translations.

This is a polynomial ring in generators $D_1, \dots, D_\ell$: the rank of the group.

$$G = SL(n), \qquad \ell = n - 1,$$

one oper. of each degree, beginning with the second.

Gelfand: E_{ik}

$$x = (\dots), \qquad f(x(1 + te_{ik})) = \\ = f(x) + tE_{ik}f + \dots$$

In the full group, $\sum E_{ii}$ (it is triv. in SL)

$$\sum E_{ik}E_{ki}$$
$$\sum E_{ij}E_{jk}E_{ki}$$
$$\dots$$

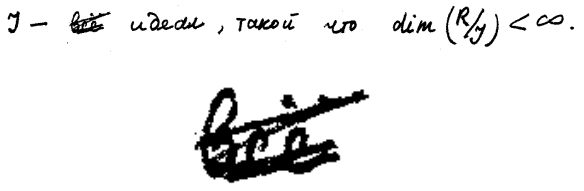
The rest are expressed through the first n.

(see Gelfand, 1947)

K. Consider generalized functions on the group that are “sols. of all these eqs.”.

R: the ring of all polynomials in $D_1, \dots, D_\ell$.

J: [cancelled: illegible] an ideal such that $dim\,(R/J) < \infty$.



Uncertain reading. Before “ideal”, a short word is cancelled: a high initial loop and several crossed lower strokes. The readings “vse” / “vsyo” (“all”) are undetermined.

Original source details: GELFAND-P000187

GELFAND-P000188

[source number: 51]

Consider generalized functions T such that

$$T^x = T \quad \text{for } x \in G$$

(x: an inner autom.)—const. on classes of conj. elements.

$$\langle T^x, f\rangle = \langle T, f^{x^{-1}}\rangle, \qquad f^{x^{-1}}(y) = f(x^{-1}yx)$$

G.: $D_1, \dots, D_\ell$: Laplace operators on the group.
Let $D_iT = \lambda_i T$; then T is an eigenfunction. (generalized)
Among them, seek "spherically symm." (zonal) ones

$$T(y) = T(x^{-1}yx).$$

Then it is an actual function and, moreover, summable (locally)
[In the right margin: Schwarz[?]]



Original source details: GELFAND-P000188

Uncertain reading. An underlined surname in the margin: "Schwartz" / "Schwarz"; the middle joined letters are compressed, and long strokes cross and surround the inscription. Its exact spelling is undetermined.

One can consider generalized eigenfunctions:

$$(D - \lambda)^2 T = 0$$

(not R, but $JT = 0$).

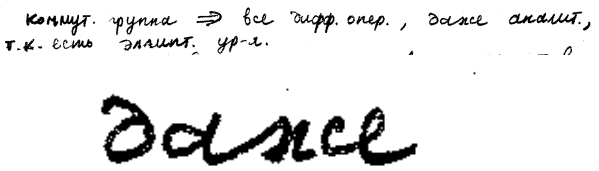
$$r(D_1, \dots, D_\ell)T = 0, \qquad r \in J.$$

For a max ideal, these are simply eigenfunctions (λ_i)
It is not necessarily continuous!
K.

$$\begin{aligned} &1) \quad T^x = T \quad \text{for } x \in G \\ &2) \quad DT = 0 \quad \text{for } D \in J \end{aligned}$$

Then it is locally summable.

Abelian group ⇒ all diff. operators, even[?] analytic, since there is an elliptic eq.



Uncertain reading. The joined word before "analytic": "even" / "further" (source dazhe / dalee). The converging middle strokes admit both readings; the literal reading is unresolved.

Original source details: GELFAND-P000188

General idea: in a nbhd., functions actually depending on ℓ params., satisfying ℓ eqs., construct an elliptic eq., away from singular points.

Characters become 1 at ∞—may have discontinuities.

How is this proved?

The set of classes of conj. elements is arranged thus: open subsets that are subvarieties $GL(n)$: distinct eigenvalues.

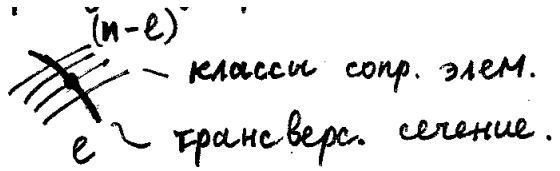
Diagon. of codim. 2 or 3, in turn, has subsets, etc. (2 eigenvalues coincide, etc.)—

—stratification.

GELFAND-P000189

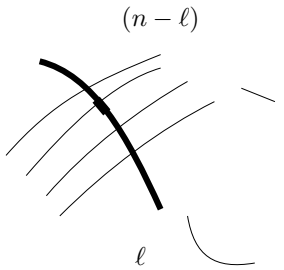
At a reg. point $y \in G$, this is an analytic function.

In a nbhd. of y, construct a new coord. system.



Original source details: GELFAND-P000189

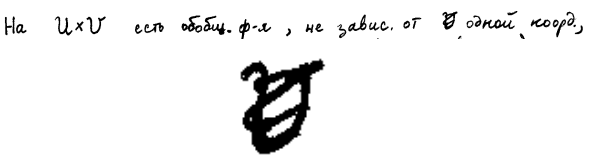
Typeset diagram



classes of conj. elements

transv. section.

On $U \times V$, there is a generalized function independent of [cancelled: illegible] one coord.,



Uncertain reading. Between "of" and "one coord.", a small sign is heavily cancelled: V / $\forall$ / another sign. Its literal reading is undetermined; the word "one" follows the cancelled sign.

Original source details: GELFAND-P000189

reducing to a generalized function on the section. (see L. Schwarz)

$$D_1, \ldots, D_\ell$$

generate almost all diff. operators on the section

(already of finite codim. in the ring of all diff. operators)

$$\Delta = D_1^2 + D_2^2 + \ldots + D_\ell^2$$

is already elliptic on the section.

here

$$\Delta^n + c_1 \Delta^{n-1} + \ldots + c_n \in J$$

Thus, $\Delta T = 0$, Δ is elliptic.

The arbitrary case reduces to this by induction on the dim. of the group.

3 different cases.

x — 1) semisimple
2) nilpotent
3) identity.

1)—reduces to the centralizer of y: a reductive group, of smaller dim. if y is not central. (then the whole group)

2) nilpotent—a special device

3) $x = e$: ln, exp reduce to the study of the Lie algebra—

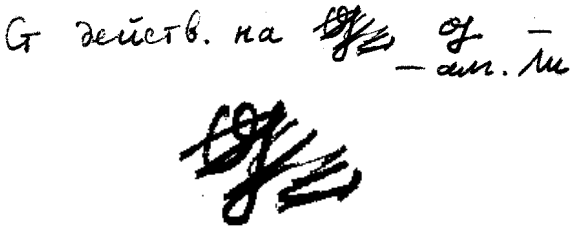
—2 separate works.

The idea for semisimple elements: passage to polar coords.

(radial part of the Laplace operator)

a: a semisimple element

G acts on [cancelled: ?] g—
—Lie alg.



Uncertain reading. After “on”, before the legible g, a symbol with loops and a long lower stroke is cancelled: g / another sign. The original inscription is undetermined.

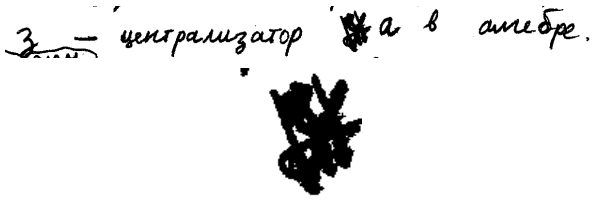
Original source details: GELFAND-P000189

if Ad_a $(z \longrightarrow aza^{-1})$

is reduced to diagonal form, i.e. without Jordan blocks

(For $SL(n)$, the matrix itself has no Jordan blocks)

z: the centralizer of [cancelled: illegible] a in the algebra.

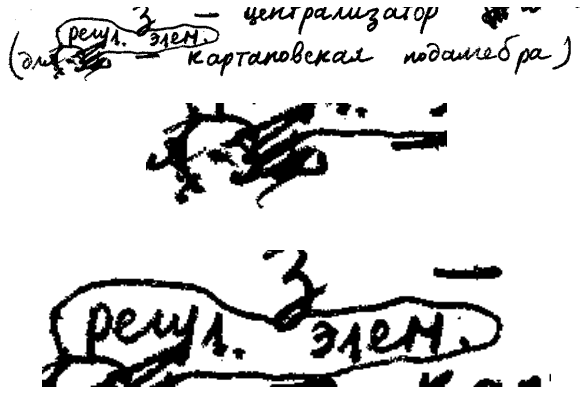


Uncertain reading. Before the separate legible a, a small symbol with several loops is heavily overwritten / cancelled. Its original reading is undetermined.

Original source details: GELFAND-P000189

reg. elem.

(for [cancelled: ?] —a Cartan subalgebra)



Original source details: GELFAND-P000189

Uncertain reading. After “for”, beneath the boxed insertion “reg. elem.”, an inscription is cancelled: a / x_0 / another symbol. Several original strokes merge with cancellation lines; the reading is undetermined.

GELFAND-P000190

[source number: 52]

Take a section (still redundant for now)

Berezin's thm. for reg. elements: restriction of operators to a Cartan subalgebra.

Ξ: the Lie group corresponding to $\mathfrak{z}$.

Construct a map

$$\begin{aligned} \varphi\colon G \times \Xi &\longrightarrow G \\ (x, y) &\longmapsto (ay)^x = xayx^{-1} \end{aligned}$$

Further notation:

$$V_a(y) = \det\bigl(Ad(ay)^{-1} - 1\bigr)_{\mathfrak{g}/\mathfrak{z}}$$

$$\Xi' = \{y \in \Xi : V_a(y) \neq 0\} \quad \text{— "reg. points" of } \Xi$$

($V_a \neq 0$, since the eigenvalue 1 is no longer present; it is on $\mathfrak{z}$)

Thm. 1 (Lemma 14)

$$\operatorname{rang} d\varphi = n = \dim G \qquad \text{on } G \times \Xi'$$

(locally this is the max of the rank: projection type)

Proof. X, Z: a pair of tangent vectors.

$$d\varphi(X, Z) = Z + \bigl(Ad(ay)^{-1} - 1\bigr)X,$$

from which everything follows.

Nondegenerate on the quotient.

Z provides full rank on the whole space.

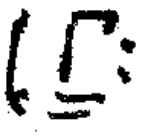
Result from another work.

$$d\mu \quad M^m \xrightarrow{\pi} N^n \quad d\nu \quad \text{— measures}, \qquad m \geq n$$

$\operatorname{rg} d\pi = n$.

One can transfer generalized functions from M to N (

Можно с M на N переносить обобщ. ф-и (Γ: локально) **Uncertain reading. Γ?**



Original source details: GELFAND-P000190

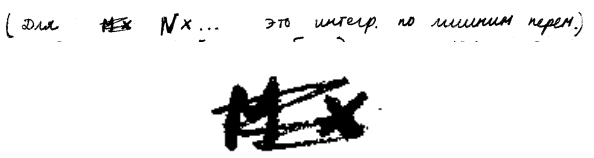
: locally)

$$\alpha \in C_0^\infty(M) \quad \longrightarrow \quad f_\alpha \in C_0^\infty(N),$$

so that $\forall F$ on N there will be

$$\int_M \alpha \cdot F \circ \pi \, d\mu = \int_N F f_\alpha \, d\nu$$

(For [cancelled:



Uncertain reading. $M\times$?

Original source details: GELFAND-P000190

] $N \times \ldots$, this is integration over the extra variables.)

This is done linearly, a partition of unity; locally this is a direct product.

GELFAND-P000191

Perspective: apply to φ.

Useful notation. (for illegibility)

On $M \times N$, there is a function (smooth)

D_1, D_2 —diff. operators on M, N.

$$f(x; D_1 : y; D_2) = (D_{1x} \times D_{2y} f)(x, y)$$

(apply D_1 in x and D_2 in y)

The identity operator is omitted.

$$f(x : y; D_2) \quad —$$

—explained in only one work.

Another notation.

$$G, \quad f(x) \text{ on } G, \qquad f(x : y) = f(xyx^{-1})$$

Next time, generalize Berezin's theorem.

For identity operators D_1 and D_2:

$$f(x : y)$$

and everything applies to it.

Dynin.

X —a smooth compact manifold

E_1, E_2 —bundles on X

$$A \colon \Gamma(E_1) \longrightarrow \Gamma(E_2) \quad \text{— lin. diff. operator}$$

Pseudodiff. operators.

$\{U_i\}$ $\quad \varphi_i$ — subord. partition of 1

$$\psi_i \varphi_i = \varphi_i$$

1) $\psi_i A(\varphi_i \cdot)$ — an ordinary pseudodiff. operator in the space of vector-valued functions

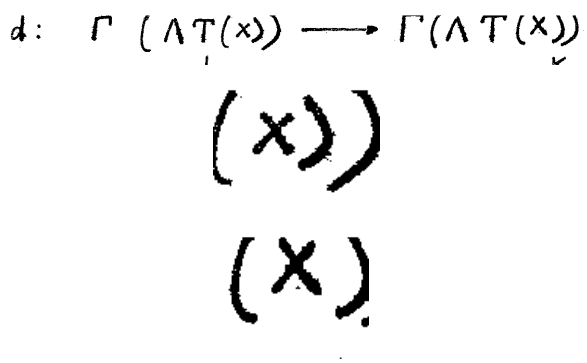
2) $\varphi\psi = 0 \;\Rightarrow\; \psi A(\varphi \cdot)$ a smoothing operator.

GELFAND-P000192

[source number: 53]

Example:

$$d\colon \Gamma(\Lambda T(x)) \longrightarrow \Gamma(\Lambda T(X))$$

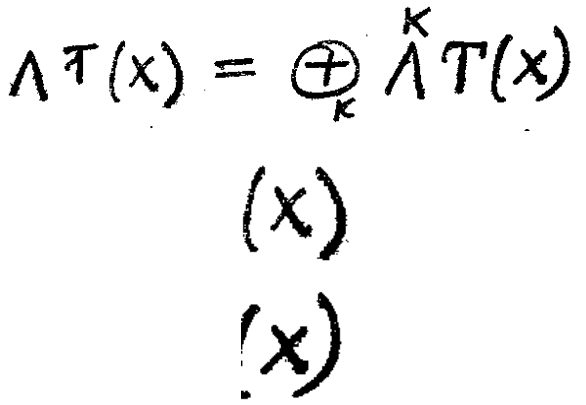


Uncertain reading. The case x/X in both arguments of T is not fully distinguishable; lower-case x at left and capital X at right are typographic candidates.

Original source details: GELFAND-P000192

all forms (inhomog.)

$$\Lambda T(X) = \bigoplus_k \Lambda^k T(x)$$



Uncertain reading. In both T arguments, crossed x/X signs of varying size; their case is undetermined. Candidates: X at left and x at right.

Original source details: GELFAND-P000192

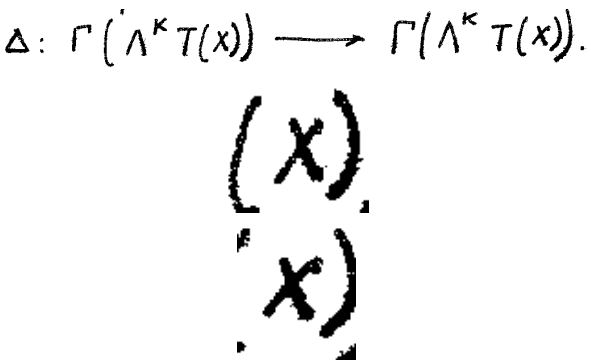
δ: —the adjoint operator (in the Riem. metric)

$$\delta = d^*.$$

$d + \delta$—the Cauchy-Riemann operator

$$\Delta = (d + \delta)^2 \quad \text{—the Laplace operator.}$$

$$\Delta\colon \Gamma(\Lambda^k T(X)) \longrightarrow \Gamma(\Lambda^k T(X)).$$

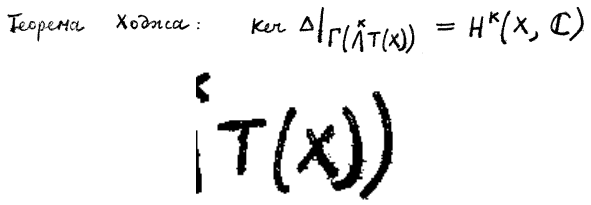


Uncertain reading. The undetermined case x/X affects both T arguments on this line; capital X signs are candidates, not established readings.

Original source details: GELFAND-P000192

Hodge's theorem:

$$\ker \Delta\big|_{\Gamma(\Lambda^k T(X))} = H^k(X, C)$$



Uncertain reading. In the argument of T beneath the restriction sign, a small crossed x/X sign; its case is undetermined. Candidate: X.

Original source details: GELFAND-P000192

(a very simple fact, see Moren, connection with the heat eq.)

Next time.

6–7 —Kirillov
7–8 —Anosov.

GELFAND-P000193

Gelfand sem. $20/\overline{\text{II}}$.
Kirillov. Lecture 2.

The case where the fibre has positive dimension. (xyx^{-1})—
—y is not central.
Let a be a semisimple element.
Take a section intersecting each class of conj. elements in 1 element.

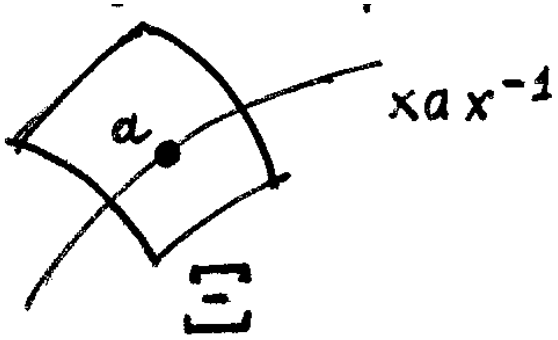
$$\mathfrak{g} \text{ — Lie alg.}, \qquad \mathfrak{z} = \{X : \operatorname{ad}_a X = X\}, \qquad \underset{\text{subgroup corresp. to } \mathfrak{z}.}{\Xi} \subset G.$$

The section is the conn. component of the centralizer of a.
Everything reduces to the group Ξ.
We shall construct a new generalized function and diff. operator.
How does one specify diff. operators on a Lie group?
Consider left-invariant diff. operators ($\exists$ a left-invariant one at $\forall$ point, agreeing with ours up to lower-order terms).

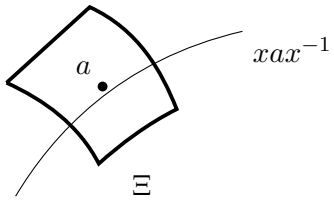
D_y: the left-invariant operator agreeing with D at y.

$D_y \in \mathfrak{g}$ describes the whole operator (on the whole group at once).

We shall restrict a diff. operator to a subset.



Typeset diagram



Original source details: GELFAND-P000193

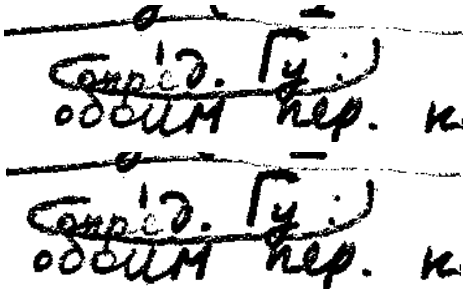
We shall regard diff. operators along the fibres as zero.

Fibres close to a may have greater dimension.

$$f(x; D_1 : y; D_2) = \underline{f(x : y; \Gamma_y(D_1 \otimes D_2))}$$

(D_1 in x, D_2 in y)

[def.?] Γ_y [?]

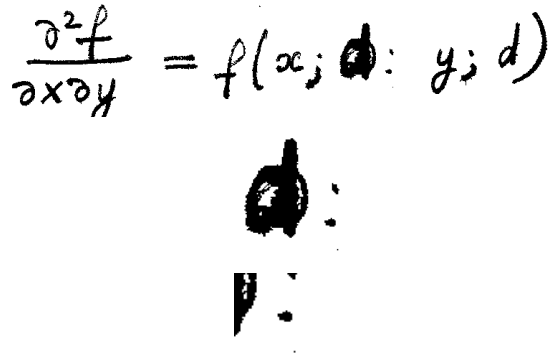


Uncertain reading. Inside the enclosure, candidate “def. Γ_y”; some strokes of the word and the final punctuation (; / ! / :) are undetermined.

Original source details: GELFAND-P000193

The explicit form of replacing operators in both variables by an operator in y.

$$\frac{\partial^2 f}{\partial x \partial y} = f(x; [d \,/\, \partial\, ?] : y; d)$$



Uncertain reading. The first operator after $x;$ is heavily overwritten: d / ∂. The square brackets give candidates; the following separator is a colon, and the second operator d is legible.

Original source details: GELFAND-P000193

Γ_y depends bilinearly on D_1, D_2, or—linearly on $D_1 \otimes D_2$.

GELFAND-P000194

[source number: 57]

Explicit expression for Γ_y. (g_1, g_2: elements of the enveloping algebra.

$$f(x; g_1 : y; g_2) = f(x : y; \Gamma_y(g_1 \otimes g_2)).$$
$$\Gamma_y(g_1 \otimes g_2) = \sigma_y(g_1) \cdot g_2,$$

where $\sigma_y(g_1)$ is a representation of the enveloping algebra, i.e.

$$\sigma_y(gh) = \\ = \sigma_y(g)\sigma_y(h).$$

Now it suffices to calculate σ_y on the generators of the enveloping algebra.

$$\sigma_y(X) = L_{X^{y^{-1}}} - R_X \qquad (X \text{ from the Lie algebra}).$$

$$\sigma_y(X) : g \longrightarrow X^{y^{-1}} g - gX$$

Derivation.

[Cancelled: (le...?]

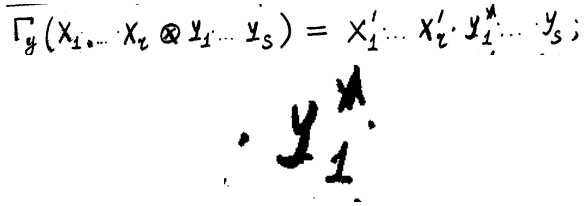


Uncertain reading. A short beginning, "(ст..." ("(le..."), beneath two cancelling lines; the further reading is undetermined.

Original source details: GELFAND-P000194

(leading term)—on a basis:

$$\Gamma_y(X_1 \cdots X_r \otimes Y_1 \cdots Y_s) = X_1' \cdots X_r' Y_1^{[*\,?]} \cdots Y_s;$$



Uncertain reading. Above Y_1, a sign with several rays: an asterisk or a cancelled prime. The brackets and question mark qualify the candidate.

Original source details: GELFAND-P000194

$$X_i' = (\operatorname{ad} y^{-1} - 1) \cdot X_i$$

[Cancelled: $f(x; g_1$?]



Uncertain reading. Candidates for the unfinished fragment: $f(x; g_1$ / $f(x, g_1$. The sign after $f($ and the separator are crossed out; no closing parenthesis is supplied.

Original source details: GELFAND-P000194

G.

$$X_1X_2\cdots X_rY_1\cdots Y_r\, f(xyx^{-1})$$

The operators must be transferred to y; Y_i remains, while
[cancelled: if?]



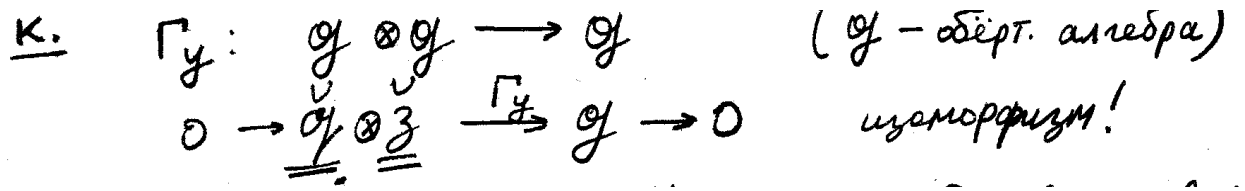
Uncertain reading. One heavily crossed-out word before the next line; candidate “if” is undetermined.

Original source details: GELFAND-P000194

if X commutes with y, nothing changes.
(It is easy to check all [inserted: this])

— · —

K.



Original source details: GELFAND-P000194

Typeset diagram

$$\begin{array}{lccccc} \Gamma_y: & \mathfrak{g}\otimes\mathfrak{g} & \longrightarrow & \mathfrak{g} & & (\mathfrak{g}\text{—enveloping algebra}) \\ & \cup \quad \cup & & & & \\ 0 \longrightarrow & \underline{\mathfrak{q}}\otimes\underline{\mathfrak{z}} & \xrightarrow{\Gamma_y} & \mathfrak{g} & \longrightarrow 0 & \text{isomorphism!} \end{array}$$

$\mathfrak{g} = \mathfrak{z} + \mathfrak{q}$,
[cancelled: where? / sub.?]



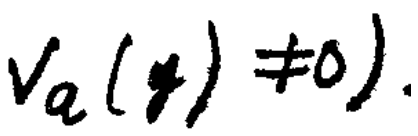
Uncertain reading. A short crossed-out word after the equality: “where” / “sub.”; middle and final strokes merge with the cancelling line.

Original source details: GELFAND-P000194

$\mathfrak{q}$: a complementary subspace, invariant under $\mathrm{Ad}\,\mathfrak{z}$: $[\mathfrak{z},\mathfrak{q}]\subset\mathfrak{q}$.

$\underline{\mathfrak{z}}$, $\underline{\mathfrak{q}}$ are now generated by them in the enveloping algebra.

Choose y regular (i.e. $[\nu/v/V]_a(y)\neq 0$). Then the map



Uncertain reading. The first sign in the condition: ν / v / V. Subscript a, argument y and $\neq 0$ are legible; the case and alphabet of the first sign are undetermined.

Original source details: GELFAND-P000194

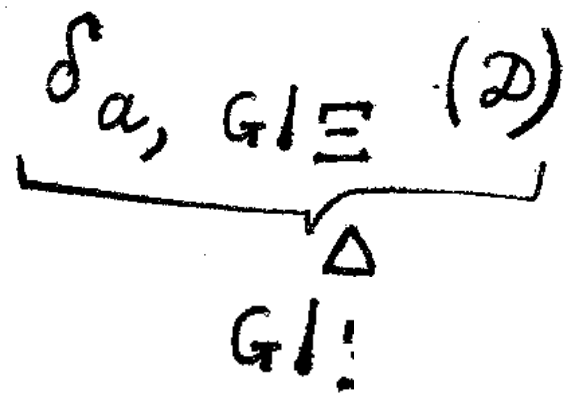
Γ_y, by the formula, is nondegenerate and unique in its leading term,

GELFAND-P000195

D: an operator on the group,

$$D \longrightarrow \underbrace{\delta_{a,\, G[|\,;\,/]\Xi}(D)}_{\Delta}$$

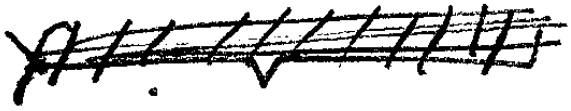
—an operator on $a\Xi$— —a submanifold.



Uncertain reading. Between G and Ξ in the subscript of δ, a nearly vertical stroke: | or /. The square brackets and semicolon in the typesetting separate candidates for one sign.

Original source details: GELFAND-P000195

The correspondence is constructed so that for functions constant on the orbits it is all the same.
[Cancelled formula: at the beginning, φ?; at the end, (D)?]



Uncertain reading. Possible separate remnants of the cancelled formula: φ at the beginning and (D) at the end. The complete formula is undetermined.

Original source details: GELFAND-P000195



Undeciphered passage. The middle and remaining undetermined part of the cancelled formula are illegible.

Original source details: GELFAND-P000195

$$\Delta_{ay} = \alpha(D_{ay})$$
$$D - \alpha(D) \in \Gamma_y(\mathfrak{q}^+ \otimes \mathfrak{z})$$

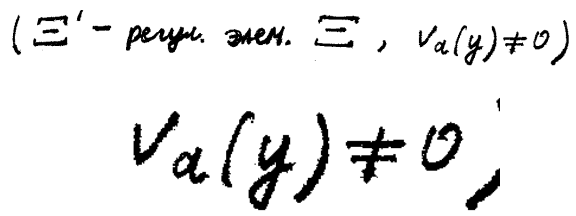
($\alpha(D)$: the constant term of D. $\mathfrak{q}^+$: elements having no constant term).
Next time, the transfer of generalized functions again.

$27/\overline{\text{II}}$ Gelfand Seminar.
Kirillov, Lecture 3.
Transfer of generalized functions.

Consider an auxiliary map:

$$\varphi : G \times \Xi' \longrightarrow G$$
$$(x, y) \longrightarrow xayx^{-1}$$

(Ξ': the regular elements of Ξ, $[\nu/v/V]_a(y) \neq 0$)



Uncertain reading. The V-shaped sign in the regularity condition: ν, v or V. Subscript a, argument y and condition $\neq 0$ are legible.

Original source details: GELFAND-P000195

$$\operatorname{rank} \varphi = \dim G.$$

Now generalized functions are transferred from G to Ξ. First they are transferred to $G \times \Xi'$.

T: generalized functions on G. We must learn to transfer ordinary functions in the forward direction. Roughly speaking, this is integration (locally).

GELFAND-P000196

[source number: 58]

$$M \longrightarrow N, \qquad C_c^\infty(M) \longrightarrow C_c^\infty(M)$$

Volumes dx and dy are given on M and N. (differential forms)

$$\alpha(x) \longrightarrow f_\alpha(y)$$

The condition must be satisfied: $\forall F(y)$

$$\int_M \alpha(x) F(\varphi(x))\, dx = \int_N F(y) f_\alpha(y)\, dy$$

Theorem: such a map exists and is continuous in the topology of these spaces.

Example.

$$\begin{matrix} M & = & N & \times & P \\ & & dy & & dz \end{matrix} \qquad (dx = dy\, dz).$$

Then the map is integration with respect to z.

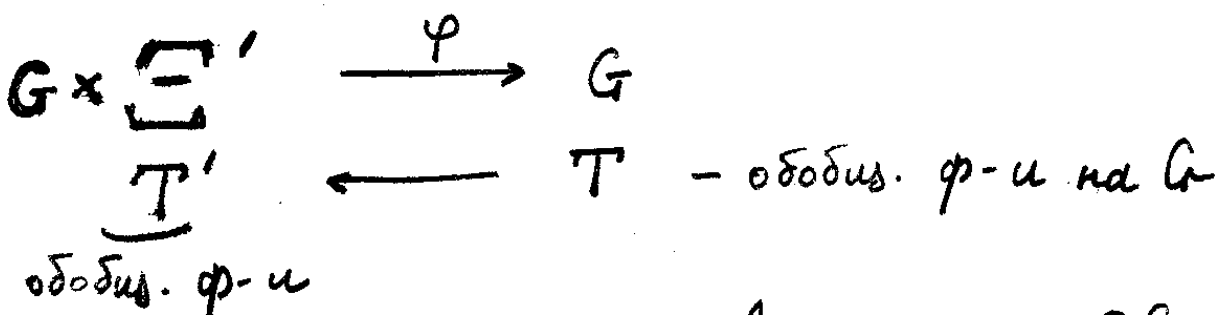
— · —

[Cancelled: M? / $\forall M$?]



Uncertain reading. Angular strokes before $G \times \Xi'$: M or $\forall M$; the cancelling lines cannot confidently be separated from the original signs.

Original source details: GELFAND-P000196



Original source details: GELFAND-P000196

Typeset diagram

$$G \times \Xi' \xrightarrow{\varphi} G$$

$$\underbrace{T'} \longleftarrow T \quad \text{—generalized functions on } G$$

generalized functions

T' is invariant under translations in its first factor

$$\langle T', f(x,y)\rangle = \langle T', f(x'x,y)\rangle$$

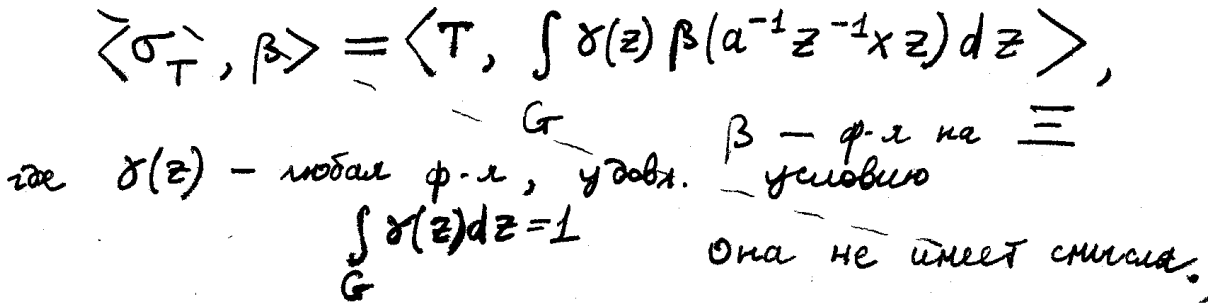
Lemma. A generalized function invariant under translations is $\int$ with respect to Haar measure (const)

$$— \cdot —$$

T' depends only on the second argument (integrate over the first).

$$T' = 1 \otimes \sigma_T$$

An explicit formula expressing σ_T through T can be given (K.)



Original source details: GELFAND-P000196

Typeset diagram

$$\langle \sigma_T, \beta \rangle = \left\langle T, \int_G \gamma(z)\beta(a^{-1}z^{-1}xz)\,dz \right\rangle,$$

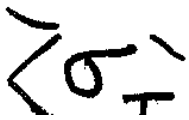
β: a function on Ξ

where $\gamma(z)$ is any function satisfying the condition

$$\int_G \gamma(z)\,dz = 1$$

It has no meaning.

Diagram and typed description. A sloping dashed line runs through the formula region toward “It has no meaning.” Its purpose—cancellation or connection with the note—is undetermined.



Original source details: GELFAND-P000196

GELFAND-P000197

Thus, we have:

$$D \longrightarrow \Delta \qquad = \delta_{a,G/\Xi}(D)$$
$$T \longrightarrow \sigma_T^{[/?]}$$

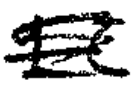


Original source details: GELFAND-P000197

Uncertain reading. A thin interrupted stroke at the upper right of σ: possibly σ'_T, or a separate nonmathematical stroke beside σ_T. The possible prime is qualified by brackets and a question mark.

$a \in U_G \subset G \qquad \Xi \supset U_\Xi \ni 1$

[Between them, cancelled: Ξ?]



Original source details: GELFAND-P000197

Uncertain reading. One cancelled sign with three horizontal levels; candidate Ξ is undetermined.

Lemma.

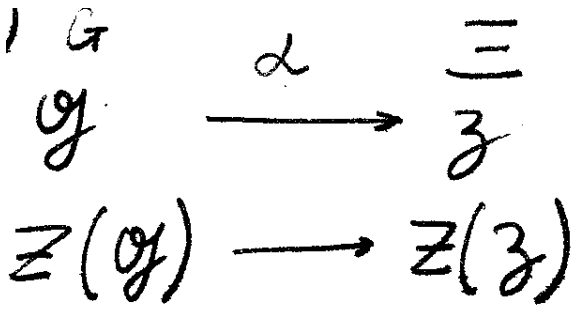
$$\sigma_{DT} = \Delta\sigma_T$$

(Essential use of $\Gamma_{ay} : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}$.)

Facts.

1) The operators Δ generate an ideal of finite codimension in the center of the group Ξ.

(On the Cartan subalgebra, these are operators symmetric under the Weyl group)

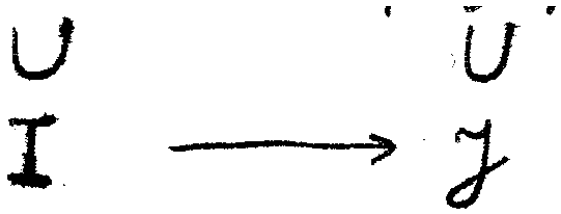


Typeset diagram

$$\begin{array}{ccc} G & & \Xi \\ & \overset{\alpha}{\longrightarrow} & \\ \mathfrak{g} & & \mathfrak{z} \\ & & \\ Z(\mathfrak{g}) & \longrightarrow & Z(\mathfrak{z}) \end{array}$$

Original source details: GELFAND-P000197

$Z(\mathfrak{g})$ is the center of the enveloping algebra; the image of $Z(\mathfrak{g})$ lies in the center $Z(\mathfrak{z})$ (can be checked). and the ideal it generates is of finite codimension.



Typeset diagram

$$\begin{array}{ccc} \cup & & \cup \\ & & \\ I & \longrightarrow & J \end{array}$$

Original source details: GELFAND-P000197

Then a complete reduction to Ξ is obtained

$$\sigma_T \text{ is locally integrable} \quad \Longrightarrow \quad T \text{ is locally integrable too}$$

a in the center $\Longrightarrow$ reduce by translation



Original source details: GELFAND-P000197

Undeciphered passage. The cancelled text after “reduce by translation” is completely covered by strokes and has not been read.

to the case $a = e$,

there is a special technique (3 papers



Original source details: GELFAND-P000197

Undeciphered passage. The cancelled text between “3 papers” and the name is completely covered by strokes and has not been read.

by Harish-Chandra)

GELFAND-P000198

[source number: 59]

6/III. Vishik Seminar. 1. Continuation of Agranovich's report. 2. Continuation of G. I. Eskin's report.

Gelfand Seminar.

Kirillov
$x^2 + y^2 - z^2 = c$, T is invariant under
[Cancelled: this gr...? / this f...?]



Original source details: GELFAND-P000198

Uncertain reading. A cancelled word and the beginning of the next: "this gr..." / "this f..."; the continuation is undetermined.

the group preserving this function.

$$P\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}\right) T = 0,$$

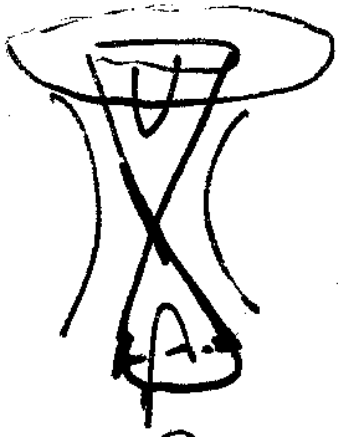
will T be locally integrable?



Original source details: GELFAND-P000198

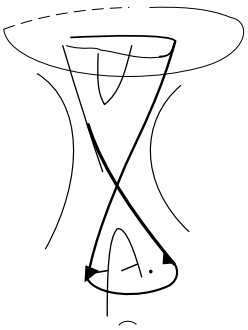
Undeciphered passage. One cancelled word before the following parenthesis is illegible; a period remains at the right.

(T: the closure of local functions constant on the layers)



Original source details: GELFAND-P000198

Typeset diagram





Original source details: GELFAND-P000198

Uncertain reading. The large upper oval, smaller upper loop, lower oval, crossing diagonals, concave side curves and internal arcs. The purpose of the internal A-/U-shaped strokes and dark lower endpoints is undetermined; no letter labels are assigned to them.

Outside the cone, the system is elliptic.

Arnold.

C. J. Earle, J. Eells.

X^2: a two-dimensional compact manifold (oriented)

$\mathrm{Diff}_0(X)$: the group of its diffeomorphisms homotopic to the identity.

Its topology has been calculated!

$$\mathrm{Diff}_0 X = \begin{cases} X = S^2\ [:\ ?] & \underset{\text{conformal maps}}{G} \times \underset{\text{contractible}}{\mathrm{Diff}(S^2 - a, b, c)} \quad (a, b, c \text{ fixed}) \\ X = T^2 & T^2 \times \underset{\text{contractible}}{\mathrm{Diff}(T^2 - a)} \\ \text{genus} \geq 2 & D \sim 0 \quad \text{contractible} \end{cases}$$



Original source details: GELFAND-P000198

Uncertain reading. After $X = S^2$, two dots at right and a separate sloping stroke at left. A colon with an additional stroke / a three-dot sign; the typeset colon is qualified by brackets and a question mark.

The proof is based on work by [Alfers / Alfors] and Bers on quasiconformal maps, to which any diffeomorphism contracts.



Original source details: GELFAND-P000198

Uncertain reading. The vowel after “ф” in the surname: “Альферса” / “Альфорса”, rendered as Alfers / Alfors.

Diff is a Lie group in the new understanding of smooth structure.

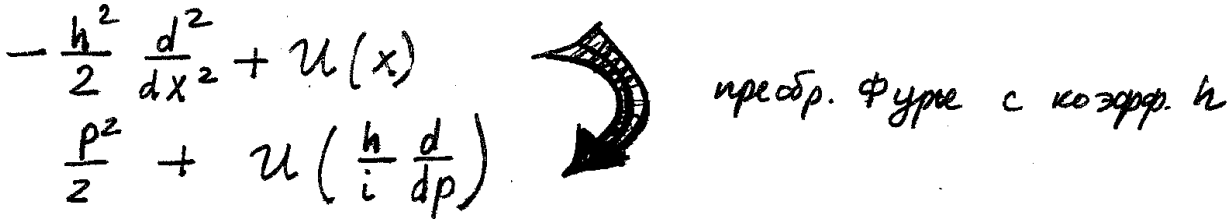
GELFAND-P000199

Anosov. Statement of the problem.

$Lu = f$, $u(x) \in H$, H is Hilbert.

In H there is an operator A; 0 is not an eigenvalue; A is self-adjoint.

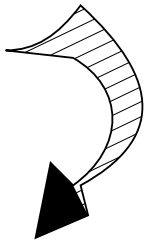
L may depend on A as well.



Original source details: GELFAND-P000199

Typeset diagram

$$-\frac{h^2}{2}\frac{d^2}{dx^2}+U(x)$$



Fourier transform with coefficient h

$$\frac{p^2}{2}+U\left(\frac{h}{i}\frac{d}{dp}\right)$$

(pseudodifferential operator)— $\overline{\mathcal{F}}_{p\to x}\,\sigma(x,p)\,\mathcal{F}_{x\to p}\,f(x)$

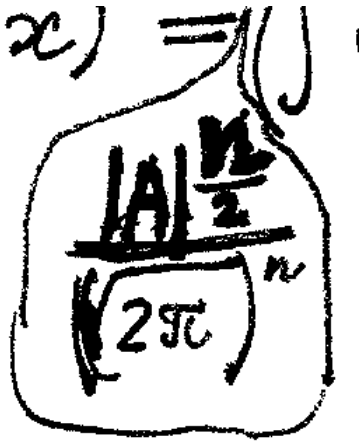
$$e^{\pm i\frac{1}{h}(p,x)}=$$

(h for dimensionlessness).

According to Maslov, $\frac{1}{h}$ is an operator; since everything is a function of h

$$=e^{\pm iA(p,x)}$$

A-Fourier transform:

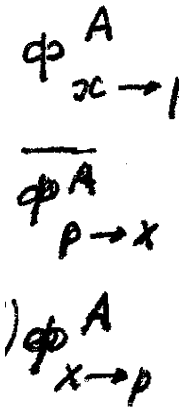


Original source details: GELFAND-P000199

Typeset diagram

$$\phi^{A}_{x\to p} f(x) = \left(\int e^{iA(x,p)} f(x)\, dx \right).$$

$$\frac{|A|^{n/2}}{(\sqrt{2\pi})^{n}}$$



Uncertain reading. The oval sign with a vertical stroke in this formula and the two final transforms: ϕ / Φ. Typeset using candidate ϕ; the case is undetermined.

Original source details: GELFAND-P000199

Consider

$$\mathcal{L}(x, \hat{p}, A^{-1}) f(x) = \overline{\phi^{A}}_{p\to x}\, L(x, p, A^{-1})\, \phi^{A}_{x\to p}\, f(x)$$

GELFAND-P000200

[source number: 60]

13/III Gelfand Seminar.
Parshin on Mumford.

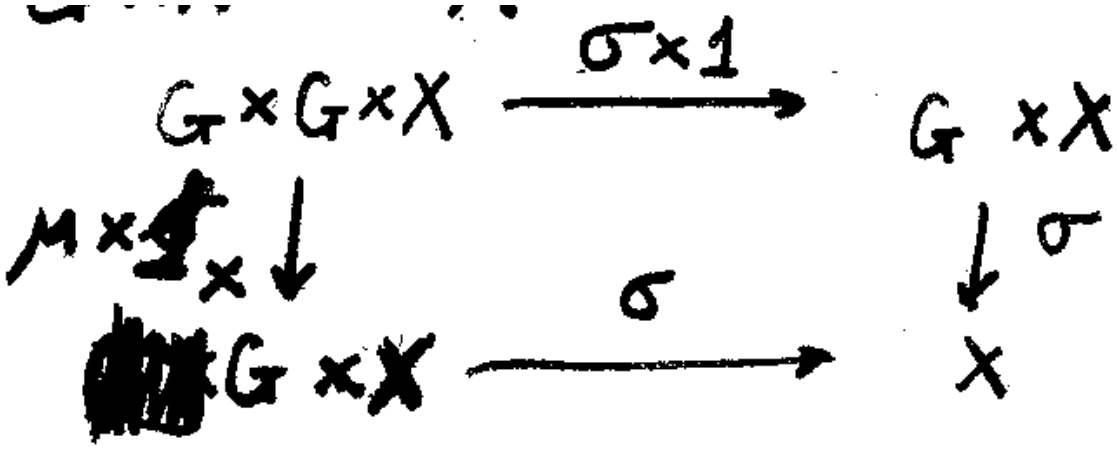


Uncertain reading. A separate blackened mark to the right of the heading. Its sign and purpose are undetermined.

Original source details: GELFAND-P000200

G is an algebraic group, X is a variety, G acts regularly on X,
if

$$G \times X \longrightarrow X$$

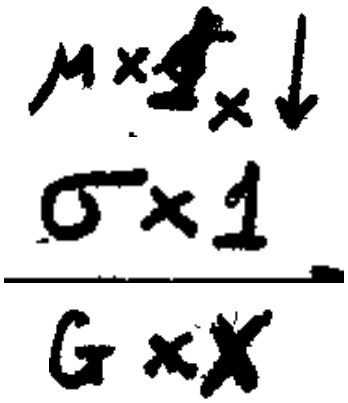


Original source details: GELFAND-P000200

Typeset diagram

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{\sigma \times 1} & G \times X \\ \downarrow{\scriptstyle \mu \times [1?]_X} & & \downarrow{\scriptstyle \sigma} \\ G \times X & \xrightarrow{\sigma} & X \end{array}$$

[cancelled: ?]



Uncertain reading. The left label: $\mu \times [1?]_X$; the overwritten middle sign is undetermined. Candidate 1 is qualified by a question mark. The top label reads $\sigma \times 1$; the lower X is overwritten.

Original source details: GELFAND-P000200



Original source details: GELFAND-P000200

Undeciphered passage. The completely cancelled former entry before the lower-left $G \times X$ is illegible.

Equivalent points—obtained by the action of G—an orbit.

The set of orbits—? When is it an algebraic variety, and in what sense?

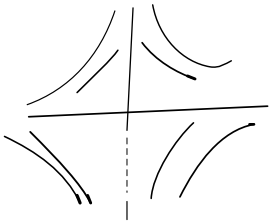
$$\underset{\text{orbits}}{X/G} \longleftarrow X$$

Property of regular maps of algebraic varieties.



Original source details: GELFAND-P000200

Typeset diagram





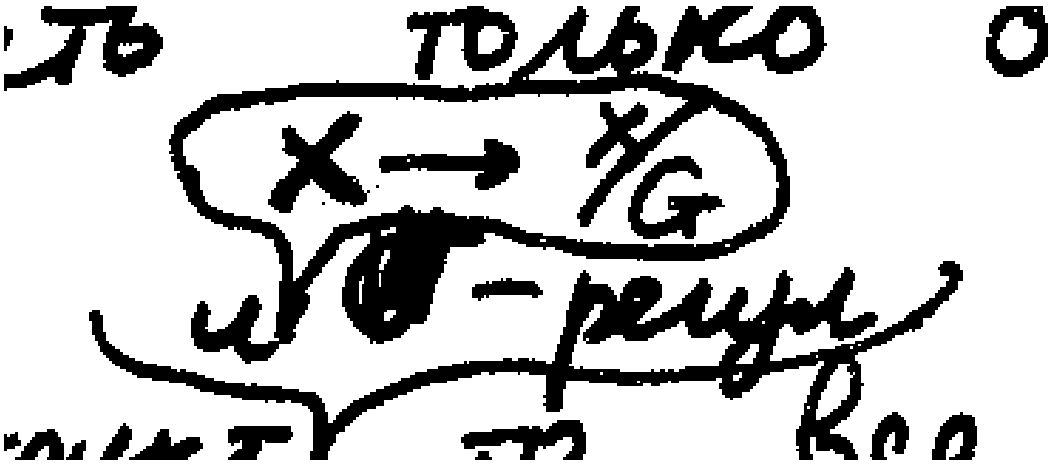
Original source details: GELFAND-P000200

Uncertain reading. Intersecting horizontal and vertical lines and two curved branches in each sector. There are no labels. Some ends are thickened and hooked; the presence and directions of possible arrows are undetermined.

A structure cannot be introduced; this is possible only after removing a point.

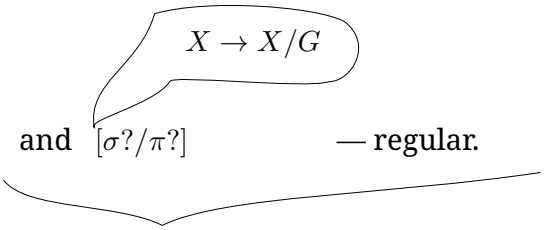
The dim of the stabilizer can only be larger (in comparison with the generic stabilizer).

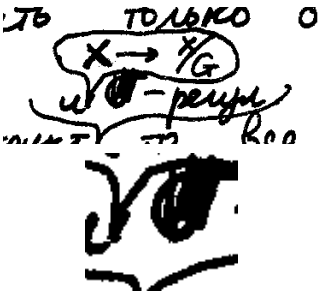
Theorem. If X/G has an algebraic structure



Original source details: GELFAND-P000200

Typeset diagram





Original source details: GELFAND-P000200

Uncertain reading. The insertion above the line, “and [overwritten sign]—regular,” is connected by an enclosure to $X \to X/G$. The loop-shaped sign: σ / π; the original reading beneath the heavy strokes is undetermined.

then all orbits must have the same dim. (in partic., be closed).

Reductive group (k of char. 0)

The quotient exists $\Longleftrightarrow$ all orbits are closed.

GELFAND-P000201

Dynin, Lecture 2.

$\Sigma(\mathbb{R}^n)$ from maps

$$\sigma\colon \mathbb{R}^n \times S^{n-1} \longrightarrow GL = \varinjlim GL(m, \mathbb{C})$$

with compact support

Outside $K_\sigma \times S^{n-1}$, $\sigma = 1$.

Theorem. ∃ one and only one map

$$\Sigma(\mathbb{R}^n) \xrightarrow{\text{ind}} \mathbb{Z},$$

with the following properties:

(Ī) Homot. stability.

(II̅)

$$\text{ind}(\sigma_0 \oplus \sigma_1) = \text{ind}\,\sigma_0 + \text{ind}\,\sigma_1$$

$\Sigma(x)$ from maps



Original source details: GELFAND-P000201

Uncertain reading. The argument after Σ is a crossed x/X sign; its case is undetermined. Candidate in the line: x.

$$\sigma\colon T_0(X) \longrightarrow GL = \varinjlim GL(m, \mathbb{C}),$$

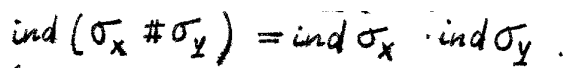
such that outside $K_\sigma \subset X$ we have $\sigma = 1$.

In the class of X whose tangent bundle is stably trivial, there exists one and only one function ind with properties Ī, II̅ and

(III̅)

$$\Sigma(X) \times \Sigma(Y) \xrightarrow{\#} \Sigma(X \times Y)$$

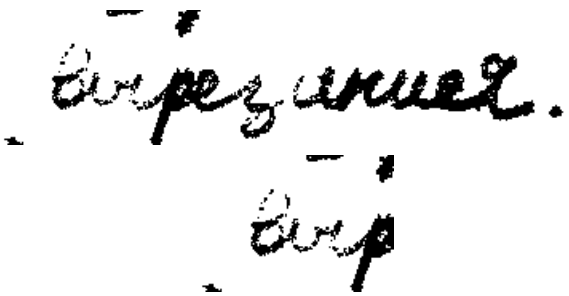
$$\text{ind}(\sigma_x \# \sigma_y) = \text{ind}\,\sigma_x \cdot \text{ind}\,\sigma_y.$$



Original source details: GELFAND-P000201

Uncertain reading. The four σ subscripts contain crossed x/X and forked y/Y signs; their case cannot be established from the strokes. Candidates in the line: x, y.

(IV̅) Invariant with respect to [excision / cutting off / trimming].

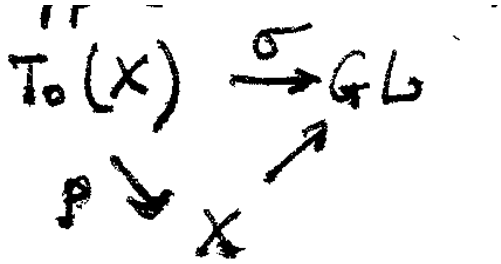


Original source details: GELFAND-P000201

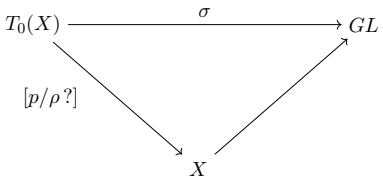
Uncertain reading. The beginning is faint; the ending “резания” is legible. Candidates: “вырезания” / “отрезания” / “обрезания” (excision / cutting off / trimming).

$$\text{supp}\,\sigma \subset U \subset X \Longrightarrow \text{ind}(\sigma|U) = \text{ind}\,\sigma$$

(V)



Typeset diagram



$\Longrightarrow \quad \operatorname{ind}\sigma = 0$

Original source details: GELFAND-P000201



Uncertain reading. The left sloping arrow label has a loop and a descending stroke: p / ρ. The candidate in the line is qualified by brackets and a question mark.

Original source details: GELFAND-P000201

(VI) For $X = \mathbb{R}^1$ there is the Noether-Muskhelishvili formula

$$\operatorname{ind}(\sigma) = \frac{1}{2\pi} \Delta_{\mathbb{R}^1 \times S^0} \arg\det\sigma.$$

GELFAND-P000202

20/III Continuation of Parshin's report.
The case of project. varieties.
27/III.

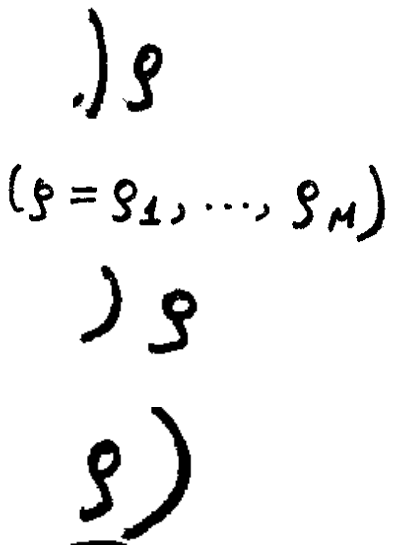
$$A(\omega, \eta_n) = A_-(\omega, \eta_n)\ M_- \underbrace{\qquad}_{(\eta_n+i)^{n_j(\omega)}\delta_{ij}} M_+ \qquad A_+(\omega, \eta_n)$$

But $n_j(\omega)$ depend on ω.
And under this assumpt. a regularizer is constructed.

One-dimensional problem

$$\mathcal{A}(\omega, \eta_n)u_+(\eta_n) + G(\omega, \eta_n)\varrho = \ell f(\eta_n) + u_-(\eta_n)$$

$$(\varrho = \varrho_1, \dots, \varrho_M)$$



Uncertain reading. The repeated auxiliary-unknown glyph has a closed upper loop and a lower curl: ϱ / s. Candidate at every corresponding occurrence: ϱ.

Original source details: GELFAND-P000202

$$\big(B(\omega, \eta_n), u_+(\ )\big)_{\eta_n} + E(\ )\varrho = g\big|_\ell$$



Uncertain reading. The lower glyph at the vertical stroke after g has a small loop: ℓ / e. Candidate in the line: ℓ; the literal letter is undetermined.

Original source details: GELFAND-P000202

(restr. to the bndry)

$$\ell - M = m_+(\omega) - m_-(\omega)$$

$\mathcal{A}_\omega(u_+, \varrho)$ — one-dimensional problem



Uncertain reading. The initial glyph in the lower underlined line: $\mathcal{A}_\omega(u_+, \varrho)$ / $\alpha_\omega(u_+, \varrho)$. The first candidate is in the line; the entire line has a wavy underline.

Original source details: GELFAND-P000202

It is solvable (there is exist. and uniqueness)

$R_\omega(f, g)$ depends cont. on ω.

GELFAND-P000203

The same in the n-dimensional case: $A(\xi', \xi_n)$ also has a right and a left regularizer. (factor out $|\xi'|$).

A_+, A_- — homog. of degree 0,

ξ_-^α enters M_-.

A regul. $R^\omega(\ell f; g)$ is constructed, and then localization. (with respect to ω)

See the work of Simonenko [In the left margin: Supposedly]



Original source details: GELFAND-P000203

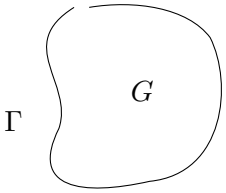
Uncertain reading. Immediately after the surname is a heavily inked vertical mark: ! / an overwritten illegible sign. A separate dot is indistinguishable.

(in abstr. form, a disc. of inverse operators of ellipt. oper., depending on a parameter) Eskin.



Original source details: GELFAND-P000203

Typeset diagram



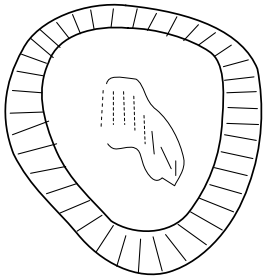
$A(x,\xi)$ — an ellipt. symbol.

A condition on the variation of $A(x,\xi)$ along the boundary.



Original source details: GELFAND-P000203

Typeset diagram





Original source details: GELFAND-P000203

Uncertain reading. Two irregular oval outlines with transverse hatching; the thin interior contour is visible with breaks. Its closure is undetermined. There are no labels or definite arrows inside.

$$\Gamma_\varepsilon = \Gamma \times [0,\varepsilon), \qquad \text{coord. } (x', \xi', x_n, \xi_n)$$

$$A(x,\xi)$$

$$a_+(x') = A(x', 0, 0, 1)$$

$$a_-(x') = A(x', 0, 0, -1)$$

$b(x') = a_+(x')^{-1} a_-(x')$ — measure of nonellipt.

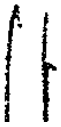
$$\gamma(x') = \frac{1}{2\pi i} \cdot \ln b(x')$$

Typeset diagram





Original source details: GELFAND-P000203



Original source details: GELFAND-P000203

Uncertain reading. To the left of a), b) are two nearly parallel long lines with lower arrowheads. The upper left branch and right side tick are visible; whether the left branch is an upper arrowhead is undetermined.

a) Req. that the branches of ln can be chosen so that [crossed out: на / их / illegible] $\gamma_j(x')$ — are cont. on all [Γ ?]



Original source details: GELFAND-P000203

Uncertain reading. After “чтобы” (so that), approximately two letters lie beneath two thick strokes: “на” (on) / “их” (their) / illegible. The writing is crossed out.



Original source details: GELFAND-P000203

Uncertain reading. The last heavily overwritten sign in a): Γ / an overwritten illegible sign. Candidate Γ in the line is qualified by a question mark.

(locally this is always poss.)

b) $|\operatorname{Re}\gamma_i(x') - \operatorname{Re}\gamma_j(x')| < 1$ on Γ

This is the class $\widehat{\mathcal{E}}_\alpha$.



Original source details: GELFAND-P000203

Uncertain reading. The main letter beneath the hat: $\widehat{\mathcal{E}}_\alpha$ / $\widehat{\varepsilon}_\alpha$. Subscript α and upper hat are visible; the first candidate is in the line.

(For fixed x' one can — argum. into one interv. $[1/\lambda/\uparrow ?]$ — they must not spread apart)



Original source details: GELFAND-P000203

Uncertain reading. The sign after "интерв." (interv.): 1 / λ / an upward arrow. Its shape does not establish an interval length.

$$\mathcal{A} = \begin{pmatrix} a_1(x,\xi) & & 0 \\ & \ddots & \\ 0 & & a_p(x,\xi) \end{pmatrix}$$

GELFAND-P000204

[source number: 65]

65

a: factorize — $\kappa_j(x')$.



Original source details: GELFAND-P000204

Uncertain reading. The looped glyph with subscript j and argument (x'): $\kappa_j(x')$ / $\chi_j(x')$ / $x_j(x')$. The first candidate is in the line.

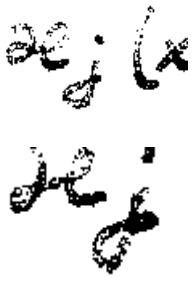
γ_j will be κ_j and this class is not obtained.



Original source details: GELFAND-P000204

Uncertain reading. The sign after "будут" (will be): κ_j / χ_j / x_j. Candidate in the line: κ_j.

$$b = e^{2\pi i \kappa_j(x')}, \qquad \gamma_j = \kappa_j$$



Original source details: GELFAND-P000204

Uncertain reading. Both right-hand looped glyphs: κ_j / χ_j / x_j. Candidate in the exponent and in the right-hand equality: κ_j.

This is for oper. in ordinary spaces.

Main question: when does there globally exist $[G \in E \,/\, G \notin E \,/\, G \,\&\, E \,/\, G$ and $E\,?]$ (and B)?



Uncertain reading. A heavily overwritten sign between G and E: $\in$ / $\notin$ / an ampersand / "and". All four readings remain open.

Original source details: GELFAND-P000204

27/III Atiyah's work as presented by S. Gelfand.
(following Petrovskii)
(Report No.319 in the Bourbaki sem.).
Consider a homog. hyperb. eqn with const. coefficients

$$Pu = \sum_{|\alpha|=m} a_\alpha D^\alpha u$$

in $\mathbb{R}^n$.
a_α — real.
[Crossed out at the left: R^n / Ha(On)]



Uncertain reading. An overwritten or crossed-out fragment: R^n / "Ha" (On). Neither reading is established.

Original source details: GELFAND-P000204

Hyperbolicity: (with respect to ξ_0):

$$\sum a_\alpha \xi^\alpha = a(\xi_0, \ldots, \xi_n)$$

$\forall$ nonzero tuple $(\xi_1, \ldots, \xi_n) \neq 0$

$$a(\xi_0, \ldots, \xi_n) = 0$$

has exactly m real distinct roots.
We seek the sol. of the problem

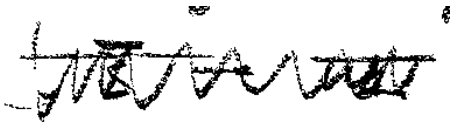
$$\begin{cases} Pu = \delta(x_0, \ldots, x_n), \\ \quad \text{with the addit. condition} \\ u = 0 \quad \text{for } x_0 < 0. \end{cases}$$

It is denoted E and is unique by the uniqueness thm for the sol. of the Cauchy problem.

It is known that it is arranged as follows:

$$\text{Cone} \qquad a(\xi_0, \ldots, \xi_n) = 0 \qquad A$$

Its dual cone B in the space $(x_0, \ldots, x_n)$.



Original source details: GELFAND-P000204

Undeciphered passage. An entire crossed-out entry is illegible. Possible candidate: “гиперпл.” (hyperpl.).

cone of hyperpl.,
tang. to A.

$$b(x_0, \ldots, x_n) = 0$$

One must take cone B, the solution is analyt. in $\forall$ connected comp. of the [complement (genitive) / complement (nominative)] of B.



Original source details: GELFAND-P000204

Uncertain reading. The ending of the last word before B: “дополнения” (complement, genitive) / “дополнение” (complement, nominative).

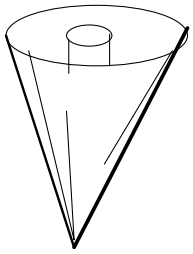
GELFAND-P000205

Domain $\overset{C}{/}$ — a lacuna if $E(x) \equiv 0$ on C.
(conn. comp.)



Original source details: GELFAND-P000205

Typeset diagram





Original source details: GELFAND-P000205

Uncertain reading. A downward-pointing cone, the upper oval, small looped contour, two short vertical strokes, and generators. The inner loop may be part of a section or a letter φ/ρ; its literal identity and the exact joins of the strokes are undetermined. No arrow directions are marked.

Stable lacuna — if it — is a lacuna for nearby eqns.
There is a nec. and suff. condition for the existence of a stable lacuna —
— certain cycles on alg. varieties are homologous to zero.
Eqn $a = 0$ — [м / и]



Uncertain reading. The short glyph after the dash: “м” (m) / “и” (and).

Original source details: GELFAND-P000205

$$(\xi_0, \ldots, \xi_n) \subset \mathbb{C}P^n$$



Uncertain reading. The rounded sign open to the right: $\subset$ / $\in$. No middle stroke is distinguishable; candidate in the line: $\subset$.

Original source details: GELFAND-P000205

$a(\xi) = 0$ — defines the variety A (complex)

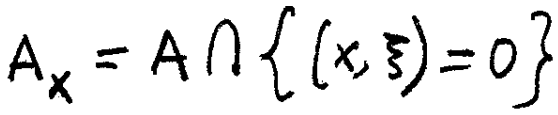
It is assumed that already in $\mathbb{C}P^n$ it is nonsingular (there are no complex singular points).
This can always be achieved by a real change of coeff. (small).

$$A_{\mathbb{R}} = A \cap \mathbb{R}P(n)$$

Let there be a point $\begin{matrix} x \\ \| \\ (x_0, \ldots, x_n) \end{matrix}$. We shall construct 2 cycles from it

1.

$$x_0\xi_0 + x_1\xi_1 + \ldots + x_n\xi_n = 0 \quad \text{— a plane (complex)}$$



Original source details: GELFAND-P000205

Typeset diagram

$$\begin{matrix} A_x & = A \cap \{(x, \xi) = 0\} \\ | & \\ \text{also has} & \\ \text{a real part} & \end{matrix}$$

$$(A_x)_{\mathbb{R}} = A_x \cap \mathbb{R}P(n); \qquad \text{its dim} = n - 2$$

This is a chain in homology (a cycle) — a sum of spheres:

$$\beta_x \in H_{n-2}(A_x)$$

For the wave eqn.

$$\frac{\partial^2}{\partial x_0^2} - \Delta u$$

$$x = (1, 0, \ldots, 0).$$

$$A = \{\xi_0^2 - \xi_1^2 - \ldots - \xi_n^2 = 0.\}$$

$$A_x : \quad \xi_0 = 0. \Rightarrow A_x = \{\xi_1^2 + \ldots + \xi_n^2 = 0\} \quad —$$

— in project. space this will be 0 (no points)

GELFAND-P000206

[source number: 66]

2. Another cycle.

Consider the map

$$\xi \mapsto \text{sign}(x, \xi),$$
$$\xi \in A_R,$$



Original source details: GELFAND-P000206

Typeset diagram

~~$\alpha =$ a cycle.~~



Original source details: GELFAND-P000206

Typeset diagram

~~$\partial\alpha \in (A_R)_x$~~

It gives a cycle.

> Split A_R by the plane
> (x, ξ) into 2 parts
> and one with $+$, the other
> with $-$.

$$\{(x, \xi) > 0\} - \{(x, \xi) < 0\}$$
$$(a \in C_{n-1}(A))$$

We obtain an $(n-1)$-dim. chain on $A \quad (C_{n-1}(A))$

$$\partial a \in (A_x)_R \quad \text{— a cycle}$$

Thus an elem. $\alpha_x \in H_{n-1}(A, Ax)$ is defined, and



Original source details: GELFAND-P000206

Uncertain reading. The second group argument looks like Ax on a common baseline; candidate A_x is also possible. In the line: $H_{n-1}(A, Ax)$.

from the exact homology seq.:

$\partial\alpha_x \in H_{n-2}(A_x)$

[Crossed out: and / H / another sign],



Uncertain reading. The densely crossed-out sign between the group and comma: “и” (and) / H / another sign.

Original source details: GELFAND-P000206

$\partial\alpha_x = 2\beta_x$.

If x varies in a connected comp. of the cone’s complement, then α_x, β_x do not change.



Uncertain reading. The crossed-out parenthesis: “(in $H_{n-\ldots}$)” / “(in H_n)”; the exact index beneath the dense strokes is undetermined.

Original source details: GELFAND-P000206

α_c, β_c — the corresp. cycles. (homology elem.).

$$\alpha_c \in H_{n-1}(A, A_c), \qquad \beta_c \in H_{n-2}(A_c)$$

Theorem. m — order of the eqn, $n+1$ — dim. of the space, m, $n+1$ are even.

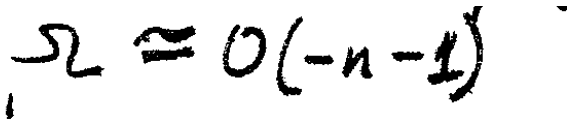
a) $m \geq n+1 \Rightarrow \alpha_c = 0$ — a nec. and suff. condition for c to be a stable lacuna

b) $m < n+1 \Rightarrow \beta_c = 0$ — ——— // ——————————— // ———

——— // ———————————————

Proof: the explicit form of E.

On $CP(n)$ consider a form of type $(n,0)$ (holomorphic) with a single pole of order $(n+1)$ (it is unique)



Typeset diagram

$\Omega \quad \cong O(-n-1)$

|

holom. forms

Original source details: GELFAND-P000206

There are no global reg. sections. (hence uniqueness)

$$\omega = x_0\, dx_1 \wedge \ldots \wedge dx_n - x_1\, dx_0 \wedge dx_2 \wedge \ldots$$

— (in affine space) — area of the sphere.

GELFAND-P000207

Use the residue form ω_a on the surface $a = 0$.

$$\frac{\omega}{a} = \frac{da}{a} \wedge \omega_a + \Theta$$



Uncertain reading. The last sign and its repetition in the parenthesis have an oval with a horizontal bar: Θ / θ. Case is undetermined; candidate in the line: Θ.

Original source details: GELFAND-P000207

($\Theta = 0$ on $[a = 0 / a = \varphi ?]$)



Uncertain reading. The last heavily overwritten sign in the underlined parenthesis: $a = 0$ / $a = \varphi$. Both readings remain open.

Original source details: GELFAND-P000207

$m \geq n + 1$ — the Herglotz—Petrovskii formula

$$E(x) = c \int_{A_R} \operatorname{sgn}(x, \xi)(x, \xi)^{m-n-1} \omega_a$$

(see the 1st issue)

gen. func.